



UNIVERSITEIT VAN AMSTERDAM

Portfolio Theory

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INTRODUCTION. In this course, we delve into the stochastic foundations of mathematical finance, focusing on the general problem of pricing and hedging European contingent claims and portfolio optimisation. We rigorously define the concept of a financial market and establish two fundamental relationships: one between arbitrage-free markets and the existence of equivalent martingale measures, and the other concerning the completeness of a market and the uniqueness of the equivalent martingale measure. We explore various methodologies for pricing and hedging contingent claims in complete and incomplete markets, including risk-neutral, variance-optimal, and utility and risk-indifferent approaches. We also examine how market participants' preferences can be quantified using utility functions and risk measures, and how these concepts are applied in portfolio optimisation. Furthermore, we discuss the formulation of consumption–investment problems as stochastic optimal control problems and introduce dynamic programming as a general solution method. This course provides students with a thorough understanding of the core principles of mathematical finance and their application in financial decision-making and risk management under uncertainty.

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Modelling Financial Markets

SECTION 1.1

Financial Markets in Finite Discrete-Time

Basic Stochastic Concepts of Mathematical Finance

The Probability Space

We work within a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω represents the set of all *outcomes* $\omega \in \Omega$; $\mathcal{F}$ is a σ -algebra on Ω containing all *events* $A \in \mathcal{F}$; and $\mathbb{P}$ denotes a probability measure on the measurable space $(\Omega, \mathcal{F})$. We sometimes refer to $\mathbb{P}$ as the *physical probability measure* since it describes the real-world probabilities associated with the occurrence of events $A \in \mathcal{F}$. A *finite probability space* is any probability space $(\Omega, \mathcal{F}, \mathbb{P})$ for which there exists an integer $N \in \mathbb{N}$ and outcomes $\omega_1, \omega_2, \dots, \omega_N$ such that $\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$, and $\mathbb{P}(\{\omega_i\}) > 0$ for all $i \in \{1, 2, \dots, N\}$.

Time and the Flow of Information

We consider time to progress in discrete steps over a finite horizon. This means that there exists a $T \in \mathbb{N}$ and real numbers $0 \leq t_0 < t_1 < \dots < t_T < \infty$ representing the *trading times*. Depending on the application, the trading times $t_0, t_1, t_2, \dots, t_T$ could correspond to days, months, or years, as well as seconds, minutes, or hours. For notational simplicity, we use the index of the trading times instead of the trading days themselves, i.e., the set of trading times is given as $\mathbf{T} := \{0, 1, \dots, T\} \subset \mathbb{N}_0$ with finite *time horizon* $T \in \mathbb{N}$.

When modelling time-dependent random phenomena, the natural chronological order implied by the flow of time must be respected. This means that an event observed yesterday reveals information that we still remember today, but today we do not necessarily know what will happen tomorrow. In probability theory, this chronological ordering of information is represented by a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbf{T}}$, with each $\mathcal{F}_t \subseteq \mathcal{F}$ for $t \in \mathbf{T}$ being a sub- σ -algebra such that $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $0 \leq s \leq t \leq T$. The filtration $\mathbb{F}$ is interpreted as a flow of information with $\mathcal{F}_t$ containing all information available up to and including the t -th trading time. Usually, we assume that:

- (i) $\mathcal{F}_0$ is $\mathbb{P}$ -trivial, i.e., $\mathbb{P}(A) \in \{0, 1\}$ for all $A \in \mathcal{F}_0$, i.e., all $\mathcal{F}_0$ -measurable random variables are constant $\mathbb{P}$ -a.s.;
- (ii) $\mathcal{F} = \mathcal{F}_T$, i.e., all events occur in the finite time horizon.

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a filtration $\mathbb{F}$, we call the tuple $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F})$ a *filtered probability space*.

Stochastic Processes and Financial Markets

An $\mathbb{R}^d$ -valued stochastic process in finite discrete-time $\{0, 1, \dots, T\}$ is any family of $\mathbb{R}^d$ -valued random variables $X_0, X_1, \dots, X_T$ defined on $(\Omega, \mathcal{F}, \mathbb{P})$. We denote a stochastic process by $(X_t)_{t \in \mathbf{T}}$ or simply X .

DEFINITION 1.1.

A stochastic process X is said to be *adapted* to $\mathbb{F}$ if X_t is $\mathcal{F}_t$ -measurable for all $t \in \mathbf{T}$ and it is called *predictable* with respect to $\mathbb{F}$ if X_t is $\mathcal{F}_{t-1}$ -measurable for all $t \in \{1, \dots, T\}$.

In mathematical finance, stochastic processes are employed to describe the random evolution of financial quantities, including asset prices and dynamic trading strategies. An asset may encompass various entities, from a bank account to stocks, bonds, commodities, options, and futures, as long as the asset's price process $(S_t)_{t \in \mathbf{T}}$ is *observable* at every trading time. In mathematical terms, this translates to $(S_t)_{t \in \mathbf{T}}$ being adapted. Suppose there are $d + 1 \in \mathbb{N}$ tradable assets. Given a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F})$ and an $\mathbb{R}^{d+1}$ -valued asset price process $X = (S^{(0)}, S^{(1)}, \dots, S^{(d)})$ adapted to $\mathbb{F}$, then we call the tuple $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ a *financial market*.

EXAMPLE 1.2 (Interpretation of Adapted and Predictable Processes in Finance). An adapted process in mathematical finance may be the price process $(S_t)_{t \in \mathbf{T}}$ of some stock. Here, adaptedness can be interpreted as follows: at each trading day $t \in \mathbf{T}$, the asset price S_t is known given the information in $\mathcal{F}_t$, which means that it can be *measured* by checking, e.g., your brokerage account. This literally translates to S_t being $\mathcal{F}_t$ -measurable for all $t \in \mathbf{T}$, which is precisely the definition of adaptedness. For most risky assets, e.g., stocks or commodities, their future price is uncertain, indicating that S_t is not measurable with respect to $\mathcal{F}_{t-1}$, hence the price process of a risky asset $(S_t)_{t \in \mathbf{T}}$ is usually *not* predictable.

In contrast, if $(B_t)_{t \in \mathbf{T}}$ represents a bank account, its value can be measured at every trading day t , and is actually already foreseeable at time $t - 1$ as the interest rate to be received the following trading period, say over a month, is already known today, hence the bank account process $(B_t)_{t \in \mathbf{T}}$ is usually assumed to be predictable and we say that $(B_t)_{t \in \mathbf{T}}$ is *locally risk-free*. If we knew the evolution of the interest rate for the entire period, then B_t is $\mathcal{F}_0$ -measurable for all $t \in \mathbf{T}$, i.e., it is entirely riskless and deterministic.

Note that so far we have not imposed any specific model for the evolution of the asset price processes $\left\{ \left(S_t^{(i)} \right)_{t \in \mathbf{T}} : i = 0, 1, \dots, d \right\}$. However, in the finite discrete-time setting, the following model is popular.

EXAMPLE 1.3 (The Multinomial and Cox–Ross–Rubinstein Model). Simple yet rather flexible discrete-time models can be constructed as *multinomial models*, in which asset prices are modelled as multiplicative cumulative processes as follows.

Let $r_1, r_2, \dots, r_T$ denote random variables representing interest rates for periods $(0, 1], \dots, (T - 1, T]$, respectively, and let $R_1^{(i)}, R_2^{(i)}, \dots, R_T^{(i)}$ for every $i = 1, 2, \dots, d$ be sets of random variables describing the returns of the i -th risky asset with price process

$(S_t^{(i)})_{t \in \mathbf{T}}$ in the same periods. We then model the bank account process B as

$$B_t := B_0 \prod_{j=1}^t (1 + r_j), \quad \forall t = 0, 1, \dots, T, \quad (1.1)$$

with initial capital $B_0 \geq 0$ and the asset price processes $(S_t)_{t \in \mathbf{T}}$ as

$$S_t^{(i)} := S_0^{(i)} \prod_{j=1}^t (1 + R_j^{(i)}), \quad \forall t = 0, 1, \dots, T, \quad i = 1, 2, \dots, d, \quad (1.2)$$

with $S_0^{(i)} > 0$ being the initial price of the i -th (risky) asset observed today. Moreover, we shall assume that $r_t > -1$ and $R_t^{(i)} > -1$ holds $\mathbb{P}$ -a.s. for all $t = 1, 2, \dots, T$ and $i = 1, 2, \dots, d$. If $r_t = r$ for all $t = 1, 2, \dots, T$ for some $r > -1$ and all $R_1^{(i)}, R_2^{(i)}, \dots, R_T^{(i)}$ are independent and only assume finitely-many values, we call $(S_t)_{t \in \mathbf{T}}$ the *multinomial model*.

If $d = 1$ and the returns are identically distributed according to

$$R_t^{(1)} = \begin{cases} U : U \in \mathbb{R} & p : 0 < p < 1 \\ D : U > D > -1 & 1 - p \end{cases}, \quad \forall t = 1, \dots, T,$$

then we call the model the *binomial* or *Cox–Ross–Rubinstein (CRR) model*.

Portfolios and Trading Strategies

Throughout this section we fix a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F})$ in finite discrete-time $\mathbf{T}$ and let $d \in \mathbb{N}$ be the total number of tradable risky assets in the market (or at least the ones that we want to trade). This is exclusive of the bank account, which is tradable as well and which we henceforth denote by $S^{(0)} = (S_t^{(0)})_{t \in \mathbf{T}}$. As before, it is reasonable to assume that the bank account process $S^{(0)}$ is predictable with respect to $\mathbb{F}$. Next, we denote by $S_t^{(i)}$ the price of the i -th asset at trading time $t \in \mathbf{T}$ and assume that for every $i = 1, \dots, d$ and $t = 0, 1, \dots, T$ the random variable $S_t^{(i)}$ is $\mathcal{F}_t$ -measurable.

DEFINITION 1.4 (Vector-Valued Asset Price Processes).

We define the *vector-valued asset price process* as

$$X_t := \left(S_t^{(0)}, S_t^{(1)}, S_t^{(2)}, \dots, S_t^{(d)} \right)^\top, \quad t = 0, 1, \dots, T.$$

Remark. The process $X = (X_t)_{t \in \mathbf{T}}$ is an $\mathbb{R}^{d+1}$ -valued adapted stochastic process giving rise to the financial market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$.

DEFINITION 1.5 (Trading Strategies).

We call any $\mathbb{R}^{d+1}$ -valued process $(\varphi_t)_{t \in \mathbf{T}}$ a *trading strategy* in the market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$, if it is predictable with respect to $\mathbb{F}$. The $(d + 1)$ -components of a trading strategy φ will be denoted by $\varphi^{(i)}$ for $i = 0, 1, \dots, d$, i.e.,

$$\varphi_t = \left(\varphi_t^{(0)}, \varphi_t^{(1)}, \dots, \varphi_t^{(d)} \right)^\top, \quad t = 0, 1, \dots, T. \quad (1.3)$$

DEFINITION 1.6 (Wealth Processes).

For a trading strategy φ we define the associated *wealth process* $(W_t(\varphi))_{t \in \mathbf{T}}$ as

$$W_t(\varphi) := \varphi_t^\top S_t = \sum_{i=0}^d \varphi_t^{(i)} S_t^{(i)}, \quad t = 0, 1, \dots, T. \quad (1.4)$$

A trading strategy φ describes a dynamically evolving portfolio consisting of the $d + 1$ assets available for trade, i.e., d risky assets with price processes $S^{(1)}, S^{(2)}, \dots, S^{(d)}$ and the bank account $S^{(0)}$. For every fixed trading day $t \in \mathbf{T}$, the random variable $\varphi_t^{(0)}$ describes the quantity (measured in some unit or currency) held in the bank account at time t , whereas $\varphi_t^{(i)}$ represents the quantity we hold of the i -th asset at time t . Intuitively, a trading strategy must be predictable, as the portfolio allocation for the trading day t is set up (hence known) at trading day $t - 1$ and $W_t(\varphi)$, i.e., the wealth at time t using strategy φ , is the value of our portfolio *before* we adjust the portfolio at trading day t .

It is noteworthy that our portfolio weights at time t following strategy φ are $\mathbb{R}^{d+1}$ -valued. Indeed, our discourse will navigate within the bounds of an idealised market framework, more formally known as a *frictionless market*. In a frictionless market, we assume that:

- (i) *No transaction costs.* The buy price is equal to the sell price for any asset (no bid-ask spread, no exchange commission and no taxation), which justifies the use of a single price process for every asset.
- (ii) *Money can be lent and borrowed with the same interest rate.* We can lend and borrow capital for the same interest rate, which justifies the use of a single interest rate r_t .
- (iii) *Assets are available in arbitrary quantities.* We are able to buy and sell all tradable assets in arbitrarily large or small fractions.
- (iv) *Short-selling is possible.* We can purchase negative quantities of an asset (called *short selling*), hence φ is $\mathbb{R}^{d+1}$ -valued.
- (v) *Trading strategies do not impact prices.* Our trading does not affect the asset price, i.e., the price process X in a frictionless market is *exogenous*.

Any deviation to incorporate market frictions would necessitate the imposition of additional constraints on the trading strategies under consideration. We leave this intricacy for more advanced explorations and confine our discourse to *frictionless financial markets*.

Gains and Costs of a Trading Strategy

For simplicity we assume that $\varphi_0^{(0)} B_0^{(0)} = W_0$ for some *initial capital* $W_0 \in \mathbb{R}$ and $\varphi_0^{(i)} = 0$ for all $i = 1, \dots, d$. This means at time $t = 0$, we hold $\frac{W_0}{B_0^{(0)}}$ units of our bank account and own none of the risky assets. Then at time $t = 1$, we set up the first portfolio using the initial capital W_0 and distribute this initial investment over the d risky assets and the bank account according to the vector φ_1 , which is actually constant if $\mathcal{F}_0$ is trivial. The wealth process $W(\varphi)$ in (1.4) is a real-valued and adapted stochastic process.

DEFINITION 1.7 (Gains and Cost Process).

Let φ be a trading strategy in the financial market $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, X)$. We define the *gains process*

$(G_t(\varphi))_{t \in \mathbf{T}}$ associated with the strategy φ as

$$G_t(\varphi) := \sum_{i=1}^t \varphi_i^\top \Delta X_i, \quad t = 0, \dots, T, \quad (1.5)$$

where $\Delta X_i := X_i - X_{i-1}$ for $i = 1, 2, \dots, T$ and where we set $G_0(\varphi) = 0$ by convention. The cost process $(C_t(\varphi))_{t \in \mathbf{T}}$ of the strategy φ is defined as

$$C_t(\varphi) := W_t(\varphi) - G_t(\varphi), \quad t = 0, 1, \dots, T. \quad (1.6)$$

The gains process describes the *profit* or *loss* (depending on its sign) that we make trading according to strategy φ , while the cost process represents the *costs* associated with strategy φ . Note that by definition of the wealth process (1.4) and the cost process (1.6), the initial cost C_0 is equal to the initial capital $W_0(\varphi) = W_0$.

The gains process $(G_t(\varphi))_{t \in \mathbf{T}}$ associated with a trading strategy φ can be reformulated as a *discrete stochastic integral*, which is henceforth denoted by $(\varphi \cdot X)_{t \in \mathbf{T}}$ and by definition satisfies:

$$(\varphi \cdot X)_t = \sum_{i=1}^t \varphi_i^\top \Delta X_i = \sum_{i=1}^t \sum_{j=1}^d \varphi_i^{(j)} \left(X_i^{(j)} - X_{i-1}^{(j)} \right), \quad \forall t = 0, 1, \dots, T.$$

If our strategy involves further inflow of money, e.g., due to an active savings plan, then the cost process takes this into account. Note however, if the strategy may necessitate the injection of additional capital or the withdrawal of excess capital, then by our notation this must be transferred to or from an external capital pool, not our bank account $S^{(0)}$. Strategies that do not produce additional costs beyond the initial investment are important.

DEFINITION 1.8 (Self-Financing Strategy).

A trading strategy φ is called *self-financing* if its cost process $(C_t(\varphi))_{t \in \mathbf{T}}$ is constant over time, that is:

$$C_t(\varphi) = W_0(\varphi), \quad \forall t \in \mathbf{T}.$$

Remark. Note that according to (1.6) a trading strategy φ is self-financing, if and only if

$$W_t(\varphi) = W_0(\varphi) + G_t(\varphi), \quad \mathbb{P}\text{-a.s.}, \quad \forall t = 0, 1, \dots, T.$$

This means that the wealth process of a self-financing strategy, at every trading time $t \in \mathbf{T}$, is equal to the sum of the initial wealth $W_0(\varphi)$ and the gains up to time t .

Discounting

DEFINITION 1.9.

Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, X)$ be a (frictionless) financial market with bank account $S^{(0)}$ such that $S_t^{(0)} > 0$ $\mathbb{P}$ -a.s. for all $t \in \mathbf{T}$, and d risky assets with price processes $X = (S^{(1)}, \dots, S^{(d)})$. We define the discounted price process $\tilde{X}$ as

$$\tilde{X} := \frac{X}{S^{(0)}} = \left(1, \frac{S^{(1)}}{S^{(0)}}, \frac{S^{(2)}}{S^{(0)}}, \dots, \frac{S^{(d)}}{S^{(0)}} \right).$$

We also write $\tilde{S}^{(i)} = \frac{S_t^{(i)}}{S_t^{(0)}}$ and define the *discounted wealth process* $(\tilde{W}_t(\varphi))_{t \in \mathbf{T}}$ as

$$\tilde{W}_t(\varphi) := \frac{W_t(\varphi)}{S_t^{(0)}} = \varphi_t^\top \tilde{X}_t, \quad \forall t \in \mathbf{T}.$$

In a finite discrete-time market, discounting a price process does not affect the mathematics beyond the numbers, but it is often more convenient to work with discounted quantities as bookkeeping becomes simpler. Specifically, note that discounting implies expressing prices in quantities of the reference asset, i.e., relative to the bank account $S^{(0)}$ instead of absolute terms with respect to a reference currency.

EXAMPLE 1.10. If, in equation (1.1), we let $B_0 = 1$ and assume that the interest rates $r_1, r_2, \dots, r_T$ are all deterministic and equal to some fixed $r > -1$, then the bank account $S_t^{(0)}$ satisfies $S_t^{(0)} = (1+r)^t$ for all $t = 0, 1, \dots, T$. In this case, we have

$$\tilde{S}_t^{(i)} = \frac{S_t^{(i)}}{(1+r)^t},$$

which represents the so-called *present value* of $S_t^{(i)}$.

The gains process $\tilde{G}_t(\varphi) = \varphi \cdot \tilde{X}$ of the discounted asset price process does not depend on the bank account part $\varphi^{(0)}$ as $\Delta \tilde{X} = (0, \Delta \tilde{S}^{(1)}, \Delta \tilde{S}^{(2)}, \dots, \Delta \tilde{S}^{(d)})$ and the following lemma holds true.

LEMMA 1.11.

A strategy φ is self-financing if and only if

$$\tilde{W}_t(\varphi) = \tilde{W}_0(\varphi) + (\varphi \cdot \tilde{X})_t$$

holds $\mathbb{P}$ -a.s. for all $t = 0, 1, \dots, T$.

Proof. “ $\Rightarrow$ ” Necessity: we assume that the strategy φ is self-financing, then by definition, for all $t \in \mathbf{T}$, we have

$$(G_t(\varphi))_{t \in \mathbf{T}} = (\varphi \bullet X)_t = \sum_{i=1}^t \varphi_i^\top \Delta X_i.$$

Since the discounted wealth process is defined by

$$\tilde{W}_t(\varphi) := \frac{W_t(\varphi)}{S_t^{(0)}} = \varphi_t^\top \tilde{X}_t, \quad \forall t \in \mathbf{T},$$

we can write

$$\begin{aligned} (\varphi \bullet \tilde{X})_t &= \sum_{i=1}^t \varphi_i^\top (\tilde{X}_i - \tilde{X}_{i-1}) \\ &= \sum_{i=1}^t \varphi_i^\top \tilde{X}_i - \sum_{i=1}^t \varphi_i^\top \tilde{X}_{i-1} \\ &= \sum_{i=1}^{t-1} \varphi_i^\top \tilde{X}_i + \varphi_t^\top \tilde{X}_t - \sum_{i=0}^{t-1} \varphi_{i+1}^\top \tilde{X}_i \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^{t-1} \left(\varphi_i^\top \tilde{X}_i - \varphi_{i+1}^\top \tilde{X}_i \right) + \varphi_t^\top \tilde{X}_t - \varphi_1^\top \tilde{X}_0 \\
&= \sum_{i=1}^{t-1} \left(\varphi_i^\top \tilde{X}_i - \varphi_{i+1}^\top \tilde{X}_i \right) + \left(\varphi_0^\top \tilde{X}_0 - \varphi_1^\top \tilde{X}_0 \right) + \varphi_t^\top \tilde{X}_t \\
&= -\varphi_0^\top \tilde{X}_0 + \tilde{W}_t(\varphi) = \tilde{W}_t(\varphi) - \tilde{W}_0(\varphi).
\end{aligned}$$

“ $\Leftarrow$ ” Sufficiency: suppose that the equality (1.2) holds. Let $t = 1$ in (1.2), we have

$$\tilde{W}_1(\varphi) = \tilde{W}_0(\varphi) + \left(\varphi \bullet \tilde{X} \right)_1 = \varphi_0^\top \tilde{X}_0 + \varphi_1^\top (\tilde{X}_1 - \tilde{X}_0) = \tilde{W}_1(\varphi) + \varphi_0^\top \tilde{X}_0 - \varphi_1^\top \tilde{X}_0,$$

then $\varphi_0^\top \tilde{X}_0 = \varphi_1^\top \tilde{X}_0$. Let $t = 2$ in (1.2), we have

$$\begin{aligned}
\tilde{W}_2(\varphi) &= \tilde{W}_0(\varphi) + \left(\varphi \bullet \tilde{X} \right)_2 \\
&= \varphi_0^\top \tilde{X}_0 + \varphi_1^\top (\tilde{X}_1 - \tilde{X}_0) + \varphi_2^\top (\tilde{X}_2 - \tilde{X}_1) \\
&= \tilde{W}_2(\varphi) + \varphi_0^\top \tilde{X}_0 + \varphi_1^\top \tilde{X}_1 - \varphi_1^\top \tilde{X}_0 - \varphi_2^\top \tilde{X}_1 \\
&= \tilde{W}_2(\varphi) + \varphi_1^\top \tilde{X}_1 - \varphi_2^\top \tilde{X}_1,
\end{aligned}$$

then $\varphi_1^\top \tilde{X}_1 = \varphi_2^\top \tilde{X}_1$. Continue the similar procedure, we get $\varphi_t^\top \tilde{X}_t = \varphi_{t+1}^\top \tilde{X}_t$ for $t \in \mathbf{T}$. Therefore,

$$\begin{aligned}
\tilde{W}_0(\varphi) + \tilde{G}_t(\varphi) &= \varphi_0^\top \tilde{X}_0 + \sum_{i=1}^t \varphi_i^\top \Delta \tilde{X}_i \\
&= \varphi_0^\top \tilde{X}_0 + \varphi_1^\top (\tilde{X}_1 - \tilde{X}_0) + \cdots + \varphi_t^\top (\tilde{X}_t - \tilde{X}_{t-1}) \\
&= \varphi_0^\top \tilde{X}_0 - \varphi_1^\top \tilde{X}_0 + \varphi_1^\top \tilde{X}_1 - \varphi_2^\top \tilde{X}_1 + \cdots + \varphi_t^\top \tilde{X}_t \\
&= \varphi_t^\top \tilde{X}_t = \tilde{W}_t(\varphi), \tag*{(\varphi_t^\top \tilde{X}_t = \varphi_{t+1}^\top \tilde{X}_t)}
\end{aligned}$$

and hence φ is self-financing.

In conclusion, we have proved that φ is self-financing if and only if (1.2) holds $\mathbb{P}$ -a.s. for all $t \in \mathbf{T}$. $\square$

LEMMA 1.12.

For any $\mathbb{R}^d$ -valued predictable process $(\varphi^{(1)}, \varphi^{(2)}, \dots, \varphi^{(d)})$ and $W_0 \in \mathbb{R}$, there exists a unique predictable process $\varphi^{(0)}$ such that the $\mathbb{R}^{d+1}$ -valued process φ , defined as

$$\varphi_t := \left(\varphi_t^{(0)}, \varphi_t^{(1)}, \dots, \varphi_t^{(d)} \right), \quad t = 0, 1, \dots, T$$

is a self-financing strategy with $W_0(\varphi) = W_0$.

Remark. We may identify any strategy $(\varphi^{(1)}, \dots, \varphi^{(d)})$ with the self-financing strategy $\varphi = (\varphi^{(0)}, \varphi^{(1)}, \dots, \varphi^{(d)})$ satisfying $W_0(\varphi) = W_0$ for some initial capital $W_0 \in \mathbb{R}$, i.e., the initial capital W_0 and the trading $(\varphi^{(1)}, \dots, \varphi^{(d)})$ in the d risky assets uniquely determine the trading in the bank account $\varphi^{(0)}$. We therefore often denote a self-financing strategy φ as $\varphi = (W_0, \phi)$, where $W_0 \in \mathbb{R}$ and ϕ is an $\mathbb{R}^d$ -valued predictable process from Lemma 1.12, and write $\tilde{W}_t(\varphi) = W_0 + G_t(\phi)$.

Arbitrage and Equivalent Martingale Measures

DEFINITION 1.13 (Arbitrage Opportunities).

A self-financing strategy φ is called an *arbitrage opportunity* if the following conditions hold true:

- (i) The initial wealth is zero, i.e., $W_0(\varphi) = 0$, $\mathbb{P}$ -a.s.
- (ii) The terminal wealth is non-negative, i.e., $W_T(\varphi) \geq 0$, $\mathbb{P}$ -a.s.
- (iii) There is a positive probability that the terminal wealth is strictly positive, i.e., $\mathbb{P}(W_T(\varphi) > 0) > 0$.

We call a financial market $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, X)$ *arbitrage free*, if there exists no such arbitrage opportunity.

LEMMA 1.14 (Law of One Price).

Let φ and ψ be two self-financing strategies such that $W_T(\varphi) \leq W_T(\psi)$, $\mathbb{P}$ -a.s. If the financial market $(\Omega, \mathcal{F}, \mathbb{P}, X)$ is arbitrage free, then also $W_t(\varphi) \leq W_t(\psi)$, $\mathbb{P}$ -a.s. for all $t = 0, 1, \dots, T-1$. In particular, in an arbitrage-free market, it holds that if $W_T(\varphi) = W_T(\psi)$, $\mathbb{P}$ -a.s., then also $W_t(\varphi) = W_t(\psi)$, $\mathbb{P}$ -a.s. for all $t = 0, 1, \dots, T-1$.

Recall that two probability measures $\mathbb{P}$ and $\mathbb{Q}$, defined on the same measurable space $(\Omega, \mathcal{F})$, are called *equivalent* if for all $A \in \mathcal{F}$ we have $\mathbb{P}(A) = 0$ if and only if $\mathbb{Q}(A) = 0$. In this case we write $\mathbb{P} \sim \mathbb{Q}$ and we may define the density process $Z_t := \frac{d\mathbb{Q}}{d\mathbb{P}}|_{\mathcal{F}_t}$ for $t = 0, 1, \dots, T$, which is a martingale with respect to $\mathbb{Q}$.

DEFINITION 1.15 (Equivalent Martingale Measures).

An *equivalent martingale measure* for the market $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, X)$ is any probability measure $\mathbb{Q} \sim \mathbb{P}$ on $(\Omega, \mathcal{F}_T)$ such that the discounted price process $\tilde{X}$ is a $\mathbb{Q}$ -martingale with respect to the filtration $\mathbb{F}$.

THEOREM 1.16 (Doob's System Theorem).

For a probability measure $\mathbb{Q}$, the following conditions are equivalent.

- (i) $\mathbb{Q}$ is a martingale measure.
- (ii) If $\varphi = (W_0, \phi)$ is a self-financing strategy and ϕ is bounded, then the discounted wealth process $\tilde{W}(\varphi)$ of φ is a $\mathbb{Q}$ -martingale.
- (iii) If $\varphi = (W_0, \phi)$ is a self-financing strategy and its discounted wealth process $\tilde{W}(\varphi)$ satisfies $\mathbb{E}_{\mathbb{Q}}[\tilde{W}_t(\varphi)^-] < \infty$ for all $t \in \mathbf{T}$, then $\tilde{W}(\varphi)$ is a $\mathbb{Q}$ -martingale.
- (iv) If $\varphi = (W_0, \phi)$ is a self-financing strategy and its discounted wealth process $\tilde{W}(\varphi)$ satisfies $W_T(\varphi) \geq 0$, $\mathbb{P}$ -a.s., then $\mathbb{E}_{\mathbb{Q}}[W_T(\varphi)] = W_0$.

LEMMA 1.17.

Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, X)$ be a financial market. If there exists an equivalent martingale measure $\mathbb{Q}$, then $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, X)$ is arbitrage free.

We can now state the following dynamic version of the so-called *first fundamental theorem of asset pricing* (FFTAP), which relates the absence of arbitrage opportunities in a market to the existence of equivalent martingale measures. In the following, for a sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, we denote by $L^0(\Omega, \mathcal{G}, \mathbb{P})$ the set of $\mathcal{G}$ -measurable random variables, identifying all $\mathbb{P}$ -a.s. equal random variables.

THEOREM 1.18 (First Fundamental Theorem of Asset Pricing).

The financial market $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, X)$ is arbitrage-free if and only if there exists an equivalent martingale measure $\mathbb{Q}$ of $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}, X)$.

Proof. If there exists an equivalent martingale measure $\mathbb{Q}$ for $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$, then it follows immediately from [Lemma 1.17](#) that the market is arbitrage-free.

The reverse direction is the tricky one and we will only prove it for the case of a finite probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Therefore, assume that $\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$ for some $N \in \mathbb{N}$ and $\mathbb{P}(\{\omega_i\}) > 0$ for all $i = 1, \dots, N$.

Suppose that the market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ is free of arbitrage. Note that this means that the set

$$\mathcal{G} := \left\{ \tilde{G}_T(\varphi) : \varphi \text{ is a predictable } \mathbb{R}^d\text{-valued process} \right\}$$

of discounted gains satisfies

$$\mathcal{G} \cap \mathcal{L}^{0,+}(\Omega, \mathcal{F}_T) = \{0\}, \quad (1.7)$$

with $\mathcal{L}^{0,+}(\Omega, \mathcal{F}_T)$ denoting the set of all non-negative $\mathcal{F}_T$ -measurable random variables. Further note that

$$\mathcal{G} = \left\{ \tilde{W}_T(\varphi) : \varphi = (W_0, \varphi) \text{ is a self-financing strategy with } W_0 = 0 \right\},$$

as every self-financing strategy φ with $\tilde{W}_0(\varphi) = 0$ satisfies $W_T(\varphi) = \tilde{G}_T(\varphi)$ and (1.7) then tells us that all self-financing strategies φ with initial wealth zero and non-negative wealth at time T lead to zero terminal wealth, i.e., $W_T(\varphi) = G_T(\varphi) = 0$.

Next, note that since $\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$, every random variable Y on $(\Omega, \mathcal{F})$ is of the form

$$Y(\omega) = \sum_{i=1}^N y_i \mathbb{1}_{\{\omega_i\}}(\omega)$$

where $y_i \in \mathbb{R}$ for $i = 1, 2, \dots, N$. Hence, we identify every $Y \in L^0(\Omega, \mathcal{F}_T)$ with the vector $y = (y_1, y_2, \dots, y_N)^\top$, where in particular, $Y = 0$ is identified with $0 \in \mathbb{R}^N$ and $\mathcal{L}^{0,+}(\Omega, \mathcal{F}_T)$ with the non-negative orthant $\mathbb{R}_+^N$.

Note further, that for every trading strategy φ the discounted terminal gain $\tilde{G}_T(\varphi)$ is $\mathcal{F}_T$ measurable as well, i.e., it is also of the form

$$\tilde{G}_T(\varphi)(\omega) = \sum_{i=1}^N g_i(\varphi) \mathbb{1}_{\{\omega_i\}}(\omega).$$

Using the same identification here, means that $\mathcal{G}$ is identified with the set

$$\mathcal{G}' = \left\{ (g_1(\varphi), g_2(\varphi), \dots, g_N(\varphi))^\top : \varphi \text{ is a trading strategy} \right\}.$$

Even if this set seems to be a little obscure, it is a linear subspace of $\mathbb{R}^N$, since the gains process is linear in φ (the discrete stochastic integral is linear in its integrand), and moreover the zero is also mapped to the zero vector in $\mathbb{R}^N$. Since, we are using the same identification we see that the property (1.7) is preserved and we have

$$\mathcal{G}' \cap \mathbb{R}_+^N = \{0\}.$$

Now, the following idea to *construct* an equivalent martingale measure out of the geometric condition (1.7) is a rather nice idea and a similar (but more involved) approach works for the infinite probability space case.

First, define the convex, closed and bounded set

$$C := \left\{ y = (y_1, y_2, \dots, y_N) \in \mathbb{R}_+^N : \sum_{i=1}^N y_i = 1 \right\}.$$

We observe that $C \subseteq \mathbb{R}_+^N$, but $0 \in C^c$ and therefore also $C \cap \mathcal{G}' = \emptyset$. This actually allows us to separate the compact convex set C from the subspace $\mathcal{G}'$ by a separating hyperplane, that is, there exists a $v \in \mathbb{R}^N \setminus \{0\}$ such that

$$v^\top g = 0, \quad \forall g \in \mathcal{G}', \quad (1.8)$$

$$v^\top f > 0, \quad \forall f \in C. \quad (1.9)$$

Note further that for all $i = 1, 2, \dots, N$ the unit basis vectors $f_i = (0, \dots, 0, 1, 0, \dots, 0)^\top$ are in C and therefore by (1.9) we have $v^\top f_i = v_i > 0$ for all $i = 1, 2, \dots, N$. We thus define

$$p_i := \frac{v_i}{\sum_{i=1}^N v_i}, \quad \forall i = 1, 2, \dots, N.$$

Note that for all $i = 1, 2, \dots, N$ we have $p_i \in (0, 1)$ and $\sum_{i=1}^N p_i = 1$, i.e., we can interpret the p_i 's as probabilities, which puts us close to the desired equivalent martingale measure. Indeed, we set $\mathbb{Q}(\{\omega_i\}) = p_i$ and note that $\mathbb{Q}$ is equivalent to $\mathbb{P}$, as both measure agrees that all ω_i have a positive probability to occur.

Next, we show that $\mathbb{Q}$ is also an martingale measure, i.e., the discounted asset price vector $\tilde{X}$ is a martingale under $\mathbb{Q}$. This can be seen as follows. Note first, that given the definition of $\mathbb{Q}$ and dividing both sides of the identity (1.8) by $\sum_{i=1}^N v_i$, we see that it is equivalent to

$$\sum_{i=1}^N p_i g_i = 0 \implies \sum_{i=1}^N \mathbb{Q}(\{\omega_i\}) g_i = 0,$$

which means $\mathbb{E}_{\mathbb{Q}}[g] = 0$ for the random variable $g = \sum_{i=1}^N g_i \mathbb{1}_{\{\omega_i\}}$. In particular, this holds for the

$$\tilde{G}_T(\varphi) = \sum_{i=1}^N g_i(\varphi) \mathbb{1}_{\{\omega_i\}}$$

from above and hence for any trading strategy φ we have

$$\mathbb{E}_{\mathbb{Q}}[\tilde{G}_T(\varphi)] = 0. \quad (1.10)$$

Now, let $A \in \mathcal{F}_{k-1}$ be arbitrary and define the strategy $\tilde{\varphi}_k(\omega) := \mathbb{1}_k(t) \mathbb{1}_A(\omega) f_i$, i.e., the strategy is that at time k we buy one unit of the i -th asset if also the event $A \in \mathcal{F}_{k-1}$ occurs.

This process is indeed a trading strategy, i.e., predictable, and inserting this into (1.10) yields:

$$\mathbb{E}_{\mathbb{Q}}[\tilde{G}_T(\tilde{\varphi})] = \mathbb{E}_{\mathbb{Q}}\left[\sum_{l=1}^T \tilde{\varphi}_l(\tilde{X}_l - \tilde{X}_{l-1})\right] = \mathbb{E}_{\mathbb{Q}}\left[\tilde{\varphi}_k(\tilde{S}_k^{(i)} - \tilde{S}_{k-1}^{(i)})\right] = \mathbb{E}_{\mathbb{Q}}\left[\mathbb{1}_A(\tilde{S}_k^{(i)} - \tilde{S}_{k-1}^{(i)})\right] = 0.$$

Hence, $\mathbb{E}_{\mathbb{Q}}[\mathbb{1}_A \tilde{S}_k^{(i)}] = \mathbb{E}_{\mathbb{Q}}[\mathbb{1}_A \tilde{S}_{k-1}^{(i)}]$ and since $A \in \mathcal{F}_{k-1}$ was arbitrary, this proves the martingale property for the asset price process $\tilde{S}^{(i)}$ (and $i \in \{1, 2, \dots, d\}$ was also arbitrary) so also the discounted process $\tilde{X}$ is a $\mathbb{Q}$ -martingale. This concludes the proof as we found an equivalent martingale measure $\mathbb{Q}$ as asserted. $\square$

Remark.

SECTION 1.2

| Pricing and Hedging European Contingent Claims

In this section, we consider a finite discrete-time financial market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$, where $\mathbf{T} = \{0, 1, \dots, T\}$ represents the trading times with time horizon $T \in \mathbb{N}$, and $d \in \mathbb{N}$ indicates the total number of risky assets. In this setting, the multivariate asset price process $X = (S^{(0)}, S^{(1)}, \dots, S^{(d)})$ is often referred to as the price process of the *underlying securities* $S^{(i)}$ for $i = 0, 1, \dots, d$, as we also explore securities whose prices are contingent on the evolution of these underlying prices. Such securities are then called *derivatives* written on the underlyings $S^{(0)}, S^{(1)}, \dots, S^{(d)}$. Derivatives are typically contracts that specify particular transactions (dependent on the underlying securities) to be executed at predetermined prices in the future.

Suppose a derivative with price process denoted by $S^{(d+1)}$ is introduced into the arbitrage-free market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ as an additional tradable asset. If the market was initially arbitrage-free and the derivative is liquidly tradable, what conditions must the price process of the derivative fulfill to maintain the absence of arbitrage opportunities? To address this question, we first refine the concept of derivatives and introduce the fundamental *pricing* and *hedging problem* that we face in mathematical finance.

European Contingent Claims

DEFINITION 1.19 (European Contingent Claims).

We call any $\mathcal{F}_T$ -measurable non-negative random variable H a *European contingent claim*. If a European contingent claim H is $\sigma(X_0, X_1, \dots, X_T)$ -measurable, then it is considered a (*European*) *derivative* of X .

A European contingent claim describes the pay-off that the holder of the instrument receives at time T , which may depend on the entire information up to time T . If it solely depends on the history of the underlying price process X , it is classified as a derivative written on X . In the context of contingent claims, the *European* refers to the payoff being fixed at the terminal date T as opposed to *American contingent claims*, which can be executed at any trading time.

EXAMPLE 1.20. (i) A *European call option* written on asset i with maturity date T and strike price K grants its owner the right, but not the obligation, to purchase at time T one unit of asset i for the price K . The contingent claim of a European call option is thus given by

$$H(\omega) \triangleq \max \left(0, S_T^{(i)}(\omega) - K \right) = \left(S_T^{(i)}(\omega) - K \right)^+.$$

(ii) A *Digital Barrier option* written on asset i with maturity date T , barrier level B , and strike price K provides a fixed payoff if the asset price reaches or exceeds the barrier level B during the option's life. The contingent claim of a Digital Barrier option is given by

$$H(\omega) \triangleq \mathbb{1}_{\{S_t^{(i)}(\omega) \geq B \text{ for some } t \in \{0, \dots, T\}\}}(\omega).$$

(iii) An *Asian call option* written on asset i with maturity date T and strike price K gives its owner the right, but not the obligation, to purchase one unit of asset i for the price K at

time T , based on the average price of the asset over a specific period. The contingent claim of an Asian call option is

$$H(\omega) \triangleq \max \left(0, \frac{1}{T} \sum_{t=0}^T S_t^{(i)}(\omega) - K \right) = \left(\frac{1}{T} \sum_{t=0}^T S_t^{(i)}(\omega) - K \right)^+.$$

The Pricing and Hedging Problem

For the buyer, derivatives serve as powerful instruments within financial markets, offering both retail and institutional buyers the opportunity to manage future cash flows, hedge commodity exposures, speculate, and more. Simultaneously, a counterparty (often a financial institution such as a bank) must undertake the inverse role in the derivative trade, i.e., selling the derivative to the client. This trade positions the seller of the claim with the liability of $-H$ at time T , which could potentially be huge and should be insured (or in financial terms *hedged*). Indeed, the intricacies of this interaction lead to the two following problems:

- *Pricing Problem*: The *asset pricing problem* inquires: given a contingent claim H with maturity date T , what determines the *fair value* of the associated derivative at any time t prior to the maturity date T ? In this context, *fair value* denotes a price that precludes arbitrage opportunities in the market upon the introduction of this newly priced asset.
- *Hedging Problem*: In every derivative contract, a counterparty assumes the opposite position. For instance, in a European call option, the seller has the obligation to sell the underlying asset at the fixed price K at time T to the option holder, while also receiving a premium for selling the contract. The hedging problem for the seller can be formulated as follows: upon selling the contingent claim H , what measures can be taken to mitigate the risk associated with the obligation to pay the random and uncertain amount H at time T ? What price should the seller of the claim charge prior to maturity?

These problems are inherently intertwined. The fundamental principle for addressing both questions involves utilising only the underlying assets $S^{(0)}, S^{(1)}, \dots, S^{(d)}$ to construct a synthetic product that *replicates* H as accurately as possible. As this product's value is derived from the given assets, it should serve as a reliable approximation for the value of H in the absence of arbitrage. We formalise this idea in the next definition.

DEFINITION 1.21.

We define a European contingent claim H as *attainable* or *replicable* if there exists a self-financing strategy φ such that the terminal wealth of this strategy satisfies $W_T(\varphi) = H$, $\mathbb{P}$ -a.s. In this case, we say that the strategy φ replicates the contingent claim H , and we refer to φ as a *replicating* or *perfect hedging strategy* for H .

Risk-Neutral Pricing

In this section, we show that for any attainable European contingent claim H , the asset pricing problem can be solved in arbitrage-free markets. Furthermore, a unique fair price can be derived for each trading day $t \in \mathbf{T}$ prior to maturity, along with a pricing formula. This is demonstrated in the following theorem, where we denote the *discounted contingent claim* by

$$\tilde{H} := \frac{H}{S_T^{(0)}}.$$

THEOREM 1.22 (Risk-Neutral Pricing).

Consider an arbitrage-free financial market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ and let $\mathbb{Q}$ be an equivalent martingale measure for the market. Then, for any attainable European contingent claim H , its discounted unique fair value $\tilde{W}_t^H$ at time t is given by

$$\tilde{W}_t^H = \mathbb{E}_{\mathbb{Q}}[\tilde{H} \mid \mathcal{F}_t] = \tilde{W}_t(\varphi), \quad \mathbb{P}\text{-a.s.}, \quad \forall t = 0, 1, \dots, T, \quad (1.11)$$

for any replicating strategy φ for H .

It is crucial to observe that W_t^H represents the *fair value* or *price* of the contingent claim H at time t , which implies that the extended market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X, S^{(d+1)})$ with $S_t^{(d+1)} := W_t^H$, for $t = 0, 1, \dots, T$, is arbitrage-free. A crucial assertion of [Theorem 1.22](#) is that the fair price is uniquely determined by (1.11). In other words: assuming the absence of arbitrage opportunities in a market yields a unique price for any attainable European contingent claim. This provides an answer to the pricing problem for all attainable claims. Nonetheless, a challenge remains: determining whether a contingent claim is attainable and constructing corresponding replicating strategies φ effectively.

Arbitrage-free Prices

In this section, we show that if in an arbitrage-free market a claim is non-attainable, then there exists an entire open interval of prices with the property that the extended market remains arbitrage-free.

DEFINITION 1.23 (Arbitrage-Free Prices).

We define a real number $\pi^H \geq 0$ as an *arbitrage-free price* of a contingent claim H if there exists a stochastic process $S^{(d+1)}$ satisfying:

- (i) $S_0^{(d+1)} = \pi^H$;
- (ii) $S_t^{(d+1)} \geq 0$ $\mathbb{P}$ -a.s. for all $t = 0, 1, \dots, T$;
- (iii) $S_T^{(d+1)} = H$ $\mathbb{P}$ -a.s.,

and is such that the extended market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, (S^{(0)}, \dots, S^{(d)}, S^{(d+1)}))$ is arbitrage-free. Furthermore, we denote the set of all such arbitrage-free prices of H as $\Pi(H)$.

Remark. Note that the definition of an arbitrage-free price assumes that the extended market $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_{t \geq 0}, (S^{(0)}, \dots, S^{(d)}, S^{(d+1)}))$ is still a financial market. By our convention of frictionless markets, we implicitly assume that we can trade the derivative $S^{(d+1)}$ at any trading time in arbitrary quantities without transaction costs or market impact. This assumption is valid for plain vanilla options (such as European put and call options) on large, liquid underlyings like the S&P 500 index, where options markets are deep and liquid. However, for less liquid options, this arbitrage-free pricing criterion is not directly applicable because the derivative cannot be traded freely in arbitrary quantities without affecting the market price. In such cases, the assumption of frictionless trading breaks down. Despite this, the arbitrage-free pricing methodology is still used in practice as an approximation.

We denote by $\mathcal{P}$ the set of all *equivalent martingale measures* of the financial market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$. The following theorem is fundamental for solving the pricing problem for non-attainable contingent claims.

THEOREM 1.24.

Let $\Pi(H)$ denote the set of arbitrage-free prices of any European contingent claim H . Then $\Pi(H)$ is non-empty and can be expressed as

$$\Pi(H) = \left\{ \mathbb{E}_{\mathbb{Q}}[\tilde{H}] : \mathbb{Q} \in \mathcal{P}, \mathbb{E}_{\mathbb{Q}}[\tilde{H}] < \infty \right\}. \quad (1.12)$$

Furthermore, we have

$$\inf \Pi(H) = \inf_{\mathbb{Q} \in \mathcal{P}} \mathbb{E}_{\mathbb{Q}}[\tilde{H}], \quad \sup \Pi(H) = \sup_{\mathbb{Q} \in \mathcal{P}} \mathbb{E}_{\mathbb{Q}}[\tilde{H}].$$

Proof. Suppose $\pi^H \in \Pi(H)$. By the definition of an arbitrage-free price π^H , the market

$$\left(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, \left(S^{(0)}, \dots, S^{(d)}, S^{(d+1)} \right) \right)$$

is arbitrage-free. Consequently, by the FFTAP, there exists an equivalent martingale measure $\mathbb{Q}$ so that for all $i = 1, 2, \dots, d+1$, we have

$$\tilde{S}_t^{(i)} = \mathbb{E}_{\mathbb{Q}}[\tilde{S}_T^{(i)} | \mathcal{F}_t], \quad \forall t = 0, 1, \dots, T.$$

In particular, when $i = 1, 2, \dots, d$, we observe that $(\tilde{S}_t^{(i)})_{t \in T}$ are $\mathbb{Q}$ -martingales, indicating that $\mathbb{Q} \in \mathcal{P}$. Thus, $\mathbb{Q}$ is also an equivalent martingale measure for the original market and we have

$$\pi^H = \mathbb{E}_{\mathbb{Q}}[\tilde{S}_T^{(d+1)} | \mathcal{F}_0] = \mathbb{E}_{\mathbb{Q}}[\tilde{H} | \mathcal{F}_0] = \mathbb{E}_{\mathbb{Q}}[\tilde{H}] < \infty,$$

which implies that $\pi^H \in \left\{ \mathbb{E}_{\mathbb{Q}}[\tilde{H}] : \mathbb{Q} \in \mathcal{P} \wedge \mathbb{E}_{\mathbb{Q}}[\tilde{H}] < \infty \right\}$.

Conversely, if $\pi^H = \mathbb{E}_{\mathbb{Q}}[\tilde{H}]$ for some $\mathbb{Q} \in \mathcal{P}$, we define

$$X_t^{(d+1)} := \mathbb{E}_{\mathbb{Q}}[\tilde{H} | \mathcal{F}_t], \quad t = 0, 1, \dots, T.$$

Note that the so defined process $(X_t^{(d+1)})_{t \in T}$ satisfies all the requirements of [Definition 1.23](#). In particular, it is a $\mathbb{Q}$ -martingale, which means that $\mathbb{Q}$ is also an equivalent martingale measure for the extended market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, (S^{(0)}, S^{(1)}, \dots, S^{(d)}, S^{(d+1)}))$. Hence, by the FFTAP, the market is arbitrage-free and $\pi^H \in \Pi(H)$.

To demonstrate that $\Pi(H)$ is non-empty, let $\mathbb{P} \sim \tilde{\mathbb{P}}$ such that $\mathbb{E}_{\tilde{\mathbb{P}}}[\tilde{H}] < \infty$, e.g., we could choose $d\tilde{\mathbb{P}} = c(1 + \tilde{H})^{-1} d\mathbb{P}$ for some normalising constant c . If the original market is arbitrage-free under $\mathbb{P}$, then it is also arbitrage-free under $\tilde{\mathbb{P}}$. Therefore, by the FFTAP, there exists an equivalent martingale measure $\mathbb{Q} \sim \tilde{\mathbb{P}}$ such that the Radon–Nikodym derivative $\frac{d\mathbb{Q}}{d\tilde{\mathbb{P}}}$ is bounded, and we have

$$\mathbb{E}_{\mathbb{Q}}[\tilde{H}] = \mathbb{E}_{\tilde{\mathbb{P}}} \left[\tilde{H} \frac{d\mathbb{Q}}{d\tilde{\mathbb{P}}} \right] < \infty.$$

This confirms the non-emptiness of $\Pi(H)$.

The result $\inf \Pi(H) = \inf_{\mathbb{Q} \in \mathcal{P}} \mathbb{E}_{\mathbb{Q}}[\tilde{H}]$ follows immediately from (1.12) and the fact that $\Pi(H) \neq \emptyset$. To establish the upper bound, we need to show that if there exists $\mathbb{Q} \in \mathcal{P}$ such that $\mathbb{E}_{\mathbb{Q}}[\tilde{H}] = \infty$, then for all $R > 0$ there exists an arbitrage-free price $\pi^H \in \Pi(H)$ with $\pi^H > R$. To this end, let $n \in \mathbb{N}$ be such that

$$\tilde{\pi}_n := \mathbb{E}_{\mathbb{Q}}[\tilde{H} \wedge n] > R$$

(which exists by monotone convergence) and define

$$X_t^{(d+1)} := \mathbb{E}_{\mathbb{Q}}[\tilde{H} \wedge n | \mathcal{F}_t], \quad t = 0, 1, \dots, T.$$

Then $X_t^{(d+1)} = \tilde{\pi}_n$ and $\mathbb{Q}$ is an equivalent martingale measure for the extended market

$$\left(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, \left(S^{(0)}, \dots, S^{(d)}, S^{(d+1)}\right)\right).$$

By the FFTAP, it is arbitrage-free. In particular, by the first part of this theorem (applied to the extended market), there exists an $\mathbb{Q}' \sim \mathbb{Q}$ such that $\tilde{\pi}_n = \mathbb{E}_{\mathbb{Q}'}[\tilde{H}] < \infty$. However, $\mathbb{Q}'$ is also an equivalent martingale measure for the original market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, (S^{(0)}, S^{(1)}, \dots, S^{(d)}))$. Thus, $\tilde{\pi} := \mathbb{E}_{\mathbb{Q}}[\tilde{H}]$ is also an arbitrage-free price for the original market, and hence

$$\pi^H = \mathbb{E}_{\mathbb{Q}}[\tilde{H}] \geq \mathbb{E}_{\mathbb{Q}'}[\tilde{H} \wedge n] = \mathbb{E}_{\mathbb{Q}'}[X_T^{(d+1)}] = X_0^{(d+1)} = \tilde{\pi}_n > R,$$

which yields a price π^H with the desired properties. $\square$

Note that if H is attainable, then it follows from [Theorem 1.22](#) that $\Pi(H) = \{W_0^H\}$, i.e., the set of arbitrage-free prices is a singleton, where W_0^H is given by (1.11). In sharp contrast, the following theorem asserts that the set of arbitrage-free prices for a non-attainable contingent claim is an open interval.

THEOREM 1.25.

For any European contingent claim H , if H is not attainable, then $\Pi(H)$ is an open interval, i.e.,

$$\Pi(H) = (\inf \Pi(H), \sup \Pi(H)),$$

and moreover, $\inf \Pi(H) < \sup \Pi(H)$.

Proof. Note first that

$$\Pi(H) = \left\{ \mathbb{E}_{\mathbb{Q}}[\tilde{H}] \mid \mathbb{Q} \in \mathcal{P}, \mathbb{E}_{\mathbb{Q}}[\tilde{H}] < \infty \right\}$$

is an interval because $\mathcal{P}$ is a convex set. We will show that $\Pi(H)$ is open by constructing for any $\pi \in \Pi(H)$ two arbitrage-free prices $\tilde{\pi}$ and $\hat{\pi}$ for $\tilde{H}$ such that $\tilde{\pi} < \pi < \hat{\pi}$.

To this end, take $\mathbb{Q} \in \mathcal{P}$ such that $\pi = \mathbb{E}_{\mathbb{Q}}[\tilde{H}]$. We will first construct an equivalent martingale measure $\hat{\mathbb{Q}} \in \mathcal{P}$ such that $\mathbb{E}_{\hat{\mathbb{Q}}}[\tilde{H}] > \mathbb{E}_{\mathbb{Q}}[\tilde{H}]$. Let

$$U_t := \mathbb{E}_{\mathbb{Q}}[\tilde{H} \mid \mathcal{F}_t], \quad t = 0, \dots, T,$$

so that

$$\tilde{H} = U_0 + \sum_{t=1}^T (U_t - U_{t-1}).$$

Since H is not attainable, there must be some $t \in \{1, \dots, T\}$ such that $U_t - U_{t-1} \notin \mathcal{K}_t \cap \mathcal{L}^1(\mathbb{Q})$, where

$$\mathcal{K}_t := \left\{ \eta^\top (\tilde{S}_t - \tilde{S}_{t-1}) \mid \eta \in \mathcal{L}^0(\Omega, \mathcal{F}_{t-1}, \mathbb{Q}; \mathbb{R}^d) \right\}.$$

It can be shown that $\mathcal{K}_t \cap \mathcal{L}^1(\mathbb{Q})$ is a closed linear subspace of $\mathcal{L}^1(\Omega, \mathcal{F}_t, \mathbb{Q})$. Therefore, by the *Hahn-Banach theorem* applied to $\{U_t - U_{t-1}\}$ and $\mathcal{K}_t \cap \mathcal{L}^1(\mathbb{Q})$ yields some $Z \in L^\infty(\Omega, \mathcal{F}_t, \mathbb{Q})$ such that

$$\sup \{ \mathbb{E}_{\mathbb{Q}}[WZ] \mid W \in \mathcal{K}_t \cap \mathcal{L}^1(\mathbb{Q}) \} < \mathbb{E}_{\mathbb{Q}}[(U_t - U_{t-1})Z] < \infty.$$

From the linearity of $\mathcal{K}_t \cap \mathcal{L}^1(\mathbb{Q})$ we deduce that

$$\mathbb{E}_{\mathbb{Q}}[WZ] = 0, \quad \forall W \in \mathcal{K}_t \cap \mathcal{L}^1(\mathbb{Q}), \quad (1.13)$$

and hence that

$$\mathbb{E}_{\mathbb{Q}}[(U_t - U_{t-1})Z] > 0. \quad (1.14)$$

There is no loss of generality in assuming that $|Z| \leq \frac{1}{3}$, so that

$$\hat{Z} := 1 + Z - \mathbb{E}_{\mathbb{Q}}[Z \mid \mathcal{F}_{t-1}],$$

can be taken as the density $\frac{d\hat{\mathbb{Q}}}{d\mathbb{Q}} = \hat{Z}$ of a new probability measure $\hat{\mathbb{Q}} \sim \mathbb{Q}$. Since Z is $\mathcal{F}_t$ -measurable, the expectation of $\tilde{H}$ under $\hat{\mathbb{Q}}$ satisfies

$$\begin{aligned} \mathbb{E}_{\hat{\mathbb{P}}}[\tilde{H}] &= \mathbb{E}_{\mathbb{Q}}[\tilde{H}\hat{Z}] \\ &= \mathbb{E}_{\mathbb{Q}}[\tilde{H}] + \mathbb{E}_{\mathbb{Q}}[\mathbb{E}_{\mathbb{Q}}[\tilde{H} \mid \mathcal{F}_t]Z] - \mathbb{E}_{\mathbb{Q}}[\mathbb{E}_{\mathbb{Q}}[\tilde{H} \mid \mathcal{F}_{t-1}]\mathbb{E}_{\mathbb{Q}}[Z \mid \mathcal{F}_{t-1}]] \\ &= \mathbb{E}_{\mathbb{Q}}[\tilde{H} + \mathbb{E}_{\mathbb{Q}}[U_t Z] - \mathbb{E}_{\mathbb{Q}}[U_{t-1} Z]] > \mathbb{E}_{\mathbb{Q}}[\tilde{H}], \end{aligned}$$

where we have used (1.14) in the last step. On the other hand,

$$\mathbb{E}_{\hat{\mathbb{P}}}[\tilde{H}] \leq \frac{5}{3}\mathbb{E}_{\mathbb{Q}}[\tilde{H}] < \infty.$$

Thus, $\hat{\pi} := \mathbb{E}_{\hat{\mathbb{P}}}[\tilde{H}]$ will yield the desired arbitrage-free price larger than π if we have $\hat{\mathbb{P}} \in \mathcal{P}$.

Let us prove that $\hat{\mathbb{P}} \in \mathcal{P}$. For $k > t$, the $\mathcal{F}_t$ -measurability of $\hat{Z}$ and the property of conditional expectation under measure change yield that

$$\mathbb{E}_{\hat{\mathbb{P}}}[\tilde{S}_k - \tilde{S}_{k-1} \mid \mathcal{F}_{k-1}] = \mathbb{E}_{\mathbb{Q}}[\tilde{S}_k - \tilde{S}_{k-1} \mid \mathcal{F}_{k-1}] = 0.$$

For $k = t$, (1.13) yields $\mathbb{E}_{\mathbb{Q}}[(\tilde{S}_t - \tilde{S}_{t-1})Z \mid \mathcal{F}_{t-1}] = 0$. Thus, it follows from $\mathbb{E}_{\mathbb{Q}}[\hat{Z} \mid \mathcal{F}_{t-1}] = 1$ that

$$\mathbb{E}_{\hat{\mathbb{P}}}[\tilde{S}_t - \tilde{S}_{t-1} \mid \mathcal{F}_{t-1}] = \mathbb{E}_{\mathbb{Q}}[\tilde{S}_t - \tilde{S}_{t-1}] (1 - \mathbb{E}_{\mathbb{Q}}[Z \mid \mathcal{F}_{t-1}]) + \mathbb{E}_{\mathbb{Q}}[(\tilde{S}_t - \tilde{S}_{t-1})Z \mid \mathcal{F}_{t-1}] = 0.$$

Finally, if $k < t$ then $\mathbb{Q}$ and $\hat{\mathbb{P}}$ coincide on $\mathcal{F}_k$. Hence

$$\mathbb{E}_{\hat{\mathbb{P}}}[\tilde{S}_k - \tilde{S}_{k-1} \mid \mathcal{F}_{k-1}] = \mathbb{E}_{\mathbb{Q}}[\tilde{S}_k - \tilde{S}_{k-1} \mid \mathcal{F}_{k-1}] = 0,$$

and we may conclude that $\hat{\mathbb{P}} \in \mathcal{P}$.

It remains to construct another equivalent martingale measure $\check{\mathbb{P}}$ such that

$$\check{\pi} := \mathbb{E}_{\check{\mathbb{P}}}[H] < \mathbb{E}_{\mathbb{Q}}[H] = \pi. \quad (1.15)$$

But this is simply achieved by letting

$$\frac{d\check{\mathbb{P}}}{d\mathbb{Q}} = 2 - \frac{d\hat{\mathbb{P}}}{d\mathbb{Q}},$$

which defines a probability measure $\check{\mathbb{P}} \sim \mathcal{P}$, because the density $\frac{d\check{\mathbb{P}}}{d\mathbb{Q}}$ is bounded from above by $\frac{5}{3}$ and below by $\frac{1}{2}$ and $\check{\mathbb{P}} \in \mathcal{P}$ is then obvious as is (1.15). $\square$

EMM Consistent with Market Prices

Complete Markets

An arbitrage-free market in which every European contingent claim is attainable, would be ideal from an efficiency perspective as every claim then has a unique price process. It makes sense to define such an efficient market.

DEFINITION 1.26 (Complete Markets).

An arbitrage-free market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ is called *complete*, if every European contingent claim $H \in \mathcal{L}^{0,+}(\mathcal{F}_T)$ is attainable.

Following [Definitions 1.21](#) and [1.26](#), an arbitrage-free market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ is complete if and only if for all $H \in \mathcal{L}^{0,+}(\mathcal{F}_T)$ there exists a self-financing trading strategy $\varphi = (W_0, \phi)$ such that

$$H = W_0 + (\phi \cdot X)_T, \quad \mathbb{P}\text{-a.s.}$$

If the market is complete, then by [Theorem 1.22](#) we see that

$$\tilde{W}_t^H = \mathbb{E}_{\mathbb{Q}}[\tilde{H} \mid \mathcal{F}_t] = \tilde{W}_0 + (\phi \cdot \tilde{S})_t,$$

for any $H \in \mathcal{L}_+^0(\mathcal{F}_T)$ and $\mathbb{Q} \in \mathcal{P}$, i.e., the $\mathbb{Q}$ -martingale $(\tilde{W}_t^H)_{t \leq T}$ can be represented as a sum of a stochastic integral $(\phi \cdot \tilde{S})$ and some constant $\tilde{W}_0$.

The Second Fundamental Theorem of Asset Pricing

THEOREM 1.27 (Second Fundamental Theorem of Asset Pricing).

An arbitrage-free financial market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ is complete if and only if there exists a unique equivalent martingale measure.

Proof. If the market is complete, then every contingent claim $H \in \mathcal{L}^{0,+}(\mathcal{F}_T)$ is attainable and therefore by [Theorem 1.22](#) there exists a unique fair price process $(W_t^H)_{t \in T}$ such that for any $\mathbb{Q} \in \mathcal{P}$ we have

$$\tilde{W}_t^H = \mathbb{E}_{\mathbb{Q}}[\tilde{H} \mid \mathcal{F}_t], \quad \forall t = 0, 1, \dots, T.$$

In particular, this holds for the claims $\tilde{H} = 1_F$ for any $F \in \mathcal{F}_T$, which for $t = 0$ yields

$$\tilde{W}_0^H = \mathbb{E}_{\mathbb{Q}}[1_F \mid \mathcal{F}_0] = \mathbb{E}_{\mathbb{Q}}[1_F] = \mathbb{Q}(F)$$

for any $\mathbb{Q} \in \mathcal{P}$, i.e., all equivalent martingale measures coincide and are equal to $W_0(\varphi) = \tilde{W}_0^H$ for any replicating strategy φ .

Conversely, if there exists only one equivalent martingale measure $\mathbb{Q}'$, then by [Theorem 1.24](#) any H must be attainable, as

$$\sup_{\mathbb{Q} \in \mathcal{P}} \Pi(H) = \sup_{\mathbb{Q} \in \mathcal{P}} \Pi(H) = \mathbb{E}_{\mathbb{Q}'}[\tilde{H}],$$

where if H would be non-attainable then $\Pi(H)$ must be an open interval by [Theorem 1.25](#). $\square$

THEOREM 1.28 (Characterisation of Complete Markets).

For $\mathbb{Q} \in \mathcal{P}$ the following conditions are equivalent:

- (i) $\mathcal{P} = \{\mathbb{Q}\}$;
- (ii) $\mathbb{Q}$ is an extreme point of $\mathcal{P}$, the set of all equivalent martingale measures;
- (iii) $\mathbb{Q}$ is an extreme point of $\mathcal{Q}$, the set of all martingale measures;
- (iv) Every $\mathbb{Q}$ -martingale $(M_t)_{t \in T}$ can be represented as a discrete stochastic integral of a d -dimensional predictable process $(\phi_t)_{t \in T}$, i.e.,

$$M_t = M_0 + \sum_{k=1}^t \phi_k (\tilde{S}_k - \tilde{S}_{k-1}), \quad t = 0, \dots, T.$$

Complete Markets in Finite Discrete Time are Finite

We call $A \in \mathcal{F}$ an *atom* of $(\Omega, \mathcal{F}, \mathbb{P})$, whenever $\mathbb{P}(A) > 0$ and for each $B \in \mathcal{F}$ with $B \subseteq A$ we have either $\mathbb{P}(B) = 0$ or $\mathbb{P}(B) = \mathbb{P}(A)$.

PROPOSITION 1.29.

Let $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ be a discrete-time financial market with finite time horizon $T \in \mathbb{N}_0$ and $d \in \mathbb{N}$ assets. If $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ is complete, then $(\Omega, \mathcal{F}, \mathbb{P})$ can be decomposed into at most $(d + 1)^T$ atoms, and $\dim \mathcal{L}^0(\Omega, \mathcal{F}, \mathbb{P}) \leq (d + 1)^T$ holds true.

Proof. We do an induction over $T \in \mathbb{N}_0$. For the base case, let $T = 1$ and assume that the market is complete. Then, for every $H \in \mathcal{L}^{0,+}(\Omega, \mathcal{F}_1)$, we have

$$H = W_0(\varphi) + \varphi_1^\top \Delta S_1, \quad \mathbb{P}\text{-a.s.}$$

As $\mathcal{F}_0$ is trivial and $W_0(\varphi)$, φ_1 , and S_0 are $\mathcal{F}_0$ -measurable, they are constant, yielding

$$\mathcal{L}^0(\Omega, \mathcal{F}) = \mathcal{L}^0\left(\Omega, \sigma\left(S_1^{(1)}, \dots, S_1^{(d)}\right)\right) = \{W_0(\varphi) + \varphi_1^\top \Delta S_1\},$$

where the dimension of $\{W_0(\varphi) + \varphi_1^\top \Delta S_1\}$ is at most $d + 1$. Consequently, $\dim \mathcal{L}^0(\Omega, \mathcal{F}) \leq d + 1$ and since the dimension of $\mathcal{L}^0(\Omega, \mathcal{F})$ coincide with the number of atoms of $(\Omega, \mathcal{F}, \mathbb{P})$ the assertion follows.

Assume that we have proven the statement for $T - 1$, and proceed to show it for T . In this case, for $H \in \mathcal{L}^0(\Omega, \mathcal{F}_T)$, we can express it as

$$H = W_0(\varphi) + \sum_{k=1}^t \varphi_k^\top \Delta S_k = W_{T-1}(\varphi) + \varphi_T^\top \Delta S_k,$$

where W_{T-1} , φ_T and S_{T-1} are all $\mathcal{F}_{T-1}$ -measurable and hence constant on every of the at most $(d + 1)^{T-1}$ atoms $(A_i)_{i=1}^{(d+1)^{T-1}}$ of $(\Omega, \mathcal{F}_{T-1}, \mathbb{P})$. By similar reasoning as before, we deduce that $\dim \mathcal{L}^0(\Omega, \mathcal{F}_T, \mathbb{P}(\cdot | A_i)) \leq d + 1$ for every $i = 1, 2, \dots, (d + 1)^{T-1}$. Observe that if B is an atom of $(\Omega, \mathcal{F}_T, \mathbb{P})$, then

$$\mathbb{P}(B) = \sum_{i=1}^{(d+1)^{T-1}} \mathbb{P}(B \cap A_i). \quad (1.16)$$

Given that $A_i \in \mathcal{F}_{T-1} \subseteq \mathcal{F}_T$ and $A_i \cap B \subseteq B$, by the atom property, it must hold either $\mathbb{P}(A_i \cap B) = 0$ or $\mathbb{P}(B \cap A_i) = \mathbb{P}(B)$. However, by (1.16), $\mathbb{P}(B \cap A_i) = \mathbb{P}(B)$ can only hold for exactly one $i \in \{1, 2, \dots, (d + 1)^{T-1}\}$, which implies that B is also an atom in $(\Omega, \mathcal{F}_T, \mathbb{P}(\cdot | A_i))$. Since there are $(d + 1)^{T-1}$ possible A_i for which B is an atom in $(\Omega, \mathcal{F}_T, \mathbb{P}(\cdot | A_i))$, and the number of atoms in every $(\Omega, \mathcal{F}_T, \mathbb{P}(\cdot | A_i))$ is at most $(d + 1)$, there can be at most $(d + 1)^{T-1} \times (d + 1)$ atoms in $(\Omega, \mathcal{F}_T, \mathbb{P})$. $\square$

Superhedging

One might think that instead of searching for a replicating strategy, the seller of a contingent claim would be even better off by constructing a strategy which has a larger terminal wealth than H .

DEFINITION 1.30.

Let H be a European contingent claim. If there exists a self-financing strategy $\varphi = (W_0, \varphi)$ such that $W_0(\varphi) = W_0$ and $W_T(\varphi) \geq H$, $\mathbb{P}$ -a.s., then we call $\varphi = (W_0, \varphi)$ a *superhedge* of H and W_0 a *superhedge price*.

Although using a superhedge is “good” for the seller of the claim, the superhedge price that the seller charges is in general not an arbitrage-free price as the following proposition shows.

PROPOSITION 1.31.

For any European contingent claim H we define the cheapest superhedge price $\pi_u(H)$ of H as

$$\pi_u(H) := \inf \{V_0 \in \mathbb{R} : \exists \text{ self-financing } (V_0, \varphi) \text{ s.t. } W_0(\varphi) = V_0, W_T(\varphi) \geq H \text{ } \mathbb{P}\text{-a.s.}\}.$$

Then $\pi_u(H) \geq \sup \Pi(H)$. If there exists a self-financing strategy φ such that $W_0(\varphi) = \pi_u(H)$ and $W_T(\varphi) \geq H$, $\mathbb{P}$ -a.s., then we call $\varphi = (\pi_u(H), \varphi)$ a cheapest superhedge of the claim H .

Pricing and Hedging in the Cox–Ross–Rubinstein Model

We now construct the Cox–Ross–Rubinstein (CRR) model on a so called *canonical probability space*. In this model, we have $d = 1$ and set the bank account $S_t^{(0)} = (1 + r)^t$ for $t = 0, 1, \dots, T$, for some fixed interest rate $r > -1$. We have only one risky asset with price process $(S_t^{(1)})_{t \in T} := (S_t)_{t \in T}$. In the CRR model, we assume that the return series

$$R_t = \frac{\Delta S_t}{S_{t-1}}, \quad t = 1, 2, \dots, T$$

can only assume two possible values called U (Up) and D (Down) satisfying $-1 < D < U$. We define this model by

$$\Omega := \{-1, +1\}^T = \{\omega = (y_1, y_2, \dots, y_T) : y_t \in \{-1, +1\}\}.$$

Let $Y_t(\omega) = y_t$ for every $\omega = (y_1, y_2, \dots, y_T)$, and set

$$R_t(\omega) = D \frac{1 - Y_t(\omega)}{2} + U \frac{1 + Y_t(\omega)}{2},$$

and for some fixed $S_0 > 0$, set

$$S_t(\omega) = S_0 \prod_{k=1}^t (1 + R_k(\omega)), \quad \tilde{S}_t(\omega) = S_0 \prod_{k=1}^t \frac{1 + R_k(\omega)}{1 + r}, \quad t = 0, 1, \dots, T. \quad (1.17)$$

As a filtration $\mathcal{F} = \{\mathcal{F}_t\}_{t \geq 0}$, we choose $\mathcal{F}_t = \sigma(S_0, S_1, \dots, S_t) = \sigma(Y_1, Y_2, \dots, Y_t)$ for $t = 0, 1, \dots, T$ and note that $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and set $\mathcal{F} := \mathcal{F}_T$. Lastly, let $\mathbb{P}$ be a probability measure on $(\Omega, \mathcal{F})$ such that $\mathbb{P}(\{\omega\}) > 0$ for all $\omega \in \Omega$. We call the financial market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, (S^{(0)}, S^{(1)}))$ the *canonical CRR model*.

The Arbitrage-Free and Complete Cox–Ross–Rubinstein Model

Note that according to [Proposition 1.29](#), there is hope that the CRR model is complete since the number of atoms in $(\Omega, \mathcal{F}, \mathbb{P})$ is exactly $(d + 1)^T = 2^T$, so right at the boundary. In particular, if there were a time when the risky asset could assume three different values with non-trivial probability, then such a multinomial model could not be complete.

THEOREM 1.32.

Let $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, (S^{(0)}, S^{(1)}))$ denote the CRR model as above. Then the model is arbitrage-free if and only if $D < r < U$. Under this condition, the model is complete, and its unique equivalent martingale measure $\mathbb{Q}$ is characterised by the fact that the random variables $R_1, R_2, \dots, R_T$ are independent under $\mathbb{Q}$ and satisfy

$$\mathbb{Q}(R_t = U) = \frac{r - D}{U - D}, \quad \forall t = 1, 2, \dots, T.$$

Proof. Recall that a measure $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ is a martingale measure for the financial market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, (S^{(0)}, S^{(1)}))$ if and only if the discounted price process $\tilde{S} := (\tilde{S}_t)_{t=0}^T$ is a martingale under $\mathbb{Q}$. In the context of the CRR model, this condition implies that for all $t = 1, 2, \dots, T-1$, we have

$$\tilde{S}_t = \mathbb{E}_{\mathbb{Q}}[\tilde{S}_{t+1} | \mathcal{F}_t], \quad \mathbb{Q}\text{-a.s.}$$

By inserting (1.17), this condition reads as

$$\tilde{S}_t = \tilde{S}_t \mathbb{E}_{\mathbb{Q}}\left[\frac{1 + R_{t+1}}{1 + r} \mid \mathcal{F}_t\right],$$

which simplifies to

$$\mathbb{E}_{\mathbb{Q}}[R_{t+1} | \mathcal{F}_t] = r \quad \mathbb{Q}\text{-a.s.}$$

Since R_{t+1} takes only the values U and D , we have

$$\mathbb{E}_{\mathbb{Q}}[R_{t+1} | \mathcal{F}_t] = U\mathbb{Q}(R_{t+1} = U | \mathcal{F}_t) + D(1 - \mathbb{Q}(R_{t+1} = U | \mathcal{F}_t)).$$

Solving for $\mathbb{Q}(R_{t+1} = U | \mathcal{F}_t)$, we obtain

$$\mathbb{Q}(R_{t+1} = U | \mathcal{F}_t) = \frac{r - D}{U - D}.$$

Since the right-hand side does not depend on $\mathcal{F}_t$, it follows that $\mathbb{Q}(R_{t+1} = U | \mathcal{F}_t)$ is constant, and thus R_{t+1} is independent of $\mathcal{F}_t$ under $\mathbb{Q}$. Therefore, the R_t are independent under $\mathbb{Q}$, and

$$\mathbb{Q}(R_t = U) = \frac{r - D}{U - D}, \quad \forall t = 1, 2, \dots, T.$$

For $\mathbb{Q}$ to be a probability measure (i.e., with $0 < \mathbb{Q}(R_t = U) < 1$), we require $D < r < U$. Therefore, if the model is arbitrage-free (i.e., there exists an equivalent martingale measure $\mathbb{Q}$), then necessarily $D < r < U$. Conversely, if $D < r < U$, then we can define $\mathbb{Q}$ as above, and since $\mathbb{Q}$ is equivalent to $\mathbb{P}$ (because $\mathbb{Q}(\omega) > 0$ for all ω), and the discounted price process $\tilde{S}$ is a $\mathbb{Q}$ -martingale, the model is arbitrage-free. Moreover, since there is a unique $\mathbb{Q}$ satisfying the martingale condition, the model is complete. $\square$

Pricing in the Cox–Ross–Rubinstein Model

Consider an arbitrage-free CRR model and denote by $\mathbb{Q}$ its unique equivalent martingale measure according to [Theorem 1.32](#). As before given any European contingent claim $H \in \mathcal{L}^{0,+}(\Omega, \mathcal{F}_T)$, we define its discounted version as $\tilde{H} = \frac{H}{S_T^{(0)}}$. Observe that we can express any contingent claim $\tilde{H}$ as a function of the risky asset paths since $\mathcal{F}_T = \sigma(S_1, \dots, S_T)$, i.e., $\tilde{H} = h(S_0, \dots, S_T)$ for some function h .

PROPOSITION 1.33.

The unique fair price process $(\tilde{W}_t^H)_{t \in T}$ of the claim H can be written as

$$\tilde{W}_t^H(\omega) = w_t(S_0, S_1(\omega), \dots, S_t(\omega)),$$

for

$$w_t(x_0, \dots, x_t) = \mathbb{E}_{\mathbb{Q}}\left[h\left(x_0, \dots, x_t, x_t \frac{S_1}{S_0}, \dots, x_t \frac{S_{T-t}}{S_0}\right)\right]. \quad (1.18)$$

Observe that the wealth process $\tilde{W}^H$ is characterised by the recursion:

$$\begin{cases} \tilde{W}_T^H = \tilde{H} \\ \tilde{W}_t^H = \mathbb{E}_{\mathbb{Q}}[\tilde{W}_{t+1}^H | \mathcal{F}_t], \quad t = 0, 1, \dots, T-1 \end{cases}.$$

Consequently, for $p = \frac{r-D}{U-D}$, the functions w_t for $t = 1, \dots, T$ satisfy the recursion:

$$\begin{aligned} w_T(x_0, x_1, \dots, x_T) &= h(x_0, \dots, x_T), \\ w_t(x_0, x_1, \dots, x_t) &= p w_{t+1}(x_0, \dots, x_t, x_t(1+U)) + (1-p) w_{t+1}(x_0, \dots, x_t, x_t(1+D)), \end{aligned}$$

where $t = T-1, T-2, \dots, 0$.

Hedging in the Cox–Ross–Rubinstein Model

DEFINITION 1.34 (Delta Hedge).

Let H be a European contingent claim and let $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, (S^{(0)}, S^{(1)}))$ denote a complete CRR model. We define the *delta hedge* φ of H in this model as $\varphi_t(\omega) = \Delta_t(S_0, S_1(\omega), \dots, S_{t-1}(\omega))$, where for $t = 1, 2, \dots, T$ we set

$$\Delta_t(x_0, \dots, x_{t-1}) := (1+r)^t \frac{w_t(x_0, \dots, x_{t-1}(1+U)) - w_t(x_0, \dots, x_{t-1}(1+D))}{x_{t-1}(1+U) - x_{t-1}(1+D)}. \quad (1.19)$$

Since the CRR model is assumed complete in the definition of the delta hedge, it would be good if the delta hedge is a replicating strategy.

PROPOSITION 1.35.

The delta hedge φ in Definition 1.34 is a replicating strategy for any European contingent claim H .

Remark. Note that for a discounted claim $\tilde{H} = h(S_T)$ with payoff function that increases with S_T the wealth function, given as

$$w_t(x) = \mathbb{E}_{\mathbb{Q}} \left[h \left(\frac{S_{T-t}}{S_0} \right) \right], \quad t \in \mathbf{T},$$

is also increasing in x and therefore the delta hedge (1.19) is non-negative and hence, does not involve short selling.

EXAMPLE 1.36 (Hedging and Pricing a European Call Option in the CRR model). Suppose we have a European call option with a payoff $H = h(S_T)$, where S_T represents the stock price at maturity T . The wealth process W^H depends only on S_t for all $t = 0, 1, \dots, T$, and $W_t^H(\omega) = w_t(S_t(\omega))$ for $(w_t)_{t \in \mathbf{T}}$ as given by (1.18), i.e.,

$$w_t(x_t) = \sum_{k=0}^{T-t} h \left(x_t(1+D)^{T-t-k}(1+U)^k \right) \binom{T-t}{k} \left(\frac{r-D}{U-D} \right)^k \left(1 - \frac{r-D}{U-D} \right)^{T-t-k},$$

and in particular, the unique arbitrage-free price π^H is given by:

$$\pi^H = w_0(S_0) = \sum_{k=0}^T h \left(S_0(1+D)^{T-k}(1+U)^k \right) \binom{T}{k} \left(\frac{r-D}{U-D} \right)^k \left(1 - \frac{r-D}{U-D} \right)^{T-k}.$$

In particular, if H is the payoff of a discounted European call option, then

$$h(x) = \frac{(x-K)^+}{(1+r)^T},$$

where K is the strike price.

The unique arbitrage-free price of a European call option with strike K and maturity T in the CRR model is given by:

$$\pi_{EU}^H = \frac{1}{(1+r)^T} \sum_{k=0}^T \left(S_0(1+D)^{T-k}(1+U)^k - K \right)^+ \binom{T}{k} \left(\frac{r-D}{U-D} \right)^k \left(1 - \frac{r-D}{U-D} \right)^{T-k}.$$

The hedging strategy, φ , is given by $\varphi_t(\omega) = \Delta_t(S_0, S_1(\omega), \dots, S_{t-1}(\omega))$, where Δ_t is the hedge ratio (or the option's delta) at time t . The hedge ratio for all $t = 1, 2, \dots, T$ is determined as follows:

$$\Delta_t(x_0, \dots, x_{t-1}) = (1+r)^t \frac{w_t(x_0, \dots, x_{t-1}(1+U)) - w_t(x_0, \dots, x_{t-1}(1+D))}{x_{t-1}(1+U) - x_{t-1}(1+D)}.$$

The hedge ratio, Δ_t , represents the number of shares of the underlying stock that need to be held at each time step to replicate the option's payoff.

Pricing and Hedging in Incomplete Financial Markets

SECTION 2.1

Variance-Optimal Hedging

In an incomplete market setting, it is generally not possible to perfectly hedge every European contingent claim. As a result, the fair price of an option is not unique but spans an interval. One approach to determine a pricing and hedging criterion in this context is to find an initial price W_0 and a trading strategy ϕ that minimise the hedging error $W_0 + G_T(\phi) - H$ at the terminal time T . A common way to quantify this error is through the $\mathcal{L}^2$ -norm. This method is known as *S quadratic* or *variance-optimal hedging*. Throughout this section, we assume that H is a non-negative, square-integrable European contingent claim, i.e.,

$$H \in \mathcal{L}^2(\Omega, \mathcal{F}_T, \mathbb{P}) \cap \mathcal{L}^{0,+}(\Omega, \mathcal{F}_T).$$

Moreover, let $X_t^{(i)} \in \mathcal{L}^2(\mathbb{P})$ for all $t = 0, 1, \dots, T$ and $i = 1, 2, \dots, d$ denote the discounted asset price processes. We define the set of square-integrable strategies $\mathcal{S}^2$ by

$$\mathcal{S}^2 := \{\phi : \phi \text{ is } \mathbb{R}^d\text{-valued predictable process such that } G_T(\phi) \in \mathcal{L}^2(\Omega, \mathcal{F}_T, \mathbb{P})\}.$$

Here, $G_T(\phi)$ is the cumulative gain process up to time T , given by

$$G_T(\phi) = \sum_{t=1}^T \phi_t \Delta X_t,$$

where $\Delta X_t = X_t - X_{t-1}$.

DEFINITION 2.1 (Variance-Optimal Strategy).

A self-financing strategy $\varphi^* = (W_0^*, \phi^*)$ with $W_0^* \in \mathbb{R}$ and $\phi^* \in \mathcal{S}^2$ is called a *variance-optimal strategy* for H if

$$\mathbb{E}_{\mathbb{P}} \left[(W_0^* + G_T(\phi^*) - H)^2 \right] \leq \mathbb{E}_{\mathbb{P}} \left[(W_0 + G_T(\phi) - H)^2 \right], \quad (2.1)$$

for all $W_0 \in \mathbb{R}$ and $\phi \in \mathcal{S}^2$. That is, it minimises the expected quadratic hedging error among all self-financing strategies.

Recall that according to the self-financing condition, any pair $(W_0, \phi) \in \mathbb{R} \times \mathcal{S}^2$ defines a self-financing strategy φ with wealth process

$$W_t(\phi) = W_0 + G_t(\phi), \quad t = 0, 1, \dots, T.$$

Consequently, $W_0 + G_T(\phi)$ in equation (2.1) equals $W_T(\phi)$.

LEMMA 2.2.

Any two variance-optimal hedging strategies φ and φ' will have the same (discounted) terminal wealth, i.e., $W_T(\varphi) = W_T(\varphi')$, $\mathbb{P}$ -a.s.

Remark. While the terminal wealth $W_T(\varphi)$ is uniquely determined for variance-optimal strategies, the trading strategies ϕ themselves may not be unique. Different strategies may lead to the same terminal wealth due to the incomplete nature of the market.

To simplify the discussion, we will henceforth assume that $d = 1$. In other words, we are considering a market consisting of only one risky asset (alongside the bank account). A variance-optimal hedge is any solution to the minimisation problem:

$$\min \left\{ \mathbb{E}_{\mathbb{P}} \left[(W_0 + G_T(\phi) - H)^2 \right] : W_0 \in \mathbb{R}, \phi \in \mathcal{S}^1 \right\}.$$

Our strategy is to first optimise over all $\phi \in \mathcal{S}^2$ for a fixed $W_0 \in \mathbb{R}$ and then subsequently optimise over W_0 , hoping that this yields a variance-optimal hedge. We define the subspace:

$$\mathcal{G}_T := \{G_T(\phi) : \phi \in \mathcal{S}^2\} \subset \mathcal{L}^2(\Omega, \mathcal{F}_T, \mathbb{P}).$$

If $\mathcal{G}_T$ is a closed subset of $\mathcal{L}^2(\Omega, \mathcal{F}_T, \mathbb{P})$, then a variance-optimal strategy in the risky asset would be any ϕ^* such that $G_T(\phi^*) = \mathcal{P}(H)$, $\mathbb{P}$ -a.s., where $\mathcal{P}$ denotes the orthogonal projection onto $\mathcal{G}_T$.

However, it is not obvious whether $\mathcal{G}_T$ is closed or not. It turns out that a simple relation between the conditional variance and the conditional expectation of the increments of X yields a sufficient condition for the closedness of $\mathcal{G}_T$. For this, let us denote the conditional variance of the increments of the asset price X_t as

$$\sigma_t^2 := \text{Var}(\Delta X_t \mid \mathcal{F}_{t-1}), \quad t = 1, \dots, T.$$

Similarly, for the conditional expectation of the increments, we write

$$\mu_t := \mathbb{E}_{\mathbb{P}}[\Delta X_t \mid \mathcal{F}_{t-1}], \quad t = 1, \dots, T.$$

PROPOSITION 2.3 (Bounded Mean–Variance Trade-Off).

If there exists a constant $K > 0$ such that

$$\mu_t^2 \leq K\sigma_t^2, \quad \mathbb{P}\text{-a.s.}, \quad \forall t = 1, \dots, T, \quad (2.2)$$

then $\mathcal{G}_T$ is a closed subspace of $\mathcal{L}^2(\Omega, \mathcal{F}_T, \mathbb{P})$. The condition (2.2) is sometimes referred to as the bounded mean–variance trade-off.

Remark. This condition ensures that the variance of the asset's returns dominates its mean squared returns uniformly over time. This condition is crucial for the closedness of $\mathcal{G}_T$, as it prevents the trading strategies from becoming too “explosive” in the presence of high expected returns relative to volatility.

PROPOSITION 2.4.

Under the bounded mean–variance trade-off condition (2.2), there exists a variance-optimal strategy $\varphi^* = (W_0^*, \phi^*)$. Moreover, such a variance-optimal strategy is $\mathbb{P}$ -a.s. unique up to modifications of ϕ_t^* on the set $\{\sigma_t = 0\}$ for all $t = 1, 2, \dots, T$.

Proof. Since the bounded mean–variance trade-off condition (2.2) holds, Proposition 2.3 ensures that $\mathcal{G}_T$ is a closed subspace of $\mathcal{L}^2(\Omega, \mathcal{F}_T, \mathbb{P})$. Let

$$\mathcal{P} : \mathcal{L}^2(\mathbb{P}) \rightarrow \mathcal{G}_T$$

denote the orthogonal projection onto $\mathcal{G}_T$. For any $Z \in \mathcal{L}^2(\mathbb{P})$, $\mathcal{P}(Z)$ is the unique element in $\mathcal{G}_T$ satisfying

$$\mathbb{E}_{\mathbb{P}}[(Z - \mathcal{P}(Z))^2] = \min_{Y \in \mathcal{G}_T} \mathbb{E}_{\mathbb{P}}[(Z - Y)^2].$$

For any $W_0 \in \mathbb{R}$, there exists $\phi^{W_0} \in \mathcal{S}^d$ such that

$$G_T(\phi^{W_0}) = \mathcal{P}(H - W_0).$$

This is because $\mathcal{P}(H - W_0) \in \mathcal{G}_T$ by definition, and there exists a strategy ϕ^{W_0} such that $G_T(\phi^{W_0}) = \mathcal{P}(H - W_0)$. The expected squared hedging error is then

$$\mathbb{E}_{\mathbb{P}}[(W_0 + G_T(\phi^{W_0}) - H)^2] = \mathbb{E}_{\mathbb{P}}[(W_0 + \mathcal{P}(H - W_0) - H)^2].$$

Using the linearity of the projection operator, we have

$$\mathcal{P}(H - W_0) = \mathcal{P}(H) - W_0\mathcal{P}(1),$$

where 1 denotes the constant function equal to 1. Therefore,

$$W_0 + \mathcal{P}(H - W_0) - H = W_0 + (\mathcal{P}(H) - W_0\mathcal{P}(1)) - H = (\mathcal{P}(H) - H) + W_0(1 - \mathcal{P}(1)).$$

Thus, the expected squared hedging error becomes

$$J(W_0) = \mathbb{E}_{\mathbb{P}}[(\mathcal{P}(H) - H) + W_0(1 - \mathcal{P}(1))]^2.$$

This is a quadratic function in W_0 , since $\mathcal{P}(H) - H$ is independent of W_0 . Since $J(W_0)$ is quadratic with respect to W_0 and the coefficient of W_0^2 is non-negative (and positive unless $1 - \mathcal{P}(1) = 0$ a.s.), it attains its minimum at a unique value $W_0^* \in \mathbb{R}$. Therefore, the variance-optimal initial investment is W_0^* , and the corresponding trading strategy is $\varphi^* = \phi^{W_0^*}$.

To show the uniqueness of the variance-optimal strategy up to modifications on $\{\sigma_t = 0\}$, suppose there are two variance-optimal strategies $\varphi^* = (W_0^*, \phi^*)$ and $\tilde{\varphi}^* = (W_0^*, \tilde{\phi}^*)$ such that $\phi^* \neq \tilde{\phi}^*$ on a set of positive probability. Then,

$$G_T(\phi^*) = \mathbb{P}(H - W_0^*) = G_T(\tilde{\phi}^*).$$

This implies

$$G_T(\phi^* - \tilde{\phi}^*) = 0.$$

Since

$$G_T(\phi^* - \tilde{\phi}^*) = \sum_{t=1}^T (\phi_t^* - \tilde{\phi}_t^*) \Delta X_t,$$

we have

$$\sum_{t=1}^T (\phi_t^* - \tilde{\phi}_t^*) \Delta X_t = 0, \quad \mathbb{P}\text{-a.s.}$$

This means that the process $\phi_t^* - \tilde{\phi}_t^*$ is orthogonal to ΔX_t in $\mathcal{L}^2(\mathbb{P})$. Because $\sigma_t^2 = \mathbb{E}_{\mathbb{P}}[(\Delta X_t)^2 \mid \mathcal{F}_{t-1}]$, it follows that

$$(\phi_t^* - \tilde{\phi}_t^*) \sigma_t^2 = 0, \quad \mathbb{P}\text{-a.s.}$$

Thus, $\phi_t^* = \widetilde{\phi}_t^*$ a.s. on $\{\sigma_t > 0\}$. Therefore, the variance-optimal strategy is unique up to modifications on the sets where $\sigma_t = 0$. $\square$

Under additional assumptions, we can construct variance-optimal strategies explicitly. In particular, when the discounted asset price process X is a martingale under the probability measure $\mathbb{P}$, the variance-optimal strategy can be derived using martingale representation theorems.

Recall that in the variance-optimal hedging problem, the goal is to find a self-financing trading strategy $\varphi = (W_0, \phi)$ that minimises the expected squared hedging error $\mathbb{E}_{\mathbb{P}}[(W_T(\varphi) - H)^2]$. Note that the wealth process of a variance-optimal strategy corresponds to the orthogonal projection of the discounted payoff H onto the space of attainable terminal wealth generated by trading in the asset X . Specifically, we can decompose the martingale $\hat{W}_t = \mathbb{E}_{\mathbb{P}}[H \mid \mathcal{F}_t]$ into components driven by X and orthogonal to X .

LEMMA 2.5 (Kunita–Watanabe Decomposition).

Let $(Y_t)_{t=0}^T$ be a square-integrable $\mathbb{P}$ -martingale, and let $(X_t)_{t=0}^T$ be a square-integrable $\mathbb{P}$ -martingale. Then there exists a predictable process ϕ and a square-integrable martingale M orthogonal to X (i.e., $\mathbb{E}_{\mathbb{P}}[\Delta M_t \Delta X_t \mid \mathcal{F}_{t-1}] = 0$ for all t), such that

$$Y_t = Y_0 + \sum_{s=1}^t \phi_s \Delta X_s + M_t, \quad t = 0, 1, \dots, T.$$

Moreover, the decomposition is unique.

Proof. Since X and M are square-integrable martingales, the Cauchy–Schwarz inequality gives

$$\mathbb{E}_{\mathbb{P}}[|M_t X_t|] \leq \mathbb{E}_{\mathbb{P}}[M_t^2]^{\frac{1}{2}} \mathbb{E}_{\mathbb{P}}[X_t^2]^{\frac{1}{2}} < \infty.$$

Thus, $M_t X_t$ is integrable. From the construction above, we have $\mathbb{E}_{\mathbb{P}}[\Delta M_t \Delta X_t \mid \mathcal{F}_{t-1}] = 0$ for all t . Then for the martingale property, we verify that

$$\begin{aligned} \mathbb{E}_{\mathbb{P}}[M_t X_t \mid \mathcal{F}_{t-1}] &= M_{t-1} X_{t-1} + \mathbb{E}_{\mathbb{P}}[\Delta M_t \Delta X_t \mid \mathcal{F}_{t-1}] + M_{t-1} \mathbb{E}_{\mathbb{P}}[\Delta X_t \mid \mathcal{F}_{t-1}] + X_{t-1} \mathbb{E}_{\mathbb{P}}[\Delta M_t \mid \mathcal{F}_{t-1}] \\ &= M_{t-1} X_{t-1}. \end{aligned}$$

In conclusion, we have proved that the process $(M_t X_t)_{t \in T}$ is a martingale.

Suppose there are two decompositions with predictable processes ϕ, ϕ' and square-integrable martingales M, M' orthogonal to X . Set $\varphi := \phi - \phi'$ and $N := M - M'$. Subtracting the decompositions gives

$$\sum_{s=1}^t \varphi_s \Delta X_s + N_t = 0, \quad t = 0, 1, \dots, T.$$

Consequently, $\Delta N_t = -\varphi_t \Delta X_t$. Since N is orthogonal to X and φ_t is $\mathcal{F}_{t-1}$ -measurable,

$$0 = \mathbb{E}_{\mathbb{P}}[\Delta N_t \Delta X_t \mid \mathcal{F}_{t-1}] = -\varphi_t \mathbb{E}_{\mathbb{P}}[(\Delta X_t)^2 \mid \mathcal{F}_{t-1}].$$

Thus, $\varphi_t = 0$ whenever the conditional variance of ΔX_t is positive; on its zero set both integrands are set to zero by the convention in the construction. Hence $\phi = \phi'$ and then $M = M'$, proving uniqueness. $\square$

THEOREM 2.6.

Assume that under $\mathbb{P}$, the discounted asset price process X is a square-integrable martingale, i.e., $\mathbb{E}_{\mathbb{P}}[\Delta X_t \mid \mathcal{F}_{t-1}] = 0$ for all t . Let H be a square-integrable contingent claim, and let $\hat{W}_t = \mathbb{E}_{\mathbb{P}}[H \mid \mathcal{F}_t]$

be its $\mathbb{P}$ -martingale representation. Define the trading strategy $\varphi^* = (W_0^*, \phi^*)$ as

$$\phi_t^* := \frac{\mathbb{E}_{\mathbb{P}}[\Delta \hat{W}_t \Delta X_t \mid \mathcal{F}_{t-1}]}{\mathbb{E}_{\mathbb{P}}[(\Delta X_t)^2 \mid \mathcal{F}_{t-1}]} \mathbf{1}_{\{\mathbb{E}_{\mathbb{P}}[(\Delta X_t)^2 \mid \mathcal{F}_{t-1}] > 0\}}, \quad t = 1, \dots, T,$$

and

$$W_0^* := \hat{W}_0 = \mathbb{E}_{\mathbb{P}}[H].$$

Then, the trading strategy φ^* is variance-optimal for hedging H , and the mean squared hedging error is given by

$$\mathbb{E}_{\mathbb{P}} \left[\left(W_0^* + \sum_{t=1}^T \phi_t^* \Delta X_t - H \right)^2 \right] = \sum_{t=1}^T \mathbb{E}_{\mathbb{P}} \left[\text{Var}(\Delta \hat{W}_t \mid \mathcal{F}_{t-1}) - (\phi_t^*)^2 \text{Var}(\Delta X_t \mid \mathcal{F}_{t-1}) \right].$$

Proof. By Lemma 2.5, the Kunita–Watanabe decomposition of $\hat{W}_t$ is

$$\hat{W}_t = \hat{W}_0 + \sum_{s=1}^t \phi_s \Delta X_s + M_t, \quad t = 0, 1, \dots, T.$$

Given that the wealth process

$$W_t^* = W_0^* + \sum_{s=1}^t \phi_s^* \Delta X_s,$$

and $W_0^* := \hat{W}_0 = \mathbb{E}_{\mathbb{P}}[H]$, we have $\hat{W}_T = W_T^* + M_T$. At terminal time T , $H = \hat{W}_T \implies M_T = H - W_T^*$. By the hint, the predictable quadratic variation of M is

$$\langle M \rangle_T = \sum_{t=1}^T \mathbb{E}_{\mathbb{P}}[(\Delta M_t)^2 \mid \mathcal{F}_{t-1}].$$

Then we calculate

$$\begin{aligned} \mathbb{E}_{\mathbb{P}}[M_T^2] &= \mathbb{E}_{\mathbb{P}}[\langle M \rangle_T] \\ &= \sum_{t=1}^T \mathbb{E}_{\mathbb{P}}[\mathbb{E}_{\mathbb{P}}[(\Delta M_t)^2 \mid \mathcal{F}_{t-1}]] \\ &= \sum_{t=1}^T \mathbb{E}_{\mathbb{P}} \left[\mathbb{E}_{\mathbb{P}} \left[(\Delta \hat{W}_t - \phi_t^* \Delta X_t)^2 \mid \mathcal{F}_{t-1} \right] \right] \quad (\Delta M_t = \Delta \hat{W}_t - \phi_t^* \Delta X_t \text{ from the lecture notes}) \\ &= \sum_{t=1}^T \mathbb{E}_{\mathbb{P}} \left[\mathbb{E}_{\mathbb{P}} \left[(\Delta \hat{W}_t)^2 \mid \mathcal{F}_{t-1} \right] - 2\phi_t^* \mathbb{E}_{\mathbb{P}}[\Delta \hat{W}_t \Delta X_t \mid \mathcal{F}_{t-1}] + (\phi_t^*)^2 \mathbb{E}_{\mathbb{P}}[(\Delta X_t)^2 \mid \mathcal{F}_{t-1}] \right] \\ &= \sum_{t=1}^T \mathbb{E}_{\mathbb{P}} \left[\mathbb{E}_{\mathbb{P}} \left[(\Delta \hat{W}_t)^2 \mid \mathcal{F}_{t-1} \right] - (\phi_t^*)^2 \mathbb{E}_{\mathbb{P}}[(\Delta X_t)^2 \mid \mathcal{F}_{t-1}] \right] \\ &\quad (\phi_t^* \mathbb{E}_{\mathbb{P}}[(\Delta X_t)^2 \mid \mathcal{F}_{t-1}] = \mathbb{E}_{\mathbb{P}}[\Delta \hat{W}_t \Delta X_t \mid \mathcal{F}_{t-1}]) \\ &= \sum_{t=1}^T \mathbb{E}_{\mathbb{P}} \left[\text{Var}(\Delta \hat{W}_t \mid \mathcal{F}_{t-1}) - (\phi_t^*)^2 \text{Var}(\Delta X_t \mid \mathcal{F}_{t-1}) \right], \end{aligned}$$

where in the last step we have

$$\text{Var}(\Delta \hat{W}_t \mid \mathcal{F}_{t-1}) = \mathbb{E}_{\mathbb{P}} \left[(\Delta \hat{W}_t)^2 \mid \mathcal{F}_{t-1} \right] - \underbrace{\left(\mathbb{E}_{\mathbb{P}}[\Delta \hat{W}_t \mid \mathcal{F}_{t-1}] \right)^2}_{=0},$$

and similarly

$$\text{Var}(\Delta X_t \mid \mathcal{F}_{t-1}) = \mathbb{E}_{\mathbb{P}}[(\Delta X_t)^2 \mid \mathcal{F}_{t-1}].$$

On the other hand,

$$\mathbb{E}_{\mathbb{P}}[M_T^2] = \mathbb{E}_{\mathbb{P}}[(H - W_T^*)^2] = \mathbb{E}_{\mathbb{P}}\left[\left(H - W_0^* - \sum_{t=1}^T \phi_t^* \Delta X_t\right)^2\right] = \mathbb{E}_{\mathbb{P}}\left[\left(W_0^* + \sum_{t=1}^T \phi_t^* \Delta X_t - H\right)^2\right].$$

Therefore, we have proved that the mean squared hedging error is given by

$$\mathbb{E}_{\mathbb{P}}\left[\left(W_0^* + \sum_{t=1}^T \phi_t^* \Delta X_t - H\right)^2\right] = \sum_{t=1}^T \mathbb{E}_{\mathbb{P}}\left[\text{Var}(\Delta \hat{W}_t \mid \mathcal{F}_{t-1}) - (\phi_t^*)^2 \text{Var}(\Delta X_t \mid \mathcal{F}_{t-1})\right]. \quad (2.3)$$

Then we have to show that the trading strategy φ^* is variance-optimal for hedging H . Suppose $\varphi = (W_0^*, \phi)$ is any other self-financing strategy with the same initial capital W_0^* , the corresponding terminal wealth is

$$W_T = W_0^* + \sum_{t=1}^T \phi_t \Delta X_t.$$

Then the hedging error is

$$H - W_T = H - W_T^* + W_T^* - W_T = M_T + W_T^* - W_T = M_T + \sum_{t=1}^T (\phi_t^* - \phi_t) \Delta X_t.$$

We calculate the mean squared hedging error as

$$\begin{aligned} \mathbb{E}_{\mathbb{P}}[(H - W_T)^2] &= \mathbb{E}_{\mathbb{P}}\left[\left(M_T + \sum_{t=1}^T (\phi_t^* - \phi_t) \Delta X_t\right)^2\right] \\ &= \mathbb{E}_{\mathbb{P}}[M_T^2] + \mathbb{E}_{\mathbb{P}}\left[\left(\sum_{t=1}^T (\phi_t^* - \phi_t) \Delta X_t\right)^2\right] + 2\mathbb{E}_{\mathbb{P}}\left[M_T \sum_{t=1}^T (\phi_t^* - \phi_t) \Delta X_t\right], \end{aligned}$$

where the cross term

$$\mathbb{E}_{\mathbb{P}}\left[M_T \sum_{t=1}^T (\phi_t^* - \phi_t) \Delta X_t\right] = \sum_{s=1}^T \sum_{t=1}^T \underbrace{\mathbb{E}_{\mathbb{P}}[\Delta M_s (\phi_t^* - \phi_t) \Delta X_t]}_{=0, \text{ since orthogonal}} = 0.$$

Therefore

$$\mathbb{E}_{\mathbb{P}}[(H - W_T)^2] = \mathbb{E}_{\mathbb{P}}[M_T^2] + \mathbb{E}_{\mathbb{P}}\left[\left(\sum_{t=1}^T (\phi_t^* - \phi_t) \Delta X_t\right)^2\right] \geq \mathbb{E}_{\mathbb{P}}[M_T^2] = \mathbb{E}_{\mathbb{P}}[(H - W_T^*)^2].$$

Hence, the trading strategy $\varphi^* = (W_0^*, \phi^*)$ is variance-optimal for hedging H and its mean squared hedging error is given by (2.3). $\square$

Remark. The variance-optimal strategy φ^* replicates the component of the martingale $\hat{W}$ that is driven by X . The residual martingale M , which is orthogonal to X , represents the part of the risk that cannot be hedged by trading in X .

SECTION 2.2

| Risk Measures

A drawback of using the variance to measure the hedging error is that the variance is symmetric; that is, positive and negative deviations from being a replicating strategy are treated the same. However, a positive deviation, i.e., when $W_T(\varphi)(\omega) - H(\omega) > 0$, is actually favourable for the bank. Therefore, it would be better to use a measure that focuses on the downside risk. This leads us to the concept of risk measures, which we introduce in this section.

Monotonicity, Cash Invariance, Convexity, and Coherence

DEFINITION 2.7 (Risk Measures).

Let $\mathcal{X}$ denote a class of real-valued random variables on $(\Omega, \mathcal{F})$ representing *financial positions*. We call a map $\rho : \mathcal{X} \rightarrow \mathbb{R}$ a *risk measure* if it satisfies the following conditions for all $X, Y \in \mathcal{X}$:

- (i) *monotonicity*: if $X \leq Y$ almost surely, then $\rho(X) \geq \rho(Y)$;
- (ii) *cash invariance*: for all $m \in \mathbb{R}$, $\rho(X + m) = \rho(X) - m$.

If a risk measure ρ satisfies $\rho(0) = 0$, then we call it *normalised*. Additionally, if ρ is *convex*, i.e., if

$$\rho(\lambda X + (1 - \lambda)Y) \leq \lambda \rho(X) + (1 - \lambda)\rho(Y), \quad \forall 0 \leq \lambda \leq 1,$$

then we call ρ a *convex risk measure*.

- Remark.*
- (i) A risk measure is monotone decreasing because the downside risk of a financial position with a larger payoff profile is less than that of a position with a smaller payoff.
 - (ii) Cash invariance means that adding a riskless amount of cash m to a financial position reduces the risk measure by m . This reflects the idea that holding cash reduces risk, while owing cash (debt) increases risk.
 - (iii) Convexity of a risk measure implies that diversification reduces risk. Specifically, combining two financial positions should not increase the risk measure beyond the weighted average of their individual risks.

DEFINITION 2.8 (Coherent Risk Measures).

A convex risk measure ρ is called a *coherent risk measure* if it is also *positive homogeneous* and *subadditive*. Specifically, ρ satisfies:

- (i) *positive homogeneity*: $\rho(\lambda X) = \lambda \rho(X)$ for all $\lambda \geq 0$;
- (ii) *subadditivity*: $\rho(X + Y) \leq \rho(X) + \rho(Y)$.

Remark. Coherence allows financial institutions to split up their financial position into smaller parts, assess the risk of each part, and then bound the total risk by the sum of the risks of the parts. However, positive homogeneity means that the financial risk grows linearly with its size, which may not always hold in real markets due to economies or diseconomies of scale. Therefore, it is sometimes more appropriate to measure risk with non-coherent (convex but not coherent) risk measures.

Acceptance Sets of Risk Measures

Let $\mathcal{X}$ denote a set of *discounted financial positions* in a financial market $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_{t=0}^T, X)$; that is, $\mathcal{X}$ is a set of real-valued measurable functions defined on some set of outcomes Ω . For instance, we could think of $\mathcal{X} = \mathcal{L}^p(\Omega, \mathcal{F}, \mathbb{P})$ for $1 \leq p \leq \infty$.

DEFINITION 2.9 (Acceptance Sets).

Given a risk measure $\rho : \mathcal{X} \rightarrow \mathbb{R}$, we define the *acceptance set* of ρ as

$$\mathcal{A}_\rho := \{X \in \mathcal{X} : \rho(X) \leq 0\}.$$

We call a financial position $X \in \mathcal{A}_\rho$ *acceptable* with respect to the risk measure ρ .

From the perspective of a supervisory agency, the risk $\rho(X)$ of a financial position X can be viewed as the minimal capital requirement that, when added to the position and invested in a risk-free asset, makes the financial position acceptable. Indeed, $\rho(X)$ is the exact amount of capital that one has to add to the financial position X such that, by cash invariance,

$$\rho(X + \rho(X)) = \rho(X) - \rho(X) = 0,$$

i.e., the adjusted position $X + \rho(X)$ is acceptable (since it belongs to $\mathcal{A}_\rho$).

PROPOSITION 2.10.

Let ρ be a risk measure on the space of bounded measurable functions $\mathcal{X} = L^\infty(\Omega, \mathcal{F}, \mathbb{P})$. For the acceptance set $\mathcal{A}_\rho$, the following holds true:

- (i) $\mathcal{A}_\rho$ is non-empty and closed in $\mathcal{X}$ equipped with the supremum norm $\|\cdot\|_\infty$.
- (ii) $\inf \{m \in \mathbb{R} : m\mathbf{1} \in \mathcal{A}_\rho\} > -\infty$.
- (iii) If $X \in \mathcal{A}_\rho$ and $Y \in \mathcal{X}$ with $Y \geq X$, then $Y \in \mathcal{A}_\rho$.
- (iv) For all $X \in \mathcal{X}$, $\rho(X) = \inf \{m \in \mathbb{R} : m + X \in \mathcal{A}_\rho\}$.
- (v) ρ is convex if and only if $\mathcal{A}_\rho$ is convex.
- (vi) ρ is positive homogeneous, i.e., $\rho(\lambda X) = \lambda \rho(X)$ for all $\lambda \geq 0$, if and only if $\mathcal{A}_\rho$ is a cone; that is, $\lambda \mathcal{A}_\rho \subseteq \mathcal{A}_\rho$ for all $\lambda \geq 0$.

Proof. (i) Non-emptiness and Closedness of $\mathcal{A}_\rho$.

(ii) Boundedness of $\inf \{m \in \mathbb{R} : m\mathbf{1} \in \mathcal{A}_\rho\}$.

(iii) Monotonicity of $\mathcal{A}_\rho$.

(iv) Expression of ρ in terms of $\mathcal{A}_\rho$.

(v) Convexity of ρ and $\mathcal{A}_\rho$.

(vi) Positive homogeneity of ρ and conic property of $\mathcal{A}_\rho$.

□

Conversely, one can start with an acceptance set $A \subseteq \mathcal{X}$ and define a risk measure based on it. For a position $X \in \mathcal{X}$, define the capital requirement as the minimal capital $m \in \mathbb{R}$ that, when added to the position, makes it acceptable:

$$\rho_A(X) := \inf \{m \in \mathbb{R} : m + X \in A\}. \quad (2.4)$$

PROPOSITION 2.11.

Assume that A is a non-empty subset of $\mathcal{X}$ satisfying:

- *Monotonicity:* If $X \in A$ and $Y \in \mathcal{X}$ with $Y \geq X$, then $Y \in A$.
- *Boundedness from below:* There exists $m_0 \in \mathbb{R}$ such that $m_0 \mathbf{1} \in A$.

Then, the functional $\rho_{\mathcal{A}}$ defined in (2.4) has the following properties:

- (i) $\rho_{\mathcal{A}}$ is a risk measure.
- (ii) If A is convex, then $\rho_{\mathcal{A}}$ is a convex risk measure.
- (iii) If A is a cone, then $\rho_{\mathcal{A}}$ is positive homogeneous. In particular, $\rho_{\mathcal{A}}$ is a coherent risk measure if A is a convex cone.
- (iv) The acceptance set of $\rho_{\mathcal{A}}$ satisfies $A_{\rho_{\mathcal{A}}} = \bar{A}$, the closure of A with respect to the supremum norm $\|\cdot\|_{\infty}$. In particular, $A_{\rho_{\mathcal{A}}} = A$ if and only if A is closed in $\|\cdot\|_{\infty}$.

Examples of Risk Measures

EXAMPLE 2.12 (Value-at-Risk (VaR)). Consider a probabilistic model $(\Omega, \mathcal{F}, \mathbb{P})$. A financial position X is often considered acceptable if the probability of a loss does not exceed a certain level $\lambda \in (0, 1)$. Specifically, we define the *value-at-risk* at confidence level $1 - \lambda$ as:

$$\text{VaR}_{1-\lambda}(X) := \inf \{m \in \mathbb{R} : \mathbb{P}(X + m < 0) \leq \lambda\}.$$

Alternatively, VaR can be expressed in terms of the quantile function of X :

$$\text{VaR}_{1-\lambda}(X) = -F_X^{-1}(\lambda),$$

where F_X^{-1} denotes the quantile function of X . The Value-at-Risk $\text{VaR}_{1-\lambda}$ is a positively homogeneous risk measure; however, it is not convex and, therefore, not a coherent risk measure on $L^0(\Omega, \mathcal{F}, \mathbb{P})$.

Remark. VaR measures the minimum amount of capital that must be added to the position X so that the probability of a loss exceeding zero is at most λ . While VaR is widely used due to its simplicity and ease of calculation, it does not account for the magnitude of losses beyond the VaR threshold.

EXAMPLE 2.13 (Worst-Case Risk Measure). We define the *worst-case risk measure* $\rho_{\max}$ as

$$\rho_{\max}(X) = -\text{essinf} X = -\inf_{\omega \in \Omega} X(\omega), \quad \forall X \in \mathcal{X}.$$

The acceptance set of $\rho_{\max}$ is

$$\mathcal{A}_{\rho_{\max}} = \{X \in \mathcal{X} : X \geq 0\},$$

i.e., all financial positions that are almost surely non-negative. Thus, $\mathcal{A}_{\rho_{\max}}$ is a convex cone, and therefore, by [Proposition 2.10](#) (vi), $\rho_{\max}$ is a coherent risk measure.

Remark. The risk measure $\rho_{\max}$ is the most conservative among all normalised risk measures ρ , since for any ρ satisfying $\rho(0) = 0$, we have

$$\rho(X) \leq \rho(\text{essinf} X \cdot \mathbf{1}) = \rho(0) - \text{essinf} X = \rho_{\max}(X).$$

Moreover, $\rho_{\max}$ can be represented as the worst-case expected loss over all probability measures in $\mathcal{M}_1$:

$$\rho_{\max}(X) = \sup_{Q \in \mathcal{M}_1} \mathbb{E}_Q[-X],$$

where $\mathcal{M}_1$ is the set of all probability measures absolutely continuous with respect to $\mathbb{P}$ such that Q concentrates its mass on the worst outcomes of X .

EXAMPLE 2.14 (Sharpe Ratio and Mean–Standard Deviation Risk Measure). For an asset with (discounted) payoff $X_1 \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and today's price X_0 , the *Sharpe ratio* is defined as:

$$\text{SR}(X_1) := \frac{\mathbb{E}[X_1 - X_0]}{\sigma(X_1)},$$

where $\sigma(X_1)$ is the standard deviation of X_1 . We can define a financial position $X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ as acceptable if its Sharpe ratio is at least a certain threshold $c > 0$. The corresponding acceptance set is:

$$\mathcal{A}_{\rho_{\text{SR}}} = \left\{ X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P}) : \frac{\mathbb{E}[X] - X_0}{\sigma(X)} \geq c \right\}.$$

The associated risk measure, called the *mean–standard deviation risk measure*, is defined as:

$$\rho_{\text{SR}}(X) = -\mathbb{E}[X] + c\sigma(X), \quad \forall X \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P}).$$

Remark. While ρ_{SR} is cash-invariant, positively homogeneous, and convex, it is not monotone on all of $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$. This means that increasing the payoff X does not necessarily decrease the risk measure $\rho_{\text{SR}}(X)$, which contradicts the monotonicity property required for a risk measure as per Definition 2.7. Nevertheless, the mean–standard deviation risk measure is widely used in portfolio optimisation and risk assessment due to its connection with the Markowitz mean–variance framework.

EXAMPLE 2.15 (Conditional Value-at-Risk (CVaR) or Expected Shortfall). The *conditional value-at-risk* at confidence level $1 - \lambda$, also known as *expected shortfall*, is defined as:

$$\text{CVaR}_{1-\lambda}(X) = -\frac{1}{\lambda} \int_0^\lambda F_X^{-1}(u) du,$$

where F_X^{-1} is the quantile function of X . Alternatively, if X is a continuous random variable, CVaR can be expressed as the conditional expectation:

$$\text{CVaR}_{1-\lambda}(X) = -\mathbb{E}[X \mid X \leq -\text{VaR}_{1-\lambda}(X)].$$

CVaR is a coherent risk measure; it is convex, positively homogeneous, monotonic, and cash-invariant. It addresses some of the limitations of VaR by considering the expected loss given that the loss exceeds the VaR threshold.

Remark. CVaR provides a more comprehensive assessment of tail risk compared to VaR. It accounts for the magnitude of losses in the tail of the loss distribution, making it a preferred risk measure in regulatory frameworks and risk management practices.

EXAMPLE 2.16 (Entropic Risk Measure). The *entropic risk measure* is defined for a financial position X and a risk aversion parameter $\theta > 0$ as:

$$\rho_{\text{ent}}(X) = \frac{1}{\theta} \ln(\mathbb{E}[e^{-\theta X}]).$$

The entropic risk measure arises from exponential *utility functions* and captures the investor's risk aversion. It is convex and cash-invariant but not positively homogeneous, so it is not a coherent risk measure.

Remark. Despite not being coherent, the entropic risk measure is useful in contexts where the exponential utility function is appropriate, such as in certain insurance and risk-sensitive control

applications. It reflects the trade-off between expected return and the variability of outcomes in a way that emphasizes the exponential growth of risk with increasing losses.

EXAMPLE 2.17 (Mean–Variance Risk Measure). The mean–variance risk measure is given by:

$$\rho_{\text{MV}}(X) = -\mathbb{E}[X] + \frac{\lambda}{2} \text{Var}(X),$$

where $\lambda > 0$ is a risk aversion parameter. This risk measure penalises the variance of the financial position, reflecting the trade-off between expected return and risk.

Remark. Like the Sharpe ratio, the mean–variance risk measure is cash-invariant and convex but not monotone. It forms the basis of the Markowitz portfolio optimisation framework. While it does not satisfy all the properties of a risk measure as per [Definition 2.7](#), it is instrumental in understanding the balance between expected returns and the variability of those returns.

The following table summarises the key properties of the risk measures discussed:

Risk Measure	Monotone	Cash-Invariant	Convex	Coherent
Value-at-Risk (VaR)	Yes	Yes	No	No
Worst-Case Risk Measure	Yes	Yes	Yes	Yes
Sharpe Ratio (Mean–Std Dev)	No	Yes	Yes	No
Conditional Value-at-Risk (CVaR)	Yes	Yes	Yes	Yes
Entropic Risk Measure	Yes	Yes	Yes	No
Mean–Variance Risk Measure	No	Yes	Yes	No

Understanding the properties of different risk measures is crucial for selecting the appropriate measure in risk management and regulatory compliance. Coherent risk measures like CVaR provide desirable properties such as subadditivity, which ensures that diversification leads to a reduction in risk.

SECTION 2.3

| Deep Hedging

SECTION 2.4

| Expected Utility

Lotteries

We assume that the set $\mathcal{X}$ is a convex subset of the set of all probability measures on some measurable space $(S, \mathcal{S})$. We write $\mathcal{M}$ instead of $\mathcal{X}$. Our aim is to consider preference relations on the space of lotteries that admit a numerical representation of a special kind.

DEFINITION 2.18 (Von Neumann–Morgenstern Representation).

Let $>$ be a preference relation on $\mathcal{M}$. A numerical representation U is called a *Von Neumann–Morgenstern representation* if there is a measurable function $u : S \rightarrow \mathbb{R}$ such that

$$U(\mu) = \int u d\mu, \quad \forall \mu \in \mathcal{M}.$$

It is easy to check that a Von Neumann–Morgenstern representation U is an affine function, i.e.,

$$U(t\mu + (1-t)v) = tU(\mu) + (1-t)U(v), \quad \forall \mu, v \in \mathcal{M}, \forall t \in [0, 1].$$

If a numerical representation U of $>$ is affine, it implies two additional properties of $>$, that we define now.

DEFINITION 2.19.

Let $>$ be a preference relation on $\mathcal{M}$. It satisfies the *independence axiom* if for all $\mu > \nu$ it holds that

$$t\mu + (1-t)\lambda > t\nu + (1-t)\lambda,$$

for all $\lambda \in \mathcal{M}$ and $t \in (0, 1]$. The preference relation satisfies the *Archimedean axiom* (also called *continuity axiom*), if for all $\mu > \lambda > \nu$, there are $t, s \in (0, 1)$ such that

$$t\mu + (1-t)\nu > \lambda > s\mu + (1-s)\nu.$$

PROPOSITION 2.20.

Assume that $>$ admits an affine numerical representation. Then $>$ satisfies the axioms of [Definition 2.19](#).

THEOREM 2.21.

Suppose that a preference relation $>$ on $\mathcal{M}$ satisfies both the independence and Archimedean axioms. Then it has an affine numerical representation, say U . Moreover, for any other affine numerical representation $\bar{U}$, there exist $a > 0$ and $b \in \mathbb{R}$ such that $\bar{U} = aU + b$.

EXAMPLE 2.22. Suppose that $\mathcal{M}$ is the set of all finite mixtures of Dirac measures δ_x , and that an affine representation U exists. Define $u(x) = U(\delta_x)$. Let $\mu = \sum t_i \delta_{x_i}$, where the $t_i \geq 0$ and $\sum t_i = 1$. Affinity of U yields

$$U(\mu) = \sum t_i u(x_i) = \int u d\mu,$$

which is the desired representation. So, in this case, if there exists an affine representation, it is automatically of Von Neumann–Morgenstern type.

In the remainder of this section we assume that the set S is a separable metric space and that $\mathcal{S}$ is its Borel σ -algebra. Recall the definition of weak convergence of probability measures on S : $\mu_n \rightarrow \mu$ if and only if $\int f d\mu_n \rightarrow \int f d\mu$ for all bounded and continuous functions f on S .

LEMMA 2.23.

Consider the space $\mathcal{M}$ of all probability measures on $(S, \mathcal{S})$ endowed with the weak topology. Fix $\mu, \nu \in \mathcal{M}$ and consider $A : t \mapsto t\mu + (1-t)\nu$. Then $A : [0, 1] \rightarrow \mathcal{M}$ is continuous. If $>$ is a continuous preference ordering on $\mathcal{M}$, then it satisfies the Archimedean axiom.

THEOREM 2.24.

Consider the space $\mathcal{M}$ of all probability measures on $(S, \mathcal{S})$ endowed with the weak topology, where S is assumed to be separable. Let $>$ be a continuous preference ordering on $\mathcal{M}$, satisfying the independence axiom. Then $>$ admits a Von Neumann–Morgenstern representation

$$U(\mu) = \int u d\mu,$$

where the function $u : S \rightarrow \mathbb{R}$ is bounded, continuous and unique up to affine transformations.

Aversion

We now delve into the realm of utility and its expectation in the context of portfolio theory. Consider a set $\mathcal{M}$ of probability measures on an interval S of $\mathbb{R}$. Let $\mathcal{S}$ represent the Borel σ -algebra on S . For our setup, $\mathcal{M}$ is assumed to be convex and contains all Dirac measures on points in S , hence all simple measures as well. Typically, the fair price of a lottery $\mu \in \mathcal{M}$ is its expectation, given by

$$m(\mu) := \int_S x \mu(dx)$$

unless stated otherwise, these expectations are assumed to be finite for all $\mu \in \mathcal{M}$.

Remark. Consider concave functions $u : S \rightarrow \mathbb{R}$ satisfying the condition:

$$u(tx + (1-t)y) \geq tu(x) + (1-t)u(y)$$

for any $x, y \in S$ and $t \in [0, 1]$. It's worth noting the special properties of such concave functions which will be crucial for our further discussions. A function $u : S \rightarrow \mathbb{R}$ exhibits strict concavity if:

$$u(tx + (1-t)y) > tu(x) + (1-t)u(y)$$

for distinct $x, y \in S$ and t between 0 and 1.

Risk aversion is a key concept in portfolio theory. It's observed that individuals often prefer a guaranteed outcome over a lottery with the same expected payoff, due to personal preferences.

DEFINITION 2.25 (Monotone and Risk Averse Preference Order).

A preference order $>$ on $\mathcal{M}$ is called:

- (i) *Monotone*: if $x > y$ implies $\delta_x > \delta_y$ (with $x, y \in S$).
- (ii) *Risk averse*: if $\delta_{m(\mu)} > \mu$, unless μ is degenerate, i.e., $\mu = \delta_{m(\mu)}$.

PROPOSITION 2.26.

If a preference order $\succ$ on $\mathcal{M}$ has a Von Neumann–Morgenstern representation given by $U(\mu) = \int u d\mu$ then:

- (i) The preference order is monotone if and only if u is strictly increasing.
- (ii) The preference order is risk averse if and only if u is strictly concave.

DEFINITION 2.27.

A function $u : S \rightarrow \mathbb{R}$ is called a *utility function* if it is strictly increasing, strictly concave, and continuous on S . A preference order $\succ$ on $\mathcal{M}$ is said to have an *expected utility representation* U if there exists a utility function u such that

$$U(\mu) = \int u d\mu, \quad \forall \mu \in \mathcal{M}.$$

For any utility function u , a unique $c(\mu) \in S$ exists such that playing a lottery μ is indifferent to receiving a certain amount $c(\mu)$ under a given preference order.

DEFINITION 2.28 (Certainty Equivalent and Risk Premium).

For a given lottery μ , the number $c(\mu)$ is its *certainty equivalent* and the difference $\rho(\mu) := m(\mu) - c(\mu)$ is termed the *risk premium*.

Notice that always $c(\mu) \leq m(\mu)$ for risk averse $\succ$ and that strict inequality holds for non-degenerate μ . Hence, a risk averse person with utility function u will not pay more than $c(\mu)$ to play a lottery μ . Conversely, the risk premium is the amount of money a seller of the lottery μ has to pay to a risk averse agent to convince him to exchange the sure amount $m(\mu)$ for the random pay-off of the lottery μ .

In the present context, we consider the following optimisation problem. Find, if it exists, a lottery μ^* that is most preferred among all lotteries in a subset of $\mathcal{M}$, equivalently, the one with the highest value of U , where U is of expected utility type.

We specialise to a specific case. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be given and a random variable X defined on it, with values in S , that has a non-degenerate distribution μ . Let $c \in \mathbb{R}$ and consider the convex combination $X_\lambda = \lambda c + (1 - \lambda)X$. Note that the distribution function of X_λ is obtained by a location-scale transformation of that of X . Write μ_λ for the distribution of X_λ ($\mu_0 = \mu$). Put

$$f(\lambda) := U(\mu_\lambda) = \int u d\mu_\lambda = \mathbb{E}[u(X_\lambda)].$$

PROPOSITION 2.29.

Assume that S is an interval, $X \geq a$ for some $a \in \text{Int}S$, $\mathbb{E}_{\mathbb{P}}[X] < \infty$ and $c \in \text{Int}S$.

- (i) The function $f : [0, 1] \rightarrow \mathbb{R}$ is strictly concave and hence its maximal value is assumed for some unique $\lambda^* \in [0, 1]$.
- (ii) We have $\lambda^* = 1$ if $m(\mu) = \mathbb{E}_{\mathbb{P}}[X] \leq c$, and $\lambda^* > 0$ if $c \geq c(\mu)$.
- (iii) If moreover u is differentiable, then we even have

$$\lambda^* = 1 \iff \mathbb{E}_{\mathbb{P}}[X] \leq c, \quad \lambda^* = 0 \iff c \leq \frac{\mathbb{E}_{\mathbb{P}}[u'(X)]}{\mathbb{E}_{\mathbb{P}}[u(X)]}.$$

EXAMPLE 2.30. Consider a risky asset S_1 with price π_1 , and a riskless asset with interest rate r ($S_0 = 1 + r$). Suppose that an agent has a C^1 utility function u and a capital (initial wealth) w . Suppose that he builds a portfolio by investing a fraction λ of his capital in the riskless asset and the rest in the risky asset. The value of the portfolio (at time $t = 1$) is then

$$\lambda w(1 + r) + (1 - \lambda)w \frac{S_1}{\pi_1},$$

and the discounted net gain is

$$\frac{w S_1(1 - \lambda)}{\pi_1(1 + r)} - \pi_1.$$

The previous proposition shows that $\lambda^* = 1$ (all capital invested in the riskless asset) if and only if

$$\frac{\mathbb{E}[S_1]}{1 + r} \leq \pi_1.$$

Hence such an agent is only willing to invest in the risky asset, when the price is below the expected discounted value. Note that this holds for any risk averse investor, regardless of the special form of the utility function u . Compare this with what happens under the risk-neutral measure.

Arrow–Pratt Coefficient

Suppose that one considers a probability measure μ that has finite variance and that is concentrated on a small interval around its mean $m = m(\mu)$. Let u be a C^2 utility function on a neighbourhood of this interval and let U be the associated expected utility representation. A first order Taylor expansion of u around m gives

$$u(x) \approx u(m) + (x - m)u'(m).$$

With $x = c(\mu)$ one obtains $u(c(\mu)) \approx u(m) + (c(\mu) - m)u'(m)$. A second order Taylor expansion of u around m gives

$$u(x) \approx u(m) + (x - m)u'(m) + \frac{1}{2}(x - m)^2 u''(m).$$

Taking expectations yields

$$u(c(\mu)) = U(\mu) = \int u d\mu \approx u(m) + \frac{1}{2} \text{Var}(\mu) u''(m).$$

Hence, combining the two approximations, for the risk premium $\rho(\mu) = m - c(\mu)$ we have the approximation

$$\rho(\mu) \approx -\frac{1}{2} \frac{u''(m)}{u'(m)} \text{Var}(\mu).$$

We shall see that, in spite of the rough heuristics, the right hand side of this equation contains a useful quantity.

DEFINITION 2.31 (Arrow–Pratt Coefficients).

For u , a twice differentiable utility function on some (open) interval S , the quantity

$$\alpha(x) := -\frac{u''(x)}{u'(x)}$$

is called the *Arrow–Pratt coefficient of absolute risk aversion* of u at the level x .

Given that u is strictly concave and strictly increasing, it follows that $\alpha(x) \geq 0$ for all x . Additionally, based on the preceding discussion, for probability measures μ that are tightly clustered around their mean m , the risk premium $\rho(\mu)$ can be approximated as the product of the Arrow–Pratt coefficient of absolute risk aversion at m and half the variance of μ . This factorisation highlights that the risk premium depends on both the level of risk aversion at the mean and the variability of the outcomes, with the variance being an intrinsic property of μ that is independent of location.

The Arrow–Pratt coefficients possess the desirable property of remaining unchanged under affine transformations. In the context of Von Neumann–Morgenstern utility representations, where the utility function u is determined uniquely up to such transformations, the Arrow–Pratt coefficient emerges as an inherent characteristic of the preference ordering, rather than a particular numerical representation. This assertion holds provided that u is twice differentiable (i.e., $u \in C^2$) and non-constant; a constant u would imply a degenerate preference order devoid of any interesting structure.

EXAMPLE 2.32. Let u be such that the Arrow–Pratt function $\alpha(\cdot)$ is a (positive) constant, also denoted by α . Then, by solving a second order linear differential equation, one finds, for some constants $a \in \mathbb{R}$ and $b > 0$,

$$u_{a,b}(x) = a - be^{-\alpha x},$$

which is an affine transformation of $u(x) = 1 - e^{-\alpha x}$. Note that u is defined on all of $\mathbb{R}$. The functions $u_{a,b}$ are called *CARA functions* (from *Constant Absolute Risk Aversion*).

Here we introduce the *HARA utility functions* (from *Hyperbolic Absolute Risk Aversion*). For these functions we have that $\alpha(x) = \frac{\gamma}{x}$ for $x > 0$. For convenience we write $c = 1 - \gamma$, and hence $\gamma < 1$. Solving the corresponding differential equation for u yields

$$u_{a,b}(x) = \begin{cases} \frac{a\gamma}{\gamma x + b}, & \gamma \neq 0 \\ a \log x + b, & \gamma = 0 \end{cases}.$$

Note that $\gamma \geq 1$ is excluded by requiring that u is strictly concave and that for all $\gamma < 1$ it holds that $u'_{a,b}(x) = ax^{\gamma-1}$. The functions $u_{a,b}$ are affine transformations of $u_{1,0}$. HARA utility functions with $\gamma > 0$ are examples of utility functions $u : [0, \infty) \rightarrow \mathbb{R}$ satisfying the *Inada conditions*, i.e., $u \in C^1(0, \infty)$, with

$$\lim_{x \rightarrow 0} u'(x) = \infty, \quad \lim_{x \rightarrow \infty} u'(x) = 0.$$

PROPOSITION 2.33.

Suppose $u_1, u_2 : S \rightarrow \mathbb{R}$ are two C^2 utility functions, with corresponding risk premia $\rho_1(\cdot), \rho_2(\cdot)$, certainty equivalents $c_1(\cdot), c_2(\cdot)$ and Arrow–Pratt coefficients $\alpha_1(\cdot)$ and $\alpha_2(\cdot)$. The following are equivalent.

- (i) $\alpha_1(x) \geq \alpha_2(x), \forall x \in S$.
- (ii) There exist a strictly increasing concave function F , defined on the range of u_2 , such that $u_1 = F \circ u_2$.
- (iii) $\rho_1(\mu) \geq \rho_2(\mu), \forall \mu \in \mathcal{M}$.

Utility-Based Shortfall Risk

Utility Theory

Portfolio Optimisation

Let $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ denote a finite discrete-time financial market with $(d + 1) \in \mathbb{N}$ -tradable assets. Assume that the market is free of arbitrage, but possibly incomplete. Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be a utility function, recall that u is increasing and concave. In this section, we usually think of the following examples:

- CARA (*Constant Absolute Risk Aversion*) utility functions $u_{a,b}$ of the form

$$u_{a,b}(x) = a - be^{-\alpha x}, \quad a \in \mathbb{R}, b > 0, \alpha > 0.$$

- CRRA (*Constant Relative Risk Aversion*) utility functions u_λ of the form

$$u_\lambda(x) := \begin{cases} \frac{x^{1-\lambda}}{1-\lambda}, & x > 0 \\ -\infty, & x \leq 0 \end{cases},$$

and if $\lambda = 1$, set

$$u_1(x) := \begin{cases} \ln x, & x > 0 \\ -\infty, & x \leq 0 \end{cases}.$$

Suppose an investor with initial capital W_0 expresses his monetary preference for terminal wealth in terms of expected utility. Then he naturally wants to maximise:

$$\mathbb{E}_{\mathbb{P}}[u(W_T(\varphi))] = \mathbb{E}_{\mathbb{P}}[u(W_0 + G_T(\varphi))], \quad (3.1)$$

over all self-financing strategies φ such that the *budget constraint* $W_0(\varphi) \leq W_0$ is satisfied.

DEFINITION 3.1 (Utility Optimal Strategy).

We call a self-financing strategy $\varphi^* = (W_0^*, \varphi^*)$ a *utility optimal strategy* with respect to utility function u and capital constraint W_0^* , if φ^* maximises the expected utility of the terminal wealth over all other self-financing strategies $\varphi = (W_0, \varphi)$ with $W_0 \leq W_0^*$, i.e.,

$$\mathbb{E}_{\mathbb{P}}[u(W_0^* + G_T(\varphi^*))] \geq \mathbb{E}_{\mathbb{P}}[u(W_0 + G_T(\varphi))],$$

for all predictable $\mathbb{R}^d$ -valued processes φ .

Let us consider as second agent a bank, which is interested in selling a contingent claim $H \in \mathcal{L}^{0,+}(\mathcal{F}_T)$. Since the market is incomplete, there might be an entire interval $\Pi(H)$ of arbitrage-free prices for the claim H . Assume that also the bank's preference is given by the expected utility of some function u , then the bank faces the two options:

- (i) Selling the contingent claim for some price π_0 and trading according to a strategy $\varphi = (\pi_0, \varphi)$ that maximises (3.1) for $W_0 = \pi_0$.

- (ii) Not selling the claim H and trade according to a strategy with initial capital 0, that maximises (3.1) with $W_0 = 0$.

DEFINITION 3.2.

We call the price $\pi_u^* \in \mathbb{R}^+$ the *utility indifferent price* of the European contingent claim H , if it satisfies:

$$\sup_{\varphi} \mathbb{E}_{\mathbb{P}}[u(\pi_u^* + G_T(\varphi) - H)] = \sup_{\varphi} \mathbb{E}_{\mathbb{P}}[u(G_T(\varphi))], \quad (3.2)$$

and the suprema on both sides are attained. In the following, we denote by $\varphi_H^* = (\pi_H^*, \varphi_H^*)$ and $\varphi^* = (0, \varphi^*)$ the optimal strategies of the left- and right-hand side of (3.2), respectively. We refer to $\phi := \varphi_H^* - \varphi^*$ as the *utility-based hedging strategy*.

Remark. Following equation (3.2), the utility indifferent price represents exactly the threshold price at which the bank exhibits no preference, from an expected utility perspective, between participating in and abstaining from selling the claim H . The utility-based hedging strategy describes the adjustment to the bank's optimal portfolio necessitated by the derivative trade.

SECTION 3.1

| Static Portfolio Optimisation

In this section, we consider a *static* portfolio optimisation problem, where trading occurs only at two times: today ($t = 0$) and at a future terminal date ($t = T$). The investor wishes to set up a portfolio ϕ today that maximises their expected utility of terminal wealth $W_T(\phi)$ according to a utility function u . For simplicity, we focus on the pure maximisation problem without considering hedging positions, i.e., we set $H = 0$. We will show that this problem has at least one solution if and only if the market $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_{t=0}^T, X)$ is arbitrage-free.

Static Utility Optimal Strategies and the Absence of Arbitrage

Let X_0 denote the vector of asset prices at time $t = 0$, and X_T the vector of asset prices at time T . Let r denote the risk-free interest rate (assumed constant), so that the risk-free asset grows to $(1 + r)^T$ over the period $[0, T]$. Define the vector of *discounted net gains* as

$$Y := \frac{X_T}{(1 + r)^T} - X_0.$$

Thus, Y represents the *discounted profit or loss per unit of each asset* over the period $[0, T]$. Consider an initial investment of W_0 , with a portfolio ϕ such that the initial cost is $\phi \cdot X_0 = W_0$. The terminal wealth of the investor is

$$W_T(\phi) = \phi \cdot X_T = (1 + r)^T (W_0 + \phi \cdot Y).$$

Therefore, maximising the expected utility of terminal wealth $W_T(\phi)$ is equivalent to maximising the expected utility of $W_0 + \phi \cdot Y$. Since $(1 + r)^T$ is a constant, we can define a new utility function $\tilde{u}$ by

$$\tilde{u}(x) = u((1 + r)^T x),$$

so that

$$u(W_T(\phi)) = \tilde{u}(W_0 + \phi \cdot Y).$$

Therefore, the original optimisation problem is equivalent to maximising $\mathbb{E}_{\mathbb{P}}[\tilde{u}(W_0 + \phi \cdot Y)]$. Since W_0 is given and $\tilde{u}$ is strictly increasing, maximising $\mathbb{E}_{\mathbb{P}}[\tilde{u}(W_0 + \phi \cdot Y)]$ over ϕ is equivalent to maximising $\mathbb{E}_{\mathbb{P}}[\tilde{u}(\phi \cdot Y)]$ over ϕ .

Let $u : D \rightarrow \mathbb{R}$ be a utility function and Y the vector of discounted net gains. Assume either of the following:

- (i) $D = \mathbb{R}$ and u is bounded from above, or;
- (ii) $D = [a, \infty)$ for some $a < 0$, and we optimise over the set of ϕ such that $\phi \cdot Y \geq a$ a.s. In this case, we also assume that for those ϕ the expected utility $\mathbb{E}_{\mathbb{P}}[u(\phi \cdot Y)]$ is finite.

In both cases above, we define

$$\Xi := \{\phi \in \mathbb{R}^d : \phi \cdot Y \in D \text{ a.s.}\}.$$

Static Utility Maximisation Problem: let $u : D \rightarrow \mathbb{R}$ be a utility function. Maximise $\mathbb{E}_{\mathbb{P}}[u(\phi \cdot Y)]$ over all portfolios ϕ such that $\phi \cdot Y \in D$ a.s.

LEMMA 3.3. (i) Let $h : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{-\infty\}$ be a concave and upper semi-continuous function with $h(0) > -\infty$. Then h attains its supremum on $\mathbb{R}^d$ if for all $\phi \neq 0$,

$$\lim_{\alpha \rightarrow \infty} h(\alpha\phi) = -\infty.$$

- (ii) Let $u : D \rightarrow \mathbb{R}$ be a utility function, where $D = [a, \infty)$ with $a < 0$. Let $0 \leq b < -a$, and let $X \geq 0$ be a random variable. Then for all $\alpha \in (0, 1]$, the implication

$$\mathbb{E}_{\mathbb{P}}[u(\alpha X - b)] < \infty \implies \mathbb{E}_{\mathbb{P}}[u(X)] < \infty$$

holds.

THEOREM 3.4.

Let $u : D \rightarrow \mathbb{R}$ be a utility function satisfying the assumption, and let Y be the vector of discounted net gains. Then a maximiser in the utility maximisation problem exists if and only if the market is free of arbitrage.

THEOREM 3.5.

Let $u : D \rightarrow \mathbb{R}$ be a continuously differentiable utility function satisfying the assumption. Assume, additionally, that $\mathbb{E}_{\mathbb{P}}[|u(\phi \cdot Y)|] < \infty$ for all $\phi \in \Xi$. Let the maximising ϕ^* be an interior point of Ξ . Then $Y u'(\phi^* \cdot Y) \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and

$$\mathbb{E}_{\mathbb{P}}[Y u'(\phi^* \cdot Y)] = 0. \quad (3.3)$$

PROPOSITION 3.6.

Under the assumptions of [Theorem 3.5](#), and assuming the market is arbitrage-free, define

$$\frac{d\mathbb{P}^*}{d\mathbb{P}} = \frac{u'(\phi^* \cdot Y)}{\mathbb{E}_{\mathbb{P}}[u'(\phi^* \cdot Y)]}. \quad (3.4)$$

Then $\mathbb{P}^*$ is an equivalent martingale measure, i.e., a risk-neutral measure.

Proof. First, we show that $\mathbb{E}[u'(\phi^* \cdot Y)] < \infty$, so that $\mathbb{P}^*$ is well defined. Define

$$c := \sup \{u'(x) : x \in D \text{ and } x \in [-|\phi^*|, |\phi^*|]\}.$$

Consider first the case in which $D = \mathbb{R}$. Then, because u' is decreasing, we have $c = u'(-|\phi^*|)$. If $D = [a, \infty)$, then $c \leq \sup \{u'(x) : x \in D\} = u'(a)$. In both cases, we have $c < \infty$. By the *Cauchy–Schwarz inequality*, we have

$$|\phi^* \cdot Y| \leq |\phi^*| \cdot |Y|.$$

Hence, if $|\phi^* \cdot Y| > |\phi^*|$, then $|Y| > 1$. From this, it follows that

$$\begin{aligned} 0 &\leq u'(\phi^* \cdot Y) \\ &= u'(\phi^* \cdot Y) \mathbb{1}_{\{|\phi^* \cdot Y| \leq |\phi^*|\}} + u'(\phi^* \cdot Y) \mathbb{1}_{\{|\phi^* \cdot Y| > |\phi^*|\}} \\ &\leq c \mathbb{1}_{\{|\phi^* \cdot Y| \leq |\phi^*|\}} + u'(\phi^* \cdot Y) \mathbb{1}_{\{|Y| > 1\}} \\ &\leq c + u'(\phi^* \cdot Y) \mathbb{1}_{\{|Y| > 1\}} \\ &\leq c + u'(\phi^* \cdot Y) |Y| \mathbb{1}_{\{|Y| > 1\}} \leq c + u'(\phi^* \cdot Y) |Y|, \end{aligned}$$

which has finite expectation, by [Theorem 3.5](#). Next, we need to verify that $\mathbb{P}^*$ is a probability measure equivalent to $\mathbb{P}$. Since u' is positive (as u is strictly increasing and continuously differentiable), and $\mathbb{E}_{\mathbb{P}}[u'(\phi^* \cdot Y)] < \infty$, the Radon–Nikodym derivative is well-defined and positive, and integrates to 1:

$$\int_{\Omega} \frac{d\mathbb{P}^*}{d\mathbb{P}} d\mathbb{P} = \mathbb{E}_{\mathbb{P}} \left[\frac{u'(\phi^* \cdot Y)}{\mathbb{E}_{\mathbb{P}}[u'(\phi^* \cdot Y)]} \right] = 1.$$

Therefore, $\mathbb{P}^*$ is a probability measure equivalent to $\mathbb{P}$. Next, we need to show that $\mathbb{P}^*$ is a martingale measure, i.e., $\mathbb{E}_{\mathbb{P}^*}[Y] = 0$. Indeed,

$$\mathbb{E}_{\mathbb{P}^*}[Y] = \int_{\Omega} Y d\mathbb{P}^* = \int_{\Omega} Y \frac{d\mathbb{P}^*}{d\mathbb{P}} d\mathbb{P} = \frac{1}{\mathbb{E}_{\mathbb{P}}[u'(\phi^* \cdot Y)]} \mathbb{E}_{\mathbb{P}}[Y u'(\phi^* \cdot Y)] = 0,$$

where the last equality follows from (3.3). Therefore, under $\mathbb{P}^*$, the discounted net gains Y have zero expectation, implying that discounted asset prices are martingales under $\mathbb{P}^*$. $\square$

SECTION 3.2

Optimal Dynamic Hedging Strategies

This section addresses the problem of portfolio optimisation within a dynamic framework. In contrast to the static case, investment decisions are not limited to a single point in time and trades can be sequentially executed at all trading times $0 = t_0 < t_1 < \dots < t_n = T$.

The dynamic framework affords investors the capability to realign their portfolio allocations at various stages, adapting to the unfolding market dynamics. Nonetheless, the computation of optimal portfolios is challenging due to the high dimensionality of the space of trading strategies. A prevalent method to surmount this complexity is *dynamic programming*, which decomposes the problem into a series of simpler, recursively solved stages. In certain instances, the optimisation problem can be explicitly solved, or meaningful insights about the solutions can be deduced.

Dynamic Utility Optimal Trading Strategies

Let us come back to one of our primary objectives of this section, the maximisation of the expected utility of terminal wealth $\mathbb{E}_{\mathbb{P}}[u(G_T(\varphi))]$, $\mathbb{E}_{\mathbb{P}}[u(W_T(\varphi))]$ or $\mathbb{E}_{\mathbb{P}}[u(W_T(\varphi) - H)]$. Here, φ represents a trading strategy and H is the pay-off function of a contingent claim sold by our agent. Solving the two utility maximisation problems on both sides of

$$\sup_{\varphi} \mathbb{E}_{\mathbb{P}}[u(\pi_u^* + G_T(\varphi) - H)] = \sup_{\varphi} \mathbb{E}_{\mathbb{P}}[u(G_T(\varphi))]$$

allows for the derivation of the utility indifferent price π_u^* . In this part, we present a martingale method to tackle the optimisation problems presented on both sides of

$$\sup_{\varphi} \mathbb{E}_{\mathbb{P}}[u(\pi_u(H) + G_T(\varphi) - H)] = \sup_{\varphi} \mathbb{E}_{\mathbb{P}}[u(G_T(\varphi))],$$

with $\pi_u(H) \in \mathbb{R}^+$. This can be done using different stochastic optimisation approaches. In this section, however, we present a version of the so-called *martingale method* to construct optimal solutions through the construction of martingale measures. The derivations in this section are based on the following result, that establishes a sufficient condition for the optimality of a self-financing trading strategy in the sense of expected utility.

THEOREM 3.7.

Assume that the utility function u is differentiable and let $\varphi = (W_0, \varphi)$ be a self-financing strategy. Define a probability measure $\mathbb{Q} \sim \mathbb{P}$ by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \frac{u'(\tilde{W}_T(\varphi))}{\mathbb{E}_{\mathbb{P}}[u'(\tilde{W}_T(\varphi))]} \quad (3.5)$$

If $\mathbb{Q}$ is an equivalent martingale measure, then φ constitutes a utility optimal strategy.

Remark. Note that (3.5) can be interpreted as a first-order condition for extrema similar to (3.4) in the static case. Indeed, since the utility function u is strictly concave, any local maxima of $\varphi \rightarrow \mathbb{E}[u(W_T(\varphi))]$ is a global maxima and unique. Following the usual steps to find extremas of a function, we set the first-derivative of the function to zero. If we assume $\tilde{W}_T(\varphi) = \tilde{G}_T(\varphi)$ and by

using the chain-rule, we obtain something as follows:

$$\frac{d}{d\varphi} \mathbb{E}_{\mathbb{P}}[u(W_T(\varphi))] = e'(u(g(\varphi)))u'(g(\varphi))g'(\varphi),$$

for $e'(\cdot) = \mathbb{E}_{\mathbb{P}}[\cdot]$, $g'(\varphi) = \tilde{S}_T - \tilde{S}_0$ with $g(\varphi) = \int_0^T \varphi d\tilde{S}$. We thus obtain:

$$\begin{aligned} \frac{d}{d\varphi} \mathbb{E}_{\mathbb{P}}[u(W_T(\varphi))] &= \mathbb{E}_{\mathbb{P}}[u'(\tilde{G}_T(\varphi))(\tilde{S}_T - \tilde{S}_0)] \\ &= \mathbb{E}_{\mathbb{Q}}\left[u'(\tilde{G}_T(\varphi)) \frac{\mathbb{E}_{\mathbb{P}}[u'(\tilde{W}_T(\varphi))]}{u'(\tilde{W}_T(\varphi))} (\tilde{S}_T - \tilde{S}_0)\right] \\ &= \mathbb{E}_{\mathbb{P}}[u'(\tilde{G}_T(\varphi))] \mathbb{E}_{\mathbb{Q}}[\tilde{S}_T - \tilde{S}_0], \end{aligned}$$

which equals zero, whenever $(\tilde{S}_t)_{t \in \mathbf{T}}$ is a $\mathbb{Q}$ -martingale, i.e., if the condition in [Theorem 3.7](#) above holds true.

Utility Indifferent Pricing and Hedging with CRRA

Let us consider that $(\Omega, \mathcal{F}, \mathbb{P}, \mathcal{F}, X)$ constitutes an arbitrage-free financial market and that $u : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable CRRA utility function. In this section, we assume that the asset price process $X = (S^{(0)}, S^{(1)}, \dots, S^{(d)})$ has the following geometric form:

$$S_t^{(0)} = S_0^{(0)} e^{rt}, \quad t = 0, 1, \dots, T,$$

and for $i = 1, 2, \dots, d$ we set

$$S_t^{(i)} = S_0^{(i)} e^{R_t^{(i)}} = S_0^{(i)} \prod_{j=1}^t (1 + \Delta \hat{R}_j^{(i)}), \quad t = 0, 1, \dots, T,$$

where $\Delta \hat{R}_t^{(i)} = e^{\Delta R_t^{(i)}} - 1$ and $\Delta R_1^{(i)}, \Delta R_2^{(i)}, \dots, \Delta R_T^{(i)}$ are *i.i.d.* The discounted price processes for $i = 1, 2, \dots, d$ then assume the form

$$\tilde{S}_t^{(i)} = \tilde{S}_0^{(i)} \prod_{j=1}^t (1 + \Delta \tilde{R}_j^{(i)}),$$

with $\tilde{R}_0^{(i)} = 0$ and $\Delta \tilde{R}_j^{(i)} = \frac{1 + \Delta \hat{R}_j^{(i)}}{1+r} - 1$.

LEMMA 3.8.

Assume X to be an adapted process in discrete time T . There exists a unique adapted process Z , solving the difference equation $Z = 1 + Z^- \cdot X$ with $Z_n = Z_{n-1}$ for all $n \in \{1, 2, \dots, T\}$. The solution Z is referred to as the stochastic exponential of X , denoted by $\mathcal{E}(X)$. Moreover, the stochastic exponential satisfies

$$\mathcal{E}(X)_t = \prod_{j=1}^t (1 + \Delta X_j), \quad \forall t \in \mathbf{T}.$$

THEOREM 3.9.

Given $\lambda \in (0, \infty)$ as a risk aversion parameter, we define the utility function $u : \mathbb{R} \rightarrow \mathbb{R} \cup \{-\infty\}$, as

follows: If $\lambda \neq 1$, then

$$u(x) := \begin{cases} \frac{x^{1-\lambda}}{1-\lambda}, & x > 0 \\ -\infty, & x \leq 0 \end{cases}.$$

If $\lambda = 1$, we define $u(x)$ differently as:

$$u(x) := \begin{cases} \ln x, & x > 0 \\ -\infty, & x \leq 0 \end{cases}.$$

Further, let $\gamma \in \mathbb{R}^d$ be such that $\gamma^\top \Delta \tilde{R}_1 > -1$ and assume that it solves the following equation:

$$\mathbb{E}_{\mathbb{P}} \left[\frac{\Delta \tilde{R}_1}{(1 + \gamma^\top \Delta \tilde{R}_1)^\lambda} \right] = 0. \quad (3.6)$$

We define the process $(\tilde{W}_t)_{t \geq 0}$ as

$$\tilde{W}_t := \tilde{W}_0 \mathcal{E}(\gamma^\top \tilde{R})_t, \quad t = 0, 1, \dots, T, \quad (3.7)$$

and for all $i = 1, \dots, d$, we set

$$\begin{aligned} \phi_t^{(i)} &:= \frac{\gamma^{(i)}}{\tilde{S}_{t-1}^{(i)}} \tilde{W}_{t-1}, \quad t = 0, 1, \dots, T, \\ \phi_t^{(0)} &:= \tilde{W}_{t-1} - \left(\phi_t^{(1)}, \dots, \phi_t^{(d)} \right) \left(\tilde{S}_{t-1}^{(1)}, \dots, \tilde{S}_{t-1}^{(d)} \right), \end{aligned}$$

Then $\phi = (\phi^{(0)}, \phi^{(1)}, \dots, \phi^{(d)})$ is a utility optimal strategy for utility function u and initial wealth $W_0 := S_0^{(0)} \tilde{W}_0$. Moreover, the discounted wealth process $\tilde{W}((W_0, \phi))$ is equal to $\tilde{W}$ defined in (3.7).

Note that equation (3.6) represents a system of d equations with d unknowns, $\gamma^{(1)}, \dots, \gamma^{(d)}$. For some concrete models, these equations can be explicitly solved.

Suppose $\Delta \tilde{R}_1$ is sufficiently small. In this case, the approximation

$$(1 + \gamma^\top \Delta \tilde{R}_1)^{-\lambda} \approx 1 - \lambda \gamma^\top \Delta \tilde{R}_1$$

is valid. Consequently, equation (3.6) simplifies to

$$\mathbb{E}_{\mathbb{P}}[\Delta \tilde{R}_1] \approx \lambda \mathbb{E}_{\mathbb{P}}[\Delta \tilde{R}_1 \Delta \tilde{R}_1^\top] \gamma.$$

This leads to the following approximation for γ :

$$\gamma \approx \frac{1}{\lambda} (\mathbb{E}_{\mathbb{P}}[\Delta \tilde{R}_1 \Delta \tilde{R}_1^\top])^{-1} \mathbb{E}_{\mathbb{P}}[\Delta \tilde{R}_1],$$

where $(\mathbb{E}_{\mathbb{P}}[\Delta \tilde{R}_1 \Delta \tilde{R}_1^\top])^{-1}$ is approximately the inverse of the covariance matrix of $\Delta \tilde{R}_1$.

This result can be interpreted in the context of investment strategy. An investor, whether adhering to a power utility or log utility, tends to maintain a constant relative proportion of wealth in each asset. The degree of risk aversion influences the allocation between the non-risky bank account and the risky assets. Notably, this optimal portfolio strategy is independent of the investment horizon T , a finding that contradicts common financial advice which suggests allocating more to riskier assets for longer investment horizons due to their potential for higher returns.

Utility Indifferent Pricing and Hedging with CARA

In this section, we focus on utility indifferent pricing and hedging using a CARA utility function of the form

$$u(x) = 1 - \exp(-\lambda x),$$

where $\lambda > 0$ denotes the *risk aversion parameter*. Note that the shifted utility of terminal wealth $u(\tilde{W}_T(\phi)) - 1$ depends on the initial capital W_0 , only via the constant factor $e^{-\lambda W_0}$ and therefore W_0 does not affect the optimality of strategies and the indifferent price when considering expected utility with CARA as criterion. Therefore, without loss of generality, we can assume that $W_0 = 0$. In this section we do not impose any particular model for the asset price process $X = (S^{(0)}, \dots, S^{(d)})$. We denote the set of all trading strategies by $\mathcal{H}$. Instead, we assume directly that the utility maximisation problems on the left and right-hand side of

$$\sup_{\phi \in \mathcal{H}} \mathbb{E}_{\mathbb{P}}[u(\pi_u(H) + G_T(\phi) - H)] = \sup_{\phi \in \mathcal{H}} \mathbb{E}_{\mathbb{P}}[u(G_T(\phi))],$$

have solutions, i.e., the supremum is attained, and that the optimal strategies $\phi^* = (0, \phi_H^*)$ and $\phi_H^* = (\pi_H^*, \phi_H^*)$ are linked via [Theorem 3.7](#) to some equivalent martingale measures $\mathbb{Q}^*$ and $\mathbb{Q}_H^*$, i.e., we assume that:

- (i) $\sup_{\phi \in \mathcal{H}} \mathbb{E}[u(\tilde{G}_T(\phi))] = \mathbb{E}[u(\tilde{G}_T(\phi^*))]$ holds and that $\mathbb{Q}^*$ given by

$$\frac{d\mathbb{Q}^*}{d\mathbb{P}} = \frac{u'(\tilde{G}_T(\phi^*))}{\mathbb{E}_{\mathbb{P}}[u'(\tilde{G}_T(\phi^*))]}, \quad (3.8)$$

is an equivalent martingale measure.

- (ii) $\sup_{(\pi, \phi) \in \mathbb{R} \times \mathcal{H}} \mathbb{E}[u(\tilde{\pi} + \tilde{G}_T(\phi) - H)] = \mathbb{E}[u(\tilde{\pi}_H^* + G_T(\phi_H^*) - H)]$ holds and $\mathbb{Q}_H^*$ given by

$$\frac{d\mathbb{Q}_H^*}{d\mathbb{P}} = \frac{u'(\tilde{W}_T(\phi_H^*))}{\mathbb{E}_{\mathbb{P}}[u'(\tilde{W}_T(\phi_H^*))]}, \quad (3.9)$$

is an equivalent martingale measure.

DEFINITION 3.10.

Let $\mathbb{P} \sim \mathbb{Q}$ be two equivalent probability measures. We define the *relative entropy* or *Kullback–Leibler divergence* $D_{\text{KL}}(\mathbb{Q}, \mathbb{P})$ of $\mathbb{Q}$ and $\mathbb{P}$ as

$$D_{\text{KL}}(\mathbb{Q}, \mathbb{P}) := \mathbb{E}_{\mathbb{Q}} \left[\log \frac{d\mathbb{Q}}{d\mathbb{P}} \right] = \mathbb{E}_{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \log \frac{d\mathbb{Q}}{d\mathbb{P}} \right].$$

LEMMA 3.11.

For any two probability measures $\mathbb{P}$ and $\mathbb{Q}$ such that $\mathbb{Q} \sim \mathbb{P}$, we have $D_{\text{KL}}(\mathbb{Q}, \mathbb{Q}) = D_{\text{KL}}(\mathbb{P}, \mathbb{P}) = 0$ and $D_{\text{KL}}(\mathbb{Q}, \mathbb{P}) \geq 0$.

THEOREM 3.12.

Let the equivalent martingale measures $\mathbb{Q}^*$ and $\mathbb{Q}_H^*$ be as in (3.8) and (3.9), respectively. Next, define the probability measure $\mathbb{P}_H \sim \mathbb{P}$ by

$$\frac{d\mathbb{P}_H}{d\mathbb{P}} = \frac{\exp(\lambda \tilde{H})}{\mathbb{E}_{\mathbb{P}}[\exp(\lambda \tilde{H})]}.$$

Then the following assertions hold true:

- (i) The measure $\mathbb{Q}^*$ minimises the entropy $D_{\text{KL}}(\mathbb{Q}, \mathbb{P})$ among all equivalent martingale measures $\mathbb{Q} \in \mathcal{P}$. Moreover, $\mathbb{Q}^*$ does not depend on λ and the maximal expected utility $U^* := \mathbb{E}[u(\tilde{G}_T(\phi^*))]$ is given by

$$U^* = 1 - \exp(-D_{\text{KL}}(\mathbb{Q}^*, \mathbb{P})).$$

- (ii) The measure $\mathbb{Q}_H^*$ minimises the relative entropy

$$D_{\text{KL}}(\mathbb{Q}, \mathbb{P}_H) = D_{\text{KL}}(\mathbb{Q}, \mathbb{P}) - \lambda \mathbb{E}_{\mathbb{Q}}[\tilde{H}] + \log(\mathbb{E}[\exp(\lambda \tilde{H})])$$

among all equivalent martingale measures $\mathbb{Q} \in \mathcal{P}$. Moreover, the maximal expected utility $U_H^* := \mathbb{E}[u(\tilde{\pi}_H^* + \tilde{G}_T(\phi^*) - \tilde{H})]$ is given by

$$U_H^* = 1 - \exp\left(-D_{\text{KL}}(\mathbb{Q}_H^*, \mathbb{P}) + \lambda \mathbb{E}_{\mathbb{Q}_H^*}[\tilde{H}] - \lambda \tilde{\pi}_H^*\right).$$

THEOREM 3.13.

In the situation of [Theorem 3.12](#), the following holds true.

- (i) The utility-based hedging strategy φ and the measure $\mathbb{Q}_H^*$ depend on the premium π_H^* only through the bank account part.
- (ii) The (discounted) utility indifferent price $\tilde{\pi}_u(H)$ is given by

$$\begin{aligned} \tilde{\pi}_u(H) &= \frac{1}{\lambda} \log\left(\mathbb{E}\left[e^{\lambda \tilde{H}}\right]\right) + D_{\text{KL}}(\mathbb{Q}^*, \mathbb{P}) - D_{\text{KL}}(\mathbb{Q}_H^*, \mathbb{P}_H), \\ &= \frac{1}{\lambda} \log\left(\mathbb{E}_{\mathbb{Q}^*}\left[\exp(-\lambda(\tilde{G}_T(\varphi) - \tilde{H}))\right]\right). \end{aligned}$$

The quantities above all depend on the risk-aversion parameter $\lambda > 0$. To reflect that, we henceforth denote the respective quantities as u_λ , $U^*(\lambda)$, $U_H^*(\pi, \lambda)$, $\varphi^*(\lambda)$, $\varphi_H^*(\lambda)$, φ_λ and lastly $\pi_{u_\lambda}^*(H)$, the utility indifferent price. In the following theorem we treat the extreme cases that $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$.

LEMMA 3.14.

With the notation from above we have:

- (i) $\varphi^*(\lambda) = \lambda^{-1} \varphi^*(1)$ and $U^*(\lambda) = U^*(1)$ for any $\lambda > 0$;
- (ii) $\lambda \mapsto \pi^*(\lambda)$ is increasing on $\mathbb{R}^+$.

THEOREM 3.15.

Let the utility indifferent price be denoted by $\pi_{u_\lambda}^*(H)$. Then the following holds true.

- (i) $\pi_{u_\lambda}^*(H) \rightarrow \sup \Pi(H)$ as $\lambda \rightarrow \infty$, i.e., the utility indifferent price converges to the upper price bound of arbitrage-free prices.
- (ii) The utility based hedging strategy $\varphi(\lambda)$ is a so-called asymptotic super-hedge for H , i.e., it satisfies

$$\liminf_{\lambda \rightarrow \infty} \tilde{W}_T(\varphi(\lambda)) \geq H \quad \mathbb{P}\text{-a.s.},$$

where $\tilde{W}_T(\varphi(\lambda)) = \pi_{u_\lambda}^*(H) + (\varphi(\lambda) \cdot \tilde{S})_T$ is the terminal value of the utility based hedging strategy.

- (iii) If the cheapest superhedge ψ is unique, then $W_t(\varphi(\lambda)) \rightarrow W_t(\psi)$ for all $t \in \mathbb{T}$ as $\lambda \rightarrow \infty$.

Proof.

□

Note that for small λ , i.e., as $\lambda \rightarrow 0$, we can use the following heuristic argument: We assume that the utility-based hedge converges for $\lambda \rightarrow 0$ as

$$\varphi(\lambda) = \eta + \mathcal{O}(\lambda),$$

where η is some limiting hedge and $\mathcal{O}(\lambda)$ represents a strategy such that $\frac{\mathcal{O}(\lambda)}{\lambda}$ is bounded by a constant independent of λ . For $\varphi_H^*(\lambda)$, we have

$$\varphi_H^*(\lambda) = \varphi^*(\lambda) + \varphi(\lambda) = \frac{1}{\lambda} \varphi^*(\lambda) + \eta + \mathcal{O}(\lambda).$$

SECTION 3.3

Optimal Control Problems in Portfolio Theory

In this section, we delve into stochastic optimal control problems within the realm of portfolio theory. Our focus is on elucidating three pivotal *consumption–investment problems* (Problems 1–3 below) and exploring two primary methods for addressing these challenges.

- (i) *The Dynamic Programming Method*: This approach simplifies complex dynamic portfolio optimisation problems into more manageable static optimisation problems. It then tackles the original problem through a recursive solution strategy.
- (ii) *The Martingale Method*: This method bifurcates the dynamic portfolio optimisation problem into two parts: a static optimisation problem and a hedging problem. We then use a Lagrange Multiplier technique to solve the static optimisation problem.

Optimal Consumption–Investment Problems

Consider a financial market in finite-discrete time, represented as $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ with trading times $\mathbf{T} = \{0, 1, 2, \dots, T\}$ and an asset price vector $X = (S^{(0)}, S^{(1)}, \dots, S^{(d)})$ for $d \in \mathbb{N}$. The riskless asset $S^{(0)}$ evolves as $S_{t+1}^{(0)} = S_t^{(0)}(1 + r_t)$, where $(r_t)_{t \in \mathbf{T}}$ is the locally riskless rate. Risky assets follow $S_{t+1}^{(i)} = S_t^{(i)}(1 + R_t^{(i)})$ for return sequences $(R_t^{(i)})_{t \in \mathbf{T}}$. Define the utility function u , denoting investor preferences. The first objective is to maximise the expected utility from terminal wealth under budget restriction.

Problem 1. Expected Utility from Terminal Wealth

$$\begin{cases} \max_{(W_0, \phi)} \mathbb{E}_{\mathbb{P}}[u(W_0 + G_T(\phi))] \\ \text{s.t. } W_0 = w \in \mathbb{R} \quad \text{and} \quad \phi \text{ is predictable.} \end{cases}$$

DEFINITION 3.16 (Consumption–Investment Plans).

A *consumption–investment plan* is a pair $(K_t, \varphi_t)_{t \in \mathbf{T}}$ consisting of a nonnegative adapted process $(K_t)_{t \in \mathbf{T}}$ and a trading strategy $(\varphi_t)_{t \in \mathbf{T}}$. It is called *self-financing* if

$$W_t(\varphi) = K_t + \varphi_{t+1}^\top X_t, \quad t = 0, \dots, T-1, \quad (3.10)$$

and *admissible* if $K_T \leq W_T$ $\mathbb{P}$ -a.s.

The self-financing condition (3.10) is expressed as:

$$\Delta W_t(\varphi) = \varphi_t^\top \Delta S_t - K_{t-1}, \quad t = 1, \dots, T,$$

or in discounted terms:

$$\Delta \tilde{W}_t(\varphi) = \varphi_t^\top \Delta \tilde{S}_t - \gamma_{t-1}, \quad t = 1, \dots, T,$$

where $\gamma_{t-1} \triangleq \frac{K_{t-1}}{S_t^{(0)}}$ is the discounted consumption. The wealth evolution is then given by

$$\tilde{W}_t(\varphi) = \tilde{W}_0 + \sum_{s=1}^t (\varphi_s^\top \Delta \tilde{S}_s - \gamma_{s-1}).$$

We explore two additional optimisation problems related to expected utility from consumption.

Problem 2. Expected Utility from Consumption

$$\begin{cases} \max_{(K, \varphi)} \mathbb{E}_{\mathbb{P}} \left[\sum_{t=1}^T \beta_t u(K_t) \right], \\ \text{s.t. } W_0 = w, \beta_t \in (0, 1], \\ \text{and } (K, \varphi) \text{ is an admissible consumption-investment plan.} \end{cases}$$

Here, β_t are discount factors and u is a utility function, assumed constant across all times t , but can be time-dependent.

Problem 3. Expected Utility from Consumption and Terminal Wealth

$$\begin{cases} \max_{(K, \varphi)} \mathbb{E}_{\mathbb{P}} \left[\sum_{t=1}^T \beta_t u_c(K_t) + \beta_T u_p(W_T(\varphi) - K_T) \right], \\ \text{s.t. } \varphi \text{ is a self-financing strategy, } \beta_t \in (0, 1], \\ \text{and } K \text{ is a non-negative admissible consumption plan.} \end{cases}$$

Here, we consider two utility functions u_c and u_p .

Stochastic Optimal Control

We frame *Problems 1-3* in the context of *stochastic optimal control problems* and address them using *dynamic programming* and the *martingale method* in the forthcoming sections. Note here that the maximisation problems on both sides of the utility-based hedging problem (3.2) can be considered as *Problem 1* (with slight adjustments).

Let $(V_t)_{t \in T}$ be a given process (think of the wealth process of some strategy) that depend on the choice of a $\mathbb{R}^m$ -valued *control sequence* $(\alpha_t)_{t \in T}$ (think of a trading strategy), that is adapted to a filtration $(\mathcal{F}_t)_{t \in T}$. More precisely, we assume that $(V_t)_{t \in T}$ is of the following form:

$$V_{t+1} = G_t(V_t, \alpha_t, \varepsilon_t), \quad t = 0, \dots, T-1, \quad (3.11)$$

with initial value V_0 and given random quantities $\varepsilon_0, \varepsilon_1, \dots, \varepsilon_{T-1}$ (think of random returns of asset prices) that are assumed to be *i.i.d.* for now. If the ε_t for $t = 0, \dots, T-1$ are k -dimensional, then the functions G_t in (3.11) are defined on some appropriate subsets of $\mathbb{R}^d \times \mathbb{R}^m \times \mathbb{R}^k$, and are also assumed to be jointly measurable in their arguments.

As filtration we take the family of σ -algebras $(\mathcal{F}_t)_{t \in T}$ given by

$$\mathcal{F}_t = \sigma(V_0, \varepsilon_0, \dots, \varepsilon_{t-1}),$$

and note that the processes $(V_t)_{t \in T}$ and $(\alpha_t)_{t \in T}$ are both $(\mathcal{F}_t)_{t \in T}$ -adapted. We further assume that the m -dimensional random variables α_t are supposed to be of the form

$$\alpha_t = \alpha(t, V_0, \dots, V_t), \quad (3.12)$$

for certain measurable functions $\alpha(t, \cdot) \in \mathcal{B}(\mathbb{R}^d)^{t+1}, \mathbb{R}^m$, for which we use the notation *Markov control* and in this case we see that the process $(\alpha_t)_{t \in T}$ is a *Markov control* and the recursion (3.11) is Markovian.

LEMMA 3.17.

The process X is Markov with respect to the filtration specified above, if α_t depends on X_t only, i.e. $\alpha_t = \alpha(t, X_t)$ for all $t \in \mathbf{T}$ for some measurable functions $\alpha(t, \cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^m$.

Next, we define a general stochastic optimal control problem that includes the investment-consumption control problems as given by Problems 1–3 as special cases.

Problem 4. Stochastic Optimal Control Problem

Let $g_0, \dots, g_{T-1} : \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}$, as well as $g_T : \mathbb{R}^d \rightarrow \mathbb{R}$ be measurable functions. We refer to $g_0, \dots, g_T$ as the *reward functions* (think of them as utility functions). The *optimal control problem* is to maximise over all controls $\alpha = (\alpha_0, \dots, \alpha_{T-1})$ the additive over time criterion given by the expectation

$$J(\alpha) := \mathbb{E}_{\mathbb{P}} \left[\sum_{t=0}^{T-1} g_t(V_t, \alpha_t) + g_T(V_T) \right], \quad (3.13)$$

with each α_t for $t = 0, 1, \dots, T-1$ as in (3.12).

Remark. Under assumption (3.12), Problem 4 is equivalent to finding measurable functions $\alpha(t, \cdot)$ maximising $J(\alpha(\cdot, \cdot))$ in (3.13), where for $\alpha = (\alpha_0, \dots, \alpha_{T-1})$ we also write $J(\alpha)$! Usually, the functions $\alpha_0, \dots, \alpha_{T-1}$ have to satisfy certain admissibility constraints. These will be clear in the appropriate context and not always explicitly mentioned. For instance, it is tacitly assumed that all random quantities involved are such that the expectations exist.

DEFINITION 3.18.

A sequence of admissible functions $\alpha^* = (\alpha_0^*, \dots, \alpha_{T-1}^*)$ is called an *optimal control* for (3.13) if $J(\alpha^*) = \sup_{\alpha} J(\alpha)$ holds, where the supremum is taken over all admissible control sequences $\alpha = (\alpha_0, \dots, \alpha_{T-1})$ with $\alpha(t, \cdot) \in \mathcal{B}((\mathbb{R}^d)^{t+1}, \mathbb{R}^m)$ for $t = 0, \dots, T-1$.

DEFINITION 3.19.

Let $(\alpha_t)_{t \in \mathbf{T}}$ be a sequence of functions and let the process $(V_t^\alpha)_{t \in \mathbf{T}}$ be defined by (3.11) and (3.12), where the notation expresses the dependence of $(V_t)_{t \in \mathbf{T}}$ on $(\alpha_t)_{t \in \mathbf{T}}$. Note that $V_0^\alpha = V_0$. Set $J_T(\alpha) = g_T(V_T^\alpha)$, and for $t = 0, 1, \dots, T-1$:

$$J_t(\alpha) := \mathbb{E}_{\mathbb{P}} \left[\sum_{s=t}^{T-1} g_s(V_s^\alpha, \alpha_s) + g_T(V_T^\alpha) \mid \mathcal{F}_t \right].$$

The so defined adapted process $(J_t(\alpha))_{t \in \mathbf{T}}$ is called the *expected future reward process* of the optimal control problem (3.13).

Note, that $\mathbb{E}_{\mathbb{P}}[J_0(\alpha)] = J(\alpha)$ and for $t = 0, 1, \dots, T-2$ the expected future reward process satisfies:

$$\begin{aligned} J_t(\alpha) &= g_t(V_t^\alpha, \alpha_t) + \mathbb{E} \left[\sum_{s=t+1}^{T-1} g_s(V_s^\alpha, \alpha_s) + g_T(V_T^\alpha) \mid \mathcal{F}_t \right] \\ &= g_t(V_t^\alpha, \alpha_t) + \mathbb{E} \left[\mathbb{E} \left[\sum_{s=t+1}^{T-1} g_s(V_s^\alpha, \alpha_s) + g_T(V_T^\alpha) \mid \mathcal{F}_{t+1} \right] \mid \mathcal{F}_t \right] \\ &= g_t(V_t^\alpha, \alpha_t) + \mathbb{E}[J_{t+1}(\alpha) \mid \mathcal{F}_t]. \end{aligned}$$

Dynamic Programming

The idea to obtain the control maximising (3.13) is based on the *dynamic programming principle*, which states that if a process is optimal over the entire sequence of periods $t = 0, 1, \dots, T$, then it has to be also optimal over each single period. In the previous section's context, this means that in order to determine the optimal control sequence $\alpha_0, \dots, \alpha_T$ we aim to sequentially optimise over the individual controls α_t for $t = 0, 1, \dots, T$.

In *Theorem 3.20* below, we show that under the assumptions of Markov controls, i.e., if $\alpha_t = \alpha(t, V_t)$ for some measurable functions $u(t, \cdot)$ for all $t \in \mathbf{T}$, this idea yields an optimal control. Let us denote the class of Markov control sequences by $\mathcal{M}$ and define for certain given measurable functions $v_0, \dots, v_T : \mathbb{R}^d \rightarrow \mathbb{R}$ the following sequence of functions:

$$\hat{v}_{t+1}(x, y) = \mathbb{E}_{\mathbb{P}}[v_{t+1}(G_t(x, y, \varepsilon_t))], \quad t = 0, \dots, T-1. \quad (3.14)$$

Note that the functions $\hat{v}_{t+1} : \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}$ are measurable in x and y .

THEOREM 3.20.

Define recursively the functions $(v_t)_{t \in \mathbf{T}}$ by

$$v_T(x) = g_T(x), \quad (3.15)$$

$$v_t(x) = \sup_{y \in \mathbb{R}^m} \{g_t(x, y) + \hat{v}_{t+1}(x, y)\}, \quad t = 0, \dots, T-1. \quad (3.16)$$

Assume that the v_t are measurable functions. Then the following hold true.

- (i) For any control sequence $\alpha \in \mathcal{M}$ of functions $(\alpha(t, \cdot))_{t \in \mathbf{T}}$, one has $v_t(V_t^\alpha) \geq J_t(\alpha)$ $\mathbb{P}$ -a.s. and $\mathbb{E}_{\mathbb{P}}[v_0(V_0)] \geq J(\alpha)$.
- (ii) Let $\alpha^* \in \mathcal{M}$. Then α^* is optimal if and only if the supremum in (3.16) is attained for $y = \alpha_t^*(x)$. If this happens, then $v_t(V_t^{\alpha^*}) = J_t(\alpha^*)$ and

$$\sup_{\alpha} J(\alpha) = J(\alpha^*) = \mathbb{E}_{\mathbb{P}}[v_0(X_0)].$$

Remark. *Theorem 3.20 depends on*

$$\hat{v}_{t+1}(x, y) = \mathbb{E}[v_{t+1}(G_t(x, y, \varepsilon_t)) \mid \mathcal{F}_t],$$

which is due to the independence of V_0 and the ε_t . This assumption might be dropped.

PROPOSITION 3.21 (Optimality Principle).

Let $\alpha^* = (\alpha_0^*, \dots, \alpha_{T-1}^*)$ be the optimal control of (3.13). Then the sequence $(\alpha_t^*, \dots, \alpha_{T-1}^*)$ is optimal for the corresponding optimisation problem over the time set $\{t, \dots, T\}$, when starting in $V_t = V_t^{\alpha^*}$. In this case, the optimal value is equal to $\mathbb{E}_{\mathbb{P}}[v_t(V_t^{\alpha^*})]$.

Proof. Consider the stochastic optimal control problem defined by (95):

$$J(\alpha) := \mathbb{E}_{\mathbb{P}} \left[\sum_{t=0}^{T-1} g_t(V_t, \alpha_t) + g_T(V_T) \right],$$

where the state process (V_t) evolves according to some given stochastic dynamics:

$$V_{t+1} = G_t(V_t, \alpha_t, \xi_t),$$

for some noise ξ_t . We assume $\alpha^* = (\alpha_0^*, \dots, \alpha_{T-1}^*)$ is the optimal control that maximises $J(\alpha)$ over all admissible controls α . Define the value functions $(v_t)_{t=0}^T$ by backward recursion:

$$v_T(x) = g_T(x),$$

and for $t = T - 1, \dots, 0$:

$$v_t(x) = \operatorname{ess\,sup}_{\alpha_t} \mathbb{E}_{\mathbb{P}} [g_t(x, \alpha_t) + v_{t+1}(G_t(x, \alpha_t, \xi_t)) \mid V_t = x].$$

The optimal control α^* attains the supremum at each time step:

$$v_t(V_t^{\alpha^*}) = g_t(V_t^{\alpha^*}, \alpha_t^*) + \mathbb{E}_{\mathbb{P}} [v_{t+1}(V_{t+1}^{\alpha^*}) \mid \mathcal{F}_t],$$

where $V_t^{\alpha^*}$ is the state process under the optimal control α^* . By definition of α^* being optimal over the entire horizon $\{0, \dots, T\}$ we have:

$$v_0(V_0) = J(\alpha^*).$$

Now consider the problem starting from time t with initial state $V_t^{\alpha^*}$. The subproblem is:

$$\max_{\beta=(\beta_t, \dots, \beta_{T-1})} \mathbb{E}_{\mathbb{P}} \left[\sum_{s=t}^{T-1} g_s(V_s^{\beta}, \beta_s) + g_T(V_T^{\beta}) \mid V_t = V_t^{\alpha^*} \right].$$

Since α^* is optimal over the entire horizon, it must also be optimal over any subhorizon. Hence, the restriction $(\alpha_t^*, \dots, \alpha_{T-1}^*)$ is optimal for the subproblem starting at time t with initial state $V_t^{\alpha^*}$. In other words, no other control sequence $(\beta_t, \dots, \beta_{T-1})$ starting at $V_t^{\alpha^*}$ can yield a strictly higher expected return than $(\alpha_t^*, \dots, \alpha_{T-1}^*)$. The optimal value function for the subproblem starting at time t is $v_t(V_t^{\alpha^*})$. Since α^* remains optimal for the subproblem, it must achieve this value:

$$\max_{\beta=(\beta_t, \dots, \beta_{T-1})} \mathbb{E}_{\mathbb{P}} \left[\sum_{s=t}^{T-1} g_s(V_s^{\beta}, \beta_s) + g_T(V_T^{\beta}) \mid V_t = V_t^{\alpha^*} \right] = v_t(V_t^{\alpha^*}).$$

Thus, we have shown that the tail of the optimal strategy α^* is indeed optimal for the subproblem starting at t , and the optimal value of that subproblem is $\mathbb{E}_{\mathbb{P}} [v_t(V_t^{\alpha^*})]$ $\square$

Algorithm 1 Dynamic Programming

Suppose that the suprema in (3.16) are attained for all $t \in \mathbf{T}$. Then define

$$\begin{aligned} v_T(x) &= g_T(x) \\ \alpha_{T-1}^*(x) &= \arg \sup \{g_{T-1}(x, y) + \hat{v}_T(x, y)\}, \end{aligned}$$

and by backward recursion for $t \in \{0, \dots, T-1\}$,

$$\begin{aligned} v_t(x) &= \sup \{g_t(x, y) + \hat{v}_{t+1}(x, y)\} = g_t(x, \alpha_t^*(x)) + \hat{v}_{t+1}(x, \alpha_t^*(x)) \\ \alpha_{t-1}^*(x) &= \arg \sup \{g_{t-1}(x, y) + \hat{v}_t(x, y)\}. \end{aligned}$$

This yields the sequence of functions $v_T, \alpha_{T-1}^*, v_{T-1}, \alpha_{T-2}^*, \dots, v_1, \alpha_0^*, v_0$ where the α_t^* constitute the optimal sequence α^* and $\mathbb{E}_{\mathbb{P}}[v_0(V_0)] = J(\alpha^*)$. The functions v_t are called the *optimal value functions*.

From here on, we drop the assumption that the $\varepsilon_1, \dots, \varepsilon_{T-1}$ are independent. One can still define random functions $\hat{v}_{t+1}$ as above, but now we alter the definition in (3.14) to

$$\hat{v}_{t+1}(x, y) := \mathbb{E}[v_{t+1}(G_t(x, y, \varepsilon_t)) \mid \mathcal{F}_t].$$

Then the $\hat{v}_{t+1}(x, y)$ are in general not deterministic anymore, but become $\mathcal{F}_t$ -measurable random variables.

The following proposition uses the concept of essential supremum. We will consider $\mathcal{B}(\mathbb{R}^d) \times \mathcal{F}_t$ -measurable functions v_t , meaning that the mapping $(x, \omega) \mapsto v_t(x, \omega)$ is $\mathcal{B}(\mathbb{R}^d) \times \mathcal{F}_t$ -measurable. As usual, dependence on ω is often suppressed and then we write $v_t(x)$ for the random variable $\omega \mapsto v_t(x, \omega)$.

PROPOSITION 3.22.

Suppose that $\mathcal{B}(\mathbb{R}^d) \times \mathcal{F}_t$ -measurable functions v_t (for $t = 0, \dots, T$) $\mathbb{P}$ -a.s. satisfy

$$\begin{aligned} v_T(x) &= g_T(x), \\ v_t(x) &= \text{ess sup} \left\{ g_t(x, y) + \mathbb{E}[v_{t+1}(G_t(x, y, \varepsilon_t)) \mid \mathcal{F}_t] : y \in \mathcal{F}_t \right\}, \quad t \leq T-1. \end{aligned}$$

Then, for any admissible control α it holds that $v_t(V_t^\alpha) \geq J_t(\alpha)$ for $t \in \mathbf{T}$ and the optimum is given by $\mathcal{F}_t$ measurable random variables $\alpha_t^*(x)$ for which the supremum is attained.

Maximising Expected Utility from Terminal Wealth using Dynamic Programming

We align Problem 1 with the general stochastic optimal control problem as in (3.13). The aim is to maximise expected utility from terminal wealth using the utility function u :

$$\begin{cases} \max_{(W_0, \phi)} \mathbb{E}_{\mathbb{P}}[u(W_0 + G_T(\phi))], \\ \text{s.t. } W_0 = w \text{ and } \phi \text{ is predictable.} \end{cases} \quad (\text{Problem 1})$$

In the context of (3.13), we set $g_t = 0$ for $t = 0, 1, \dots, T-1$ and $g_T = u$, with V_t as the wealth at time t for a self-financing strategy $\varphi = (w, \phi)$.

We consider $\mathbb{R}^d$ -valued predictable processes $(\phi_t)_{t \in \mathbf{T}}$ satisfying $\phi_0 = 0$ and $W_T((w, \phi)) \geq 0$. For relative portfolio holdings, define control variables $\alpha_t = (\alpha_t^{(0)}, \alpha_t^{(1)}, \dots, \alpha_t^{(d)})$ as

$$\alpha_t^{(i)} = \frac{\phi_{t+1}^{(i)} S_t^{(i)}}{V_t}, \quad i = 0, 1, \dots, d,$$

and note that $\alpha_t^{(0)} = 1 - \sum_{i=1}^d \alpha_t^{(i)}$ with $\alpha_t^{(i)} \in (0, 1)$. The self-financing property leads to

$$V_t = \varphi_t^{(0)} S_t^{(0)} + \sum_{i=1}^d \varphi_t^{(i)} S_t^{(i)} = \varphi_{t+1}^{(0)} S_t^{(0)} + \sum_{i=1}^d \varphi_{t+1}^{(i)} S_t^{(i)}.$$

Given the predictable nature of ϕ and the evolution of $S_{t+1}^{(i)} = S_t^{(i)}(1 + R_t^{(i)})$, we derive

$$\begin{aligned} V_{t+1} &= V_t + \varphi_{t+1}^{(0)} \Delta S_t^{(0)} + \sum_{i=1}^d \varphi_{t+1}^{(i)} \Delta S_t^{(i)} \\ &= V_t + \varphi_{t+1}^{(0)} S_t^{(0)} r_t + \sum_{i=1}^d \varphi_{t+1}^{(i)} S_t^{(i)} R_t^{(i)} \\ &= V_t + V_t \left(\alpha_t^{(0)} r_t + \sum_{i=1}^d \alpha_t^{(i)} R_t^{(i)} \right). \end{aligned} \quad (3.17)$$

Comparing (3.11) with (3.17), we find that $V_{t+1} = G_t(V_t, \alpha_t, \xi_t)$, with $\xi_t = R_t$, V_t and α_t as defined above and such that

$$G_t(V_t, \alpha_t, \xi_t) = V_t \left(\alpha_t^{(0)} (1 + r_t) + \sum_{i=1}^d \alpha_t^{(i)} (1 + \xi_t^{(i)}) \right). \quad (3.18)$$

For maximising expected utility from terminal wealth only, we have

$$\begin{aligned} g_t(V_t, \alpha_t) &= 0, \quad t < T, \\ g_T(V_T, \alpha_T) &= u(V_T), \end{aligned}$$

leading to the dynamic programming algorithm:

$$\begin{cases} v_T(v) = g_T(v) = u(v), \\ v_t(v) = \max_{\alpha_t} \mathbb{E}_{\mathbb{P}}[v_{t+1}(G_t(V_t, \alpha_t, R_t)) \mid V_t = v], \quad t = 0, 1, \dots, T-1, \end{cases} \quad (3.19)$$

with no maximisation over α_T . The optimal control sequence u^* , if existent, is derived accordingly.

EXAMPLE 3.23 (Maximising Expected Utility out of Terminal Wealth in the Binomial Model). Recall the Dynamic Programming algorithm (3.19), namely

$$\begin{cases} v_T(v) = u(v) \text{ and, for } t < T, \\ v_t(v) = \max_{\alpha_t} \mathbb{E}_{\mathbb{P}}[v_{t+1}(G_t(V_t, \alpha_t, \xi_t)) \mid V_t = v] \end{cases}$$

where the dynamics $G_t(V_t, \alpha_t, \xi_t)$ were specified as above in (3.18) namely

$$G_t(V_t, \alpha_t, \xi_t) = V_t \left(\alpha_t^{(0)}(1 + r_t) + \sum_{i=1}^d \alpha_t^{(i)}(1 + \xi_t^{(i)}) \right).$$

The only particular aspect for the binomial model here is that $d = 1$ and, letting

$$\alpha_t = \alpha_t^{(1)} = \frac{\phi_{t+1}^{(1)} S_t}{V_t},$$

one has $\alpha_t^{(0)} = (1 - \alpha_t)$. With prohibition of short selling, i.e., requiring $\phi_t^{(1)} > 0$, we obtain $\alpha_t \in (0, 1)$. If $r_t = 0$, we may then write

$$G_t(V_t, \alpha_t, \xi_t) = V_t(1 + \alpha_t R_t).$$

An investment strategy is here given by the two-dimensional vector $\alpha_t = (\alpha_t^{(0)}, \alpha_t^{(1)})$ where $\alpha_t^{(0)}$ and $\alpha_t^{(1)}$ denote the number of units of the non-risky and the risky assets respectively held in the portfolio in period t . Recall also that the process α_t is supposed to be predictable, i.e., α_t is taken to be $\mathcal{F}_{t-1}$ -measurable. We now describe the backwards recursion, from period $T-1$ to 0, to determine the optimal investment strategy $(\alpha_t^*)_{t=0, \dots, T-1}$.

Take period $T-1$: we have to decide for $\alpha_{T-1}^{(0)}, \alpha_{T-1}^{(1)}$. Let $S_{T-1} = S_0 U^n D^{T-n-1}$. Recalling that we had assumed $r = 0$, we have to impose that for all outcomes, i.e., independently on whether prices go up or down,

$$\alpha_T^{(1)} S_T + \alpha_T^{(0)} = V_T = V^*. \quad (3.20)$$

This implies that the following system of equations in $\alpha_T^{(1)}, \alpha_T^{(0)}$ has to be satisfied:

$$\alpha_T^{(1)} S_{T-1} U + \alpha_T^{(0)} = v \left(\frac{p}{q} \right)^{n+1} \left(\frac{1-p}{1-q} \right)^{T-n-1}, \quad (\text{prices go up})$$

$$\alpha_T^{(1)} S_{T-1} D + \alpha_T^{(0)} = v \left(\frac{p}{q} \right)^n \left(\frac{1-p}{1-q} \right)^{T-n}, \quad (\text{prices go down})$$

where

$$p = \mathbb{P}(R_t = U), \quad q = \frac{1+r-D}{U-D},$$

from which we obtain

$$\alpha_T^{(1)} S_{T-1}(U - D) = v(1+r)^T \left(\left(\frac{p}{q} \right)^n \left(\frac{1-p}{1-q} \right)^{T-n-1} - \left(\frac{p}{q} \right)^{n+1} \left(\frac{1-p}{1-q} \right)^{T-n} \right).$$

Consequently, we have

$$\alpha_T^{(1)} = \frac{v(1+r)^T}{S_{T-1}(U - D)} \left(\frac{pq}{\frac{p}{q}(1-q)} \right)^n \left(\frac{1-p}{1-q} \right)^{T-n-1}.$$

And therefore

$$\begin{aligned} \alpha_{T-1}^{(1)} &= \frac{v(pq)^{\frac{n}{T}} (1-p)^{\frac{T-n-1}{T}}}{(p-q)S_0 U^n (U-D)(1-q)^{T-n-1}} \\ \alpha_{T-1}^{(0)} &= v \left(\frac{q}{p} \right)^n \left(\frac{1-p}{1-q} \right)^{T-n-1} \left(\frac{U(1-p)q - D(1-q)p}{(U-D)q(1-q)} \right). \end{aligned}$$

Recalling the self-financing condition, the fact that $q = \frac{1+r-D}{U-D}$, and putting

$$C(p, q) := \frac{U-D}{p-q} + (Dq - Dp) + (Dpq - Upq),$$

for the optimal value induced in period $T-1$ we thus obtain

$$V_{T-1}^* = \alpha_{T-1}^{(1)} S_{T-1} + \alpha_{T-1}^{(0)} = v \left(\frac{pq}{(1-q)} \right)^n \left(\frac{1-p}{1-q} \right)^{T-n-1} C(p, q).$$

Notice now that V_{T-1}^* has the same structure as V_T^* so that the calculations for the following period $T-2$ proceed in exactly the same way as for $T-1$ and so forth, which allows to straightforwardly complete the backwards recursion.

In the generic period $t \leq T$, with $n \leq t$ up-movement up to time t , the condition (3.20) becomes

$$\alpha_t^{(1)} S_t + \alpha_t^{(0)} = V_t^*$$

and one obtains

$$\begin{aligned} \alpha_t^{(1)} &= \frac{v(pq)^{\frac{n}{t}} (1-p)^{\frac{t-n-1}{t}}}{(p-q)S_0 U^n (U-D)(1-q)^{t-n-1}}, \\ \alpha_t^{(0)} &= v \left(\frac{q}{p} \right)^n \left(\frac{1-p}{1-q} \right)^{t-n-1} \left(\frac{U(1-p)q - D(1-q)p}{(U-D)q(1-q)} \right), \end{aligned}$$

with the optimal wealth

$$V_t^* = v \left(\frac{p}{q} \right)^n \left(\frac{1-p}{1-q} \right)^{t-n}.$$

Notice also that the ratio of the wealth invested in the risky asset is

$$\pi_t = \frac{\alpha_{t+1}^{(1)}}{\alpha_t^{(1)} S_t} = \frac{1}{(U-D)q(1-q)(p-q)},$$

and it is independent of t and S_t . It follows that the optimal strategy requires investing in the risky asset, in each period and in every state, the same fraction of wealth.

Maximising Expected Utility from Consumption using Dynamic Programming

We align *Problem 2* with the general stochastic optimal control problem as follows:

$$\begin{cases} \max_{(K, \varphi)} \mathbb{E}_{\mathbb{P}} \left[\sum_{t=1}^T \beta_t u(K_t) \right], \\ \text{s.t. } W_0 = w \text{ and } \alpha_t \in (0, 1], \\ \text{and } (K, \varphi) \text{ is an admissible consumption-investment.} \end{cases} \quad (\text{Problem 2})$$

Incorporating a consumption process into the optimal control setting, for a self-financing consumption-investment plan (K, α) , we have:

$$\begin{aligned} V_{t+1} &= V_t + \phi_{t+1}^{(0)} \Delta S_t^{(0)} + \sum_{i=1}^d \phi_{t+1}^{(i)} \Delta S_t^{(i)} - K_t, \\ &= V_t \left(\alpha_t^{(0)} r_t + \sum_{i=1}^d \alpha_t^{(i)} R_t^{(i)} \right) - K_t, \\ &= V_t \left((1 + r_t) + \sum_{i=1}^d \alpha_t^{(i)} (R_t^{(i)} - (1 + r_t)) \right) - K_t(1 + r_t), \end{aligned}$$

where $\alpha_t^{(0)}, \dots, \alpha_t^{(d)}$ denote the relative portfolio holdings $\alpha_t^{(i)} = \frac{\phi_{t+1}^{(i)} S_t^{(i)}}{V_t}$ as before. Therefore, we identify:

$$V_{t+1} = G_t(V_t, \alpha_t, \xi_t),$$

with

$$G_t(V_t, \alpha_t, \xi_t) = V_t \left((1 + r_t) + \sum_{i=1}^d \alpha_t^{(i)} (\xi_{t+1}^{(i)} - (1 + r_t)) \right) - K_t(1 + r_t).$$

Here, the control sequence is $\alpha_t = (\alpha_t^{(1)}, \dots, \alpha_t^{(d)}, K_t)$, controlling both relative portfolio holdings and consumption at each time $t = 0, 1, \dots, T-1$. The utility is solely on consumption:

$$g_t(V_t, \alpha_t) = u(K_t), \quad t = 0, \dots, T,$$

with the wealth V_t entering only through the constraint $K_T \leq V_T$. The dynamic programming algorithm is thus defined as:

$$\begin{cases} v_T(v) = \begin{cases} u(v) & K_T = v \\ -\infty & K_T > v \end{cases}, \\ v_t(v) = \max_{\alpha_t} (u(K_t) + \beta_t \mathbb{E}_{\mathbb{P}}[v_{t+1}(G_t(V_t, \alpha_t, \xi_t)) \mid V_t = v]), \end{cases}$$

where $u(K) = -\infty$ for $K < 0$ ensures non-negativity of K , and we set:

$$J_t \triangleq \max_{K_t, \dots, K_T (K_T \leq V_T)} \mathbb{E}_{\mathbb{P}} \left[\sum_{s=t}^T \beta_{s-t} u(K_s) \mid V_t = v \right],$$

with the condition $K_T \leq V_T$ enforced by setting $u(K_T) = -\infty$ for $K_T > V_T$.

Maximising Expected Utility from Consumption & Terminal Wealth using Dynamic Programming

In the previous case of expected utility only from consumption, all of V_T was consumed at the terminal time $t = T$. Here we generalize this problem by leaving $V_T - K_T$ for future

investment and maximising expected utility from consumption of K_t for $t = 0, \dots, T$, as well as from the terminal wealth $V_T - K_T$.

Letting $u_c(\cdot)$ and $u_p(\cdot)$ denote the utility functions for consumption and terminal wealth respectively, then the problem is given by:

$$\begin{cases} \max_{(K, \varphi)} \mathbb{E}_{\mathbb{P}} \left[\sum_{t=1}^T \beta_t u_c(K_t) + \beta_T u_p(W_T(\varphi) - K_T) \right], \\ \text{s.t. } \varphi \text{ is a self-financing strategy,} \\ \text{and } K \text{ is a non-negative admissible consumption plan.} \end{cases} \quad (\text{Problem 3})$$

For this problem, we can now adapt the previous solution method: we identify

$$V_{t+1} = G_t(V_t, \alpha_t, \xi_t)$$

with $\xi_t = R_t$ and

$$G_t(V_t, \alpha_t, \xi_t) = V_t \left((1 + r_t) + \sum_{i=1}^d \alpha_t^{(i)} (R_t^{(i)} - (1 + r_t)) \right) - K_t(1 + r_t),$$

and with the control sequence α given by

$$\alpha_t = (\alpha_t^{(1)}, \dots, \alpha_t^{(d)}, K_t),$$

where

$$\alpha_t^{(i)} = \frac{\phi_{t+1}^{(i)} S_t^{(i)}}{V_t}$$

as before. In the present case, however, we have

$$g_t(V_t, \alpha_t) = u_c(K_t), \quad t = 0, \dots, T-1,$$

and terminal utility

$$g_T(V_T, \alpha_T) = u_p(V_T - K_T).$$

The dynamic programming algorithm becomes

$$\begin{aligned} v_T(x) &= \max_{0 \leq K \leq x} [u_c(K) + u_p(x - K)], \\ v_t(x) &= \max_{\alpha_t} (u_c(K_t) + \beta_t \mathbb{E}_{\mathbb{P}}[v_{t+1}(G_t(V_t, \alpha_t, \xi_t)) \mid V_t = v]), \end{aligned}$$

and notice that the only difference with the pure consumption–investment problem in Section 3.3.2 is the form of $v_T(x)$ (in the previous case it is essentially given by $v_T(x) = u(x) = u_c(x)$).

The Martingale Method

Appendix

SECTION A.1

| Financial Markets in Finite Discrete-Time

We assume that $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ is a frictionless financial market in finite discrete-time $\mathbf{T} = \{0, 1, \dots, T\}$ and $(d + 1)$ -tradable assets with $d \in \mathbb{N}$. If φ_t and X_t are two $(d + 1)$ -dimensional vectors we denote the scalar product as $\varphi_t^\top X_t$, but you might also write $\varphi_t \cdot X_t$ or $\langle \varphi_t, X_t \rangle_{\mathbb{R}^d}$ if you prefer. For any stochastic process $(Y_t)_{t \in \mathbf{T}}$ we write $\Delta Y_t = Y_t - Y_{t-1}$ for all $t \geq 1$ and set $\Delta Y_0 = Y_0$. For $A, B \in \mathcal{F}$ we use the notation $\mathbb{P}(B \mid A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)}$ to denote the conditional probability.

EXERCISE A.1 (Predictable or Adapted). Consider a financial market with only two assets which are modelled as usual by the stochastic process $X = (S^{(0)}, S^{(1)})$ and that is adapted to the filtration $(\mathcal{F}_t)_{t \in \mathbf{T}}$. Decide which of the following processes φ are predictable and which in general are not:

- (i) $\varphi_t = \mathbb{1}_{\{S_t^{(1)} > S_{t-1}^{(1)}\}}$;
- (ii) $\varphi_1 = 1$ and $\varphi_t = \mathbb{1}_{\{S_{t-1}^{(1)} > S_{t-2}^{(1)}\}}$ for $t \geq 2$;
- (iii) $\varphi_t = \mathbb{1}_A \cdot \mathbb{1}_{\{t > t_0\}}$, where $t_0 \in \{0, \dots, T\}$ and $A \in \mathcal{F}_{t_0}$;
- (iv) $\varphi_t = \mathbb{1}_{\{S_t^{(1)} > S_0^{(1)}\}}$;
- (v) $\varphi_1 = 1$ and $\varphi_t = 2\varphi_{t-1} \mathbb{1}_{\{S_{t-1}^{(1)} < S_0^{(1)}\}}$ for $t \geq 2$.

SOLUTION. Given that the stochastic process $X_t = (S_t^{(0)}, S_t^{(1)})_{t \in \mathbf{T}}$ is adapted to the filtration $(\mathcal{F}_t)_{t \in \mathbf{T}}$, $S_t^{(0)}$ and $S_t^{(1)}$ are $\mathcal{F}_t$ -measurable but generally not $\mathcal{F}_{t-1}$ -measurable.

- (i) φ_t is generally not predictable, where $S_{t-1}^{(1)}$ is $\mathcal{F}_{t-1}$ -measurable, but $S_t^{(1)}$ is generally not $\mathcal{F}_{t-1}$ -measurable.
- (ii) $\varphi_1 = 1$ is a constant and hence predictable; φ_t is predictable where $S_{t-1}^{(1)}$ and $S_{t-2}^{(1)}$ are both $\mathcal{F}_{t-1}$ -measurable.
- (iii) When $t \leq t_0$: $\mathbb{1}_{\{t > t_0\}}$, therefore $\varphi_t = 0$ is a constant and hence predictable. When $t > t_0$: since $t \in \mathbf{T} = \{0, 1, \dots, T\} \ni t_0, t-1 \geq t_0$, so $A \in \mathcal{F}_{t_0} \subseteq \mathcal{F}_{t-1}$. Then φ_t is the product of $\mathcal{F}_{t-1}$ -measurable functions and hence predictable.
- (iv) φ_t is generally not predictable, where $S_0^{(1)}$ is $\mathcal{F}_0$ -measurable and thus $\mathcal{F}_{t-1}$ -measurable, but $S_t^{(1)}$ is generally not $\mathcal{F}_{t-1}$ -measurable.
- (v) For $t \geq 2$, $S_{t-1}^{(1)}$ and $S_0^{(1)}$ are $\mathcal{F}_{t-1}$ -measurable by previous arguments; φ_{t-1} is $\mathcal{F}_{t-2}$ -measurable, and we can calculate the value of φ_{t-1} at time $t-1$, since we know the initial value $\varphi_1 = 1$.

and the recurring procedure, so φ_{t-1} is $\mathcal{F}_{t-1}$ -measurable. In conclusion, φ_t is the product of $\mathcal{F}_{t-1}$ -measurable functions and hence predictable.

□

EXERCISE A.2 (Equivalent Definition of Self-Financing Strategies). Let $\varphi = (\varphi_t)_{t \in T}$ be a trading strategy such that $\varphi_t^\top X_t = \varphi_{t+1}^\top X_t$ for all $t \in \{0, \dots, T-1\}$. Show that this is equivalent to

$$\Delta W_t(\varphi) = \varphi_t^\top \Delta X_t, \quad \forall t \in \{1, 2, \dots, T\},$$

where $(W_t(\varphi))_{t \in T}$ denotes the wealth process of the strategy φ . Whenever a trading strategy satisfies one of the equivalent conditions, we call the strategy φ *self-financing*.

PROOF. " $\Rightarrow$ " Necessity: we assume that $\varphi_t^\top X_t = \varphi_{t+1}^\top X_t$ for all $t \in \{0, \dots, T-1\}$, then we have

$$\begin{aligned} \Delta W_t(\varphi) &= W_t(\varphi) - W_{t-1}(\varphi) \\ &= \sum_{i=0}^d \varphi_t^{(i)} S_t^{(i)} - \sum_{i=0}^d \varphi_{t-1}^{(i)} S_{t-1}^{(i)} \\ &= \varphi_t^\top \sum_{i=0}^d (S_t^{(i)} - S_{t-1}^{(i)}) \\ &= \varphi_t^\top \begin{pmatrix} S_t^{(0)} - S_{t-1}^{(0)} \\ S_t^{(1)} - S_{t-1}^{(1)} \\ \vdots \\ S_t^{(d)} - S_{t-1}^{(d)} \end{pmatrix} \\ &= \varphi_t^\top \begin{pmatrix} \Delta S_t^{(0)} \\ \Delta S_t^{(1)} \\ \vdots \\ \Delta S_t^{(d)} \end{pmatrix} = \varphi_t^\top \Delta X_t, \end{aligned}$$

where $X_t := (S_t^{(0)}, S_t^{(1)}, S_t^{(2)}, \dots, S_t^{(d)})^\top$.

" $\Leftarrow$ " Sufficiency: suppose that $\Delta W_t(\varphi) = \varphi_t^\top \Delta X_t$ holds for all $t \in \{1, 2, \dots, T\}$. And similarly, we can write

$$\Delta W_{t+1}(\varphi) = W_{t+1}(\varphi) - W_t(\varphi),$$

where $\Delta W_{t+1}(\varphi) = \varphi_{t+1}^\top \Delta X_{t+1} = \varphi_{t+1}^\top (X_{t+1} - X_t)$, thus

$$\varphi_{t+1}^\top \Delta X_{t+1} = \sum_{i=0}^d \varphi_{t+1}^{(i)} S_{t+1}^{(i)} - \sum_{i=0}^d \varphi_t^{(i)} S_t^{(i)} = \varphi_{t+1}^\top \sum_{i=0}^d S_{t+1}^{(i)} - \varphi_t^\top \sum_{i=0}^d S_t^{(i)} = \varphi_{t+1}^\top X_{t+1} - \varphi_t^\top X_t.$$

Subtract $\varphi_{t+1}^\top X_{t+1}$ from both sides, we get $\varphi_{t+1}^\top X_t = \varphi_t^\top X_t$.

In conclusion, we have proved that

$$\varphi_t^\top X_t = \varphi_{t+1}^\top X_t, \quad \forall t \in \{0, \dots, T-1\}$$

is equivalent to

$$\Delta W_t(\varphi) = \varphi_t^\top \Delta X_t, \quad \forall t \in \{1, 2, \dots, T\}.$$

□

EXERCISE A.3 (Martingale Measures and Densities). Let $\mathbb{Q}$ be a probability measure on $(\Omega, \mathcal{F})$ which is absolutely continuous with respect to $\mathbb{P}$. Show that the density process

$$Z_t := \frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_t}, \quad t = 0, 1, \dots, T,$$

is a martingale with respect to $\mathbb{P}$, where $\frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_t}$ denotes the Radon–Nikodym derivative of $\mathbb{Q}$ with respect to $\mathbb{P}$ restricted to the sub- σ -algebra $\mathcal{F}_t$.

SOLUTION. By definition, Z_t is $\mathcal{F}_t$ -measurable because it is a conditional expectation given $\mathcal{F}_t$. Since $\mathbb{Q} \ll \mathbb{P}$, the Radon–Nikodym derivative $\frac{d\mathbb{Q}}{d\mathbb{P}} \geq 0$ almost surely. By the *tower property*,

$$\mathbb{E}_{\mathbb{P}}[Z_t] = \mathbb{E}_{\mathbb{P}} \left[\mathbb{E}_{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \mid \mathcal{F}_t \right] \right] = \mathbb{E}_{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \right] = \mathbb{Q}(\Omega) = 1,$$

which implies Z_t is integrable. Then by the *tower property*,

$$\mathbb{E}_{\mathbb{P}}[Z_{t+1} \mid \mathcal{F}_t] = \mathbb{E}_{\mathbb{P}} \left[\mathbb{E}_{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \mid \mathcal{F}_{t+1} \right] \mid \mathcal{F}_t \right] = \mathbb{E}_{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \mid \mathcal{F}_t \right] = Z_t.$$

Therefore, the density process Z_t is a martingale with respect to $\mathbb{P}$. □

EXERCISE A.4 (A Geometric Model). Let $(S_t)_{t \in T}$ be a strictly positive price process of some risky asset, say a stock. The returns are defined by $R_t = \frac{\Delta S_t}{S_{t-1}}$ for $t = 1, 2, \dots, T$ so that

$$S_t = S_0 \prod_{k=1}^t (1 + R_k)$$

and let the filtration $\mathbb{F}$ be given by $\mathcal{F}_t = \sigma(S_0, \dots, S_t)$. Then:

- (i) Show that $(S_t)_{t \in T}$ is a $\mathbb{P}$ -martingale if $R_1, \dots, R_T$ are independent and $\mathbb{E}[R_t] = 0$.
- (ii) Give necessary and sufficient conditions on $R_1, \dots, R_T$ such that $(S_t)_{t \in T}$ is a $\mathbb{P}$ -martingale.
- (iii) Do you find an example such that $(S_t)_{t \in T}$ is a martingale, but the returns are not independent?

SOLUTION. (i) The price process evolves as:

$$S_{t+1} = S_t(1 + R_{t+1}).$$

Taking the conditional expectation:

$$\mathbb{E}[S_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[S_t(1 + R_{t+1}) \mid \mathcal{F}_t].$$

Since S_t is $\mathcal{F}_t$ -measurable, we can factor it out:

$$\mathbb{E}[S_{t+1} \mid \mathcal{F}_t] = S_t \mathbb{E}[1 + R_{t+1} \mid \mathcal{F}_t].$$

Given that $R_1, \dots, R_T$ are independent, and $S_t = S_0 \prod_{k=1}^t (1 + R_k)$, R_{t+1} is independent of $\mathcal{F}_t = \sigma(S_0, \dots, S_t) = \sigma(S_0, R_1, R_2, \dots, R_t)$. We also have $\mathbb{E}[R_{t+1}] = 0$, then:

$$\mathbb{E}[1 + R_{t+1} \mid \mathcal{F}_t] = 1 + \mathbb{E}[R_{t+1}] = 1.$$

Thus:

$$\mathbb{E}[S_{t+1} \mid \mathcal{F}_t] = S_t.$$

Since this holds for all t , $(S_t)_{t \in T}$ is a martingale under the measure $\mathbb{P}$.

- (ii) To determine the necessary and sufficient conditions for $(S_t)_{t \in T}$ to be a martingale, revisit the martingale property:

$$\mathbb{E}[R_t \mid \mathcal{F}_{t-1}] = 0, \quad \forall t.$$

The martingale property: $\mathbb{E}[S_{t+1} \mid \mathcal{F}_t] = S_t$. Given that $S_{t+1} = S_t(1 + R_{t+1})$. We compute

$$\begin{aligned} \mathbb{E}[S_t(1 + R_{t+1}) \mid \mathcal{F}_t] &= S_t, \\ \implies S_t \mathbb{E}[1 + R_{t+1} \mid \mathcal{F}_t] &= S_t, \\ \implies \mathbb{E}[1 + R_{t+1} \mid \mathcal{F}_t] &= 1, & ((S_t)_{t \in T} \text{ strictly positive}) \\ \implies \mathbb{E}[R_{t+1} \mid \mathcal{F}_t] &= 0. \end{aligned}$$

Thus, the necessary and sufficient condition for $(S_t)_{t \in T}$ to be a martingale is

$$\mathbb{E}[R_t \mid \mathcal{F}_{t-1}] = 0, \quad \forall t.$$

- (iii) Consider a return process defined by:

$$R_t = \epsilon_t \epsilon_{t-1}, \quad t \geq 1,$$

where ϵ_t are *i.i.d.* standard normal random variables, and $\epsilon_0 = 0$. In this case, each R_t depends on the previous ϵ_{t-1} , so the returns $R_1, \dots, R_t$ are not independent. However, the process S_t is still a martingale. The price process is

$$S_t = S_0 \prod_{k=1}^t (1 + R_k).$$

Since $R_t = \epsilon_t \epsilon_{t-1}$, we can write:

$$S_t = S_0 \prod_{k=1}^t (1 + \epsilon_k \epsilon_{k-1}).$$

Because

$$\begin{aligned} \mathbb{E}[R_t \mid \mathcal{F}_{t-1}] &= \mathbb{E}[\epsilon_t \epsilon_{t-1} \mid \mathcal{F}_{t-1}] \\ &= \epsilon_{t-1} \mathbb{E}[\epsilon_t \mid \mathcal{F}_{t-1}] & (\epsilon_{t-1} \text{ is } \mathcal{F}_{t-1}\text{-measurable}) \\ &= \epsilon_{t-1} \cdot 0 = 0, & (\epsilon_t \text{ is independent of } \mathcal{F}_{t-1} \text{ and } \mathbb{E}[\epsilon_t] = 0) \end{aligned}$$

despite the dependence between R_t , the process satisfies the martingale condition:

$$\mathbb{E}[S_{t+1} \mid \mathcal{F}_t] = S_t \mathbb{E}[1 + R_{t+1} \mid \mathcal{F}_t] = S_t + S_t \mathbb{E}[R_{t+1} \mid \mathcal{F}_t] = S_t,$$

where $\mathbb{E}[R_{t+1} \mid \mathcal{F}_t] = 0$.

□

EXERCISE A.5 (Equivalent Martingale Measure). Let $Z_1, \dots, Z_T$ be independent standard normal random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, and let $\mathcal{F}_t$ be the σ -field generated by $Z_1, \dots, Z_t$, where $t = 1, \dots, T$. We also let $\mathcal{F}_0 := \{\emptyset, \Omega\}$. For constants $X_0^{(1)} > 0$, $\sigma_i > 0$, and $m_i \in \mathbb{R}$, we now define the discounted price process of a risky asset as the following sequence of log-normally distributed random variables,

$$X_t^{(1)} := X_0^{(1)} \prod_{i=1}^t e^{\sigma_i Z_i + m_i}, \quad t = 0, \dots, T.$$

Construct an equivalent martingale measure for $X_t^{(1)}$ under which the random variables $X_t^{(1)}$ have still a log-normal distribution.

SOLUTION. Under the original measure $\mathbb{P}$, the logarithm of the price process can be expressed by

$$\log X_t^{(1)} = \log X_0^{(1)} + \sum_{i=1}^t (\sigma_i Z_i + m_i),$$

which implies that the price at time $t + 1$ should be

$$X_{t+1}^{(1)} = X_t^{(1)} e^{\sigma_{t+1} Z_{t+1} + m_{t+1}}.$$

This process is log-normally distributed under $\mathbb{P}$ with drift m_{t+1} and volatility σ_{t+1} .

We apply the *Girsanov's Theorem* to change the measure. Under the new measure $\mathbb{Q}$, we modify the drift term while leaving the volatility term unchanged. The new dynamics of the process under $\mathbb{Q}$ are described by:

$$Z_t^{\mathbb{Q}} = Z_t + \theta_t,$$

where θ_t is a constant shift introduced by the change of measure, ensuring that the process is a martingale under $\mathbb{Q}$.

Under the new measure $\mathbb{Q}$, the asset price dynamics become:

$$X_{t+1}^{(1)} = X_t^{(1)} e^{\sigma_{t+1}(Z_{t+1} + \theta_{t+1}) + m_{t+1}}.$$

We require that the drift term be adjusted, so that:

$$\mathbb{E}^{\mathbb{Q}} \left[e^{\sigma_{t+1} Z_{t+1}^{\mathbb{Q}} + m_{t+1}} \mid \mathcal{F}_t \right] = e^r, \implies \mathbb{E}^{\mathbb{Q}} \left[e^{\sigma_{t+1} Z_{t+1} + m_{t+1}} \right] = e^{m_{t+1} + \frac{1}{2} \sigma_{t+1}^2}, \implies m_{t+1} + \frac{1}{2} \sigma_{t+1}^2 = r,$$

which gives the adjusted drift under the martingale measure $\mathbb{Q}$:

$$m_{t+1}^{\mathbb{Q}} = r - \frac{1}{2} \sigma_{t+1}^2.$$

Under the equivalent martingale measure $\mathbb{Q}$, the price process becomes:

$$X_{t+1}^{(1)} = X_t^{(1)} e^{\sigma_{t+1} Z_{t+1}^{\mathbb{Q}} + (r - \frac{1}{2} \sigma_{t+1}^2)},$$

where $Z_{t+1}^{\mathbb{Q}}$ are standard normal variables under the measure $\mathbb{Q}$. The price process retains its log-normal distribution but with the drift adjusted to satisfy the martingale property. $\square$

EXERCISE A.6 (Insider Trading). Let $Y_1, \dots, Y_T$ be *i.i.d.* random variables on some $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{E}[Y_t] = 0$ for all t and let $\mathcal{F}_t = \sigma(Y_1, \dots, Y_t)$. Let $X_t = \sum_{k=1}^t Y_k$ for $t \leq T$. Obviously, the X_t form a martingale. Consider an insider trader, that is a trader whose information pattern is given by the σ -algebras $\tilde{\mathcal{F}}_t := \sigma(\mathcal{F}_t \cup \sigma(X_T))$, i.e., at any time $t \leq T$ she 'knows' the final value X_T .

- (i) Show that the X_t don't result in a martingale *w.r.t.* the enlarged filtration of the $\tilde{\mathcal{F}}_t$.
- (ii) Let $\hat{X}_t = X_t - \frac{\sum_{k=0}^{t-1} X_T - X_k}{T-k}$, $t \leq T$. Show that the $\left(\hat{X}_t \right)_{t \in T}$ yield a martingale *w.r.t.* enlarged filtration.
Hint: use that $\mathbb{E}[X_t \mid X_T] = \frac{t}{T} X_T$ and independence of the Y_k .
- (iii) Construct a self-financing strategy of investments ξ_t *w.r.t.* the enlarged filtration (so the ξ_t are $\tilde{\mathcal{F}}_{t-1}$ -measurable) such that $\mathbb{E} \left[\sum_{t=1}^T \xi_t (X_t - X_{t-1}) \right]$ is positive. This should follow from maximisation of the expected gain $\mathbb{E} \left[\sum_{t=1}^T \tilde{\xi}_t (X_t - X_{t-1}) \right]$ over all self-financing strategies such that $|\tilde{\xi}_t| \leq 1$.

SOLUTION. (i) A process is a martingale *w.r.t.* the enlarged filtration if it satisfies the martingale property:

$$\mathbb{E}[X_{t+1} | \tilde{\mathcal{F}}_t] = X_t, \quad \forall t \leq T-1.$$

Given that $X_t = \sum_{k=1}^t Y_k$ for $t \leq T$, we have $X_{t+1} = X_t + Y_{t+1}$, where the insider trader knows X_T at any time t . Then:

$$\mathbb{E}[X_{t+1} | \tilde{\mathcal{F}}_t] = X_t + \mathbb{E}[Y_{t+1} | \tilde{\mathcal{F}}_t].$$

Since $\tilde{\mathcal{F}}_t$ includes knowledge of X_T , Y_{t+1} is no longer independent of X_T , thus:

$$\mathbb{E}[Y_{t+1} | \tilde{\mathcal{F}}_t] \neq 0.$$

Therefore:

$$\mathbb{E}[X_{t+1} | \tilde{\mathcal{F}}_t] \neq X_t,$$

and hence X_t is not a martingale with respect to the enlarged filtration.

(ii) Given that

$$\mathbb{E}[X_t | X_T] = \frac{t}{T} X_T,$$

and the increments Y_k are independent and have zero mean, we know:

$$X_T = X_t + \sum_{k=t+1}^T Y_k \implies X_T - X_t = \sum_{k=t+1}^T Y_k.$$

This is the sum of the remaining Y_k from $t+1$ to T . Given the total sum $X_T - X_t$ of $T-t$ *i.i.d.* random variables $Y_{t+1} \dots Y_T$, and knowing that these variables are conditionally identically distributed given $X_T - X_t$, their conditional expectations are equal and sum up to $X_T - X_t$. Therefore, for each k from $t+1$ to T ,

$$\mathbb{E}[Y_k | \tilde{\mathcal{F}}_t] = \frac{X_T - X_t}{T-t}.$$

Since the Y_k are identically distributed and the total sum is $X_T - X_t$ we compute:

$$\begin{aligned} \mathbb{E}[X_{t+1} | \tilde{\mathcal{F}}_t] &= X_t + \mathbb{E}[Y_{t+1} | X_T] = X_t + \frac{X_T - X_t}{T-t}. \\ \mathbb{E}[\hat{X}_{t+1} | \tilde{\mathcal{F}}_t] &= \mathbb{E}\left[X_{t+1} - \sum_{k=0}^t \frac{X_T - X_k}{T-k} \mid \tilde{\mathcal{F}}_t\right]. \\ \mathbb{E}[\hat{X}_{t+1} | \tilde{\mathcal{F}}_t] - \tilde{X}_t &= \left(X_t + \frac{X_T - X_t}{T-t}\right) - \sum_{k=0}^t \frac{X_T - X_k}{T-k} - \left(X_t - \sum_{k=0}^{t-1} \frac{X_T - X_k}{T-k}\right) \\ &= \left(\frac{X_T - X_t}{T-t}\right) - \left(\frac{X_T - X_t}{T-t}\right) = 0. \end{aligned}$$

Therefore $(\hat{X}_t)_{t \in T}$ is a martingale *w.r.t.* the enlarged filtration $\tilde{\mathcal{F}}_t$.

(iii) Since the insider trader knows the value of X_T at all times, they can devise a strategy that leverages this information. Define the strategy as:

$$\xi_t = \text{sign}(X_T - X_{t-1}) := \begin{cases} 1, & X_T > X_{t-1} \\ 0, & X_T = X_{t-1} \\ -1, & X_T < X_{t-1} \end{cases}.$$

The trader always bets in the direction of the final value X_T , based on the current information. From (ii), we have

$$\mathbb{E}[Y_{t+1} | \tilde{\mathcal{F}}_t] = \frac{X_T - X_t}{T-t}.$$

Then the expected gain from the strategy is:

$$\begin{aligned}\mathbb{E}\left[\sum_{t=1}^T \xi_t (X_t - X_{t-1})\right] &= \mathbb{E}\left[\sum_{t=1}^T \xi_t Y_t\right] \\ &= \mathbb{E}\left[\sum_{t=1}^T \xi_t \mathbb{E}[Y_t \mid \tilde{\mathcal{F}}_{t-1}]\right] \\ &= \mathbb{E}\left[\sum_{t=1}^T \xi_t \frac{X_T - X_{t-1}}{T - (t-1)}\right] \\ &= \mathbb{E}\left[\sum_{t=1}^T \frac{|X_T - X_{t-1}|}{T - t + 1}\right] \stackrel{a.s.}{>} 0,\end{aligned}$$

where the denominator is positive, and $|X_T - X_{t-1}| > 0$ almost surely, since $\mathbb{P}(X_T = X_{t-1}) \xrightarrow{a.s.} 0$. Therefore the expected gain is positive.

□

SECTION A.2

| Martingale Measures and Other Considerations

EXERCISE A.7 (Parametrisation of Martingale Measures). Fix the time horizon at T and assume the initial σ -algebra $\mathcal{F}_0$ to be trivial. Let $\left(S_t^{(0)}\right)_{t \in T}$ be identically equal to 1 and let $Z_t := \log \frac{S_t}{S_{t-1}}$. Suppose that the market that is described by the pair of processes $S^{(0)}, S^{(1)}$ is arbitrage-free. Suppose that $\mathbb{P}$ is such that the $Z_1, Z_2, \dots, Z_T$ are *i.i.d.* with a common normal $\mathcal{N}(\mu, \sigma^2)$ distribution.

Give a relation between the parameters μ and σ^2 if $\mathbb{P}$ is a martingale measure. Is it possible that Z_t has a Gamma distribution instead of a normal one if $\mathbb{P}$ is a martingale measure?

SOLUTION. For P to be a martingale measure, the discounted price process $S_t^{(1)}$ must be a martingale. Since we have the risk-free asset $S_t^{(0)} = 1$, no discounting is required. Thus, we require the martingale property

$$\mathbb{E}_P \left[S_{t+1}^{(1)} \mid \mathcal{F}_t \right] = S_t^{(1)}.$$

From the definition

$$Z_t := \log \frac{S_t}{S_{t-1}} \quad \text{for } t \in \{1, 2, \dots, T\},$$

we can get:

$$S_t^{(1)} = S_0^{(1)} \exp \left(\sum_{i=1}^t Z_i \right).$$

The martingale condition becomes:

$$\mathbb{E}_P \left[\exp(Z_{t+1}) \mid \mathcal{F}_t \right] = 1.$$

Given that $Z_t \sim \mathcal{N}(\mu, \sigma^2)$, we compute the expected value of the exponential of a normal random variable, we have: $\mathbb{E}[\exp(Z_t)] = \exp\left(\mu + \frac{\sigma^2}{2}\right)$. Taking the natural logarithm of both sides gives the relation: $\mu + \frac{\sigma^2}{2} = 0$. Hence, the condition for P to be a martingale measure is: $\mu = -\frac{\sigma^2}{2}$.

The expectation $\mathbb{E}[\exp(Z_t)]$ must equal 1 for the martingale condition to hold. The moment-generating function of a Gamma-distributed random variable is different from that of a normal distribution. For a Gamma-distributed variable with shape parameter and rate parameter, the moment-generating function is given by:

$$\mathbb{E}[\exp(Z_t)] = \left(1 - \frac{1}{\beta}\right)^{-\alpha}, \quad \text{for } \beta > 1.$$

For this to equal 1, we would require that the shape and rate parameters be set in such a way that the exponential of the Gamma distribution results in 1, which is generally not possible for all t unless $\alpha = 0$ or $\beta = \infty$ both of which are degenerate cases. Therefore, the log returns Z_t cannot have a Gamma distribution and still satisfy the martingale property under P .

□

EXERCISE A.8 (Sensitivity of Option Prices). Consider an arbitrage-free market with one risky asset. Let $S^{(1)}$ be its price process and $S^{(0)}$ the deterministic price process of the riskless asset. Consider a European call option with discounted payoff

$$\tilde{H} = \frac{(S_T^{(1)} - K)^+}{S_T^{(0)}},$$

for some $K > 0$. Assume that $S_T^{(1)}$ has a density *w.r.t.* Lebesgue measure under any risk-neutral measure. Let π^* be an arbitrage-free price of the call option under some risk-neutral measure $\mathbb{P}^*$. Obviously π^* depends on K and $S_0^{(1)}$, so we write $\pi^* = \pi^*(K, S_0^{(1)})$. Show that

$$\frac{\partial \pi^*}{\partial S_0^{(1)}} < 1, \quad \frac{\partial \pi^*}{\partial K} = -(1 - F^*(K)) \frac{1}{S_T^{(0)}},$$

where F^* is the distribution function of $S_T^{(1)}$ under $\mathbb{P}^*$. To show the first assertion you may make additional assumptions, e.g., that $S_T^{(1)}$ is increasing in $S_0^{(1)}$, or even more explicit, $S_T^{(1)} = S_0^{(1)} R_T$, with R_T a positive random variable.

SOLUTION. For the Delta of the option, assume that is increasing with, which means if the initial stock price goes up, so does the terminal stock price. We can make the assumption that: $S_T^{(1)} = S_0^{(1)} R_T$, where R_T is positive random variable representing the return on the stock. The discounted payoff of the call option becomes:

$$\tilde{H} = \frac{(S_0^{(1)} R_T - K)^+}{S_T^{(0)}}.$$

The price of the option is the expected value of this payoff under the risk-neutral measure $\mathbb{P}^*$:

$$\pi^*(K, S_0^{(1)}) = \mathbb{E}_{\mathbb{P}^*}[\tilde{H}] = \frac{1}{S_T^{(0)}} \mathbb{E}_{\mathbb{P}^*}[(S_0^{(1)} R_T - K)^+].$$

Since $(S_0^{(1)} R_T - K)^+ = \max(S_0^{(1)} R_T - K, 0)$ is a piecewise linear function with respect to $S_0^{(1)}$, we can directly differentiate it with respect to $S_0^{(1)}$:

$$\frac{\partial}{\partial S_0^{(1)}} (S_0^{(1)} R_T - K)^+ = \begin{cases} R_T, & \text{if } S_0^{(1)} R_T > K \\ 0, & \text{if } S_0^{(1)} R_T \leq K \end{cases}.$$

This means: $\frac{\partial}{\partial S_0^{(1)}} (S_0^{(1)} R_T - K)^+ = R_T \cdot \mathbb{1}_{\{S_0^{(1)} R_T > K\}}$, substituting this into the derivative of the option price with respect to $S_0^{(1)}$, we get:

$$\frac{\partial \pi^*}{\partial S_0^{(1)}} = \frac{1}{S_T^{(0)}} \mathbb{E}_{\mathbb{P}^*} \left[R_T \mathbb{1}_{\{S_0^{(1)} R_T > K\}} \right].$$

R_T is a positive random variable, and its expected value under the risk-neutral measure is typically less than or equal to 1 (since the expected return on the stock is the risk-free rate in a risk-neutral world), we conclude: $\frac{\partial \pi^*}{\partial S_0^{(1)}} < 1$.

For the sensitivity of the option price with respect to the strike price K , the option price is given by:

$$\pi^*(K, S_0^{(1)}) = \frac{1}{S_T^{(0)}} \mathbb{E}_{\mathbb{P}^*} [(S_T^{(1)} - K)^+].$$

Take the derivative with respect to K :

$$\frac{\partial \pi^*}{\partial K} = \frac{1}{S_T^{(0)}} \mathbb{E}_{\mathbb{P}^*} \left[\frac{\partial}{\partial K} (S_T^{(1)} - K)^+ \right].$$

The derivative of $(S_T^{(1)} - K)^+$ with respect to K is $-\mathbb{1}_{\{S_T^{(1)} > K\}}$, so we have

$$\frac{\partial \pi^*}{\partial K} = -\frac{1}{S_T^{(0)}} \mathbb{E}_{\mathbb{P}^*} \left[\mathbb{1}_{\{S_T^{(1)} > K\}} \right].$$

The expected value of the indicator function is the probability that $S_T^{(1)} > K$, which is $1 - F^*(K)$, where $F^*(K)$ is the cdf under the risk-neutral measure. Therefore,

$$\frac{\partial \pi^*}{\partial K} = -\frac{1 - F^*(K)}{S_T^{(0)}}.$$

□

EXERCISE A.9 (Put–Call Parity in a Multi-Period Model). Consider an arbitrage-free market model with a single risky asset, $S^{(1)}$, and a riskless bank account, $S_t^{(0)} = (1 + r)^t$, for some $r > -1$. Suppose that an arbitrage-free price π_{call} has been fixed for the discounted claim

$$H_{\text{call}} = \frac{(S_T^{(1)} - K)^+}{S_T^{(0)}},$$

of a European call option with strike $K \geq 0$. Then there exists a nonnegative adapted process $X^{(2)}$ with $X_0^{(2)} = \pi_{\text{call}}$ and $X_T^{(2)} = H_{\text{call}}$ such that the extended market model with discounted price process $(\mathbb{1}, \tilde{S}^{(1)}, \tilde{X}^{(2)})$ is arbitrage-free.

Show that the discounted European contingent claim

$$H_{\text{put}} = \frac{(K - S_T^{(1)})^+}{S_T^{(0)}}$$

is attainable in the extended model, and that its unique arbitrage-free price is given by

$$\pi_{\text{put}} = \frac{K}{(1 + r)^T} - S_0^{(1)} + \pi_{\text{call}}.$$

SOLUTION. The *put–call parity* can be expressed by

$$H_{\text{call}} - H_{\text{put}} = \frac{1}{S_T^{(0)}} (S_T^{(1)} - K).$$

Then we can create a portfolio by investing $\varphi^{(0)} = \frac{K}{S_T^{(0)}}$ units in the bank, shorting one unit of the risky asset ($\varphi^{(1)} = -1$) and holding one call option ($\varphi^{(2)} = 1$). At time T , the wealth process is

$$W_T(\varphi) = \varphi^{(0)} S_T^{(0)} + \varphi^{(1)} S_T^{(1)} + \varphi^{(2)} X_T^{(2)} = \frac{K}{S_T^{(0)}} S_T^{(0)} - S_T^{(1)} + H_{\text{call}} = K - S_T^{(1)} + H_{\text{call}}.$$

We consider two cases:

- $S_T^{(1)} \geq K$: $H_{\text{call}} = S_T^{(1)} - K$ and $W_T = K - S_T^{(1)} - (S_T^{(1)} - K) = 0$;
- $S_T^{(1)} < K$: $H_{\text{call}} = 0$ and $W_T = K - S_T^{(1)}$.

Therefore, the terminal wealth is $W_T(\varphi) = (K - S_T^{(1)})^+$. Since

$$H_{\text{put}} = \frac{(K - S_T^{(1)})^+}{S_T^{(0)}},$$

we have

$$\tilde{W}_T(\varphi) = \frac{W_T}{S_T^{(0)}} = \frac{(K - S_T^{(1)})^+}{S_T^{(0)}} = H_{\text{put}},$$

which shows that it is attainable in the extended market. Therefore, the initial cost is

$$W_0(\varphi) = \varphi^{(0)} S_0^{(0)} + \varphi^{(1)} S_0^{(1)} + \varphi^{(2)} X_0^{(2)} = \frac{K}{S_T^{(0)}} S_0^{(0)} - S_0^{(1)} + \pi_{\text{call}} = \frac{K}{(1+r)^T} - S_0^{(1)} + \pi_{\text{call}}.$$

Hence, the unique arbitrage-free price of the discounted put option is

$$\pi_{\text{put}} = \frac{K}{(1+r)^T} - S_0^{(1)} + \pi_{\text{call}}.$$

□

EXERCISE A.10 (Additional Information in the Market). Consider a complete market model with finite time horizon $T = 2$ and a discounted price process $(X_t)_{t=0,1,2}$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t=0,1,2}, \mathbb{P})$. We denote the unique equivalent martingale measure by $\mathbb{Q}$.

Now, we extend this model by introducing two additional states, ω^+ and ω^- , describing further information that will be uncovered at time $t = 2$. We then define:

$$\begin{aligned}\widehat{\Omega} &:= \Omega \times \{\omega^+, \omega^-\}, \\ \widehat{\mathcal{F}} &:= \sigma(A \times B \mid A \in \mathcal{F}, B \in \{\omega^+, \omega^-\}), \\ \widehat{\mathbb{P}}[\{\omega, \omega^\pm\}] &:= \frac{1}{2} \mathbb{P}[\{\omega\}], \quad \omega \in \Omega.\end{aligned}$$

The unveiling of additional information at $t = 2$ is reflected by the chosen price process $\hat{X}_t(\omega, \omega^\pm) := X_t(\omega)$ and the following filtration:

$$\begin{aligned}\widehat{\mathcal{F}}_0 &:= \{\emptyset, \widehat{\Omega}\}, \\ \widehat{\mathcal{F}}_1 &:= \sigma(A \times \{\omega^+, \omega^-\} \mid A \in \mathcal{F}_1), \\ \widehat{\mathcal{F}}_2 &:= \widehat{\mathcal{F}}.\end{aligned}$$

For $0 < p < 1$, we further specify the probability measure $\widehat{\mathbb{P}}_p^*$ on $\widehat{\mathcal{F}}$ via

$$\widehat{\mathbb{P}}_p^*[A \times \{\omega^+\}] := p \cdot \mathbb{Q}[A], \quad \widehat{\mathbb{P}}_p^*[A \times \{\omega^-\}] := (1-p) \cdot \mathbb{Q}[A], \quad A \in \mathcal{F}.$$

Then solve the following exercises:

- (i) Verify that each measure $\widehat{\mathbb{P}}_p^*$ is an equivalent martingale measure for the extended model.
- (ii) Argue that the extended model is incomplete and find a nonattainable contingent claim.
- (iii) Does every equivalent martingale measure belong to the set $\{\widehat{\mathbb{P}}_p^* : 0 < p < 1\}$? If yes, give a proof, if not, find a counterexample.

SOLUTION. (i) The measure $\widehat{\mathbb{P}}_p^*$ is defined on the extended probability space $\widehat{\Omega} = \Omega \times \{\omega^+, \omega^-\}$ by

$$\widehat{\mathbb{P}}_p^*[A \times \{\omega^+\}] = p\mathbb{Q}[A] \quad \text{and} \quad \widehat{\mathbb{P}}_p^*[A \times \{\omega^-\}] = (1-p)\mathbb{Q}[A].$$

Thus, $\widehat{\mathbb{P}}_p^*$ assigns probabilities p and $1-p$ to ω^+ and ω^- , respectively, while retaining the measure $\mathbb{Q}$ on Ω .

We check that the discounted price process $\widehat{X}_t(\omega, \omega^\pm) = X_t(\omega)$ is a martingale under $\widehat{\mathbb{P}}_p^*$. At times $t = 0$ and $t = 1$, $\widehat{X}_t$ agrees with X_t ; since X is already a martingale under $\mathbb{Q}$, the martingale property holds under $\widehat{\mathbb{P}}_p^*$.

At time $t = 2$, the additional information from ω^+ or ω^- is revealed, but it does not affect the price process. Since $\widehat{X}_2 = X_2$, the martingale property remains true under the extended measure $\widehat{\mathbb{P}}_p^*$. Thus, $\widehat{\mathbb{P}}_p^*$ is a martingale measure.

To verify that $\widehat{\mathbb{P}}_p^*$ is equivalent to $\widehat{\mathbb{P}}$, observe that the two measures assign positive probability to the same events. Both are based on $\mathbb{P}$ on Ω , and they differ only in the positive probabilities assigned to ω^+ and ω^- . Hence, $\widehat{\mathbb{P}}_p^*$ is an equivalent martingale measure for the extended model.

- (ii) By the definition, in a complete market, every contingent claim can be replicated by trading in the available assets, and in an incomplete market, some claims cannot be perfectly hedged.

The original market (before introducing ω^+ and ω^-) was complete because there was only one equivalent martingale measure $\mathbb{Q}$. However, after introducing the additional information at time $t = 2$, we have a family of martingale measures $\widehat{\mathbb{P}}_p^*$ for $0 < p < 1$, indicating multiple possible risk-neutral probabilities for the additional information. This means that the extended model has more uncertainty than the original model, and there are more contingent claims that we might want to hedge. However, since we only have one underlying asset X_t and no new tradable assets corresponding to ω^+ and ω^- , not all claims can be replicated.

One such nonattainable contingent claim could be a binary option that pays 1 if ω^+ occurs and 0 otherwise. This claim depends on the additional information revealed at time $t = 2$, but since there is no tradable asset that directly relates to ω^+ or ω^- , this claim cannot be perfectly hedged in the extended market. Thus, the extended model is incomplete.

- (iii) The set $\{\widehat{\mathbb{P}}_p^* : 0 < p < 1\}$ defines a family of martingale measures parametrised by p , where the probabilities assigned to ω^+ and ω^- are p and $1-p$, respectively. Consider the following counterexample.

Consider an equivalent martingale measure $\widehat{\mathbb{P}}^*$, where the probability assigned to ω^+ is independent of the original measure $\mathbb{Q}$ on Ω . Suppose we define a measure where $\widehat{\mathbb{P}}^*[\omega^+] = \frac{1}{3}$ and $\widehat{\mathbb{P}}^*[\omega^-] = \frac{2}{3}$ regardless of $\mathbb{Q}$.

This measure $\widehat{\mathbb{P}}^*$ would still be equivalent to $\widehat{\mathbb{P}}$, as it assigns non-zero probability to the same events. However, it is not of the form $\widehat{\mathbb{P}}_p^*$, because p in $\widehat{\mathbb{P}}_p^*$ depends on $\mathbb{Q}$, while the measure here does not.

□

EXERCISE A.11 (Arbitrage-Free Prices of Non-Attainable Claims). Read the remainder of the proof of *Theorem 2.9* from the point where we introduce the density $\widehat{Z}$. Then prove that the measure $\check{\mathbb{P}}$ in the proof of *Theorem 2.9* is a martingale measure for the market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$.

SOLUTION. In *Theorem 2.9*, two equivalent martingale measures $\widehat{\mathbb{P}}$ and $\check{\mathbb{P}}$ were constructed as follows: the Radon–Nikodym derivative (density) of $\widehat{\mathbb{P}}$ with respect to $\mathbb{Q}$ is given by:

$$\frac{d\widehat{\mathbb{P}}}{d\mathbb{Q}} = \widehat{Z} := 1 + Z - \mathbb{E}_{\mathbb{Q}}[Z \mid \mathcal{F}_{t-1}],$$

where Z is a measurable function introduced to adjust the martingale property. The density of $\check{P}$ with respect to Q is defined by:

$$\frac{d\check{P}}{dQ} = 2 - \frac{d\hat{P}}{dQ} = 2 - \hat{Z}.$$

For the probability measure, P^* is a martingale measure if the discounted price process X_t is a martingale under P^* , it satisfies:

$$\mathbb{E}_{P^*}[X_{t+1} | \mathcal{F}_t] = X_t.$$

We know that $\check{P}$ and Q are related by the Radon–Nikodym derivative $Z_{\check{P}|Q} = \frac{d\check{P}}{dQ}$, and the conditional expectation under $\check{P}$ can be expressed as:

$$\mathbb{E}_{\check{P}}[X_{t+1} | \mathcal{F}_t] = \frac{\mathbb{E}_Q\left[X_{t+1} \frac{d\check{P}}{dQ} | \mathcal{F}_t\right]}{\mathbb{E}_Q\left[\frac{d\check{P}}{dQ} | \mathcal{F}_t\right]}.$$

We need to prove this conditional expectation equals X_t . Since $\hat{P}$ is already known to be a martingale measure, we have: $\mathbb{E}_{\hat{P}}[X_{t+1} | \mathcal{F}_t] = X_t$, and the Radon–Nikodym derivative $\frac{d\hat{P}}{dQ} = \hat{Z}$ satisfies the necessary conditions for this martingale property.

Substitute the Radon–Nikodym derivative of $\check{P}$ into the conditional expectation formula:

$$\frac{dQ}{d\check{P}} = 2 - \hat{Z}.$$

Using this in the conditional expectation formula under Q , we get:

$$\mathbb{E}_{\check{P}}[X_{t+1} | \mathcal{F}_t] = \frac{\mathbb{E}_Q\left[X_{t+1}(2 - \hat{Z}) | \mathcal{F}_t\right]}{\mathbb{E}_Q[2 - \hat{Z} | \mathcal{F}_t]}.$$

Using the linearity of conditional expectation, we can expand the right-hand side:

$$\begin{aligned} \mathbb{E}_Q[X_{t+1}(2 - \hat{Z}) | \mathcal{F}_t] &= 2\mathbb{E}_Q[X_{t+1} | \mathcal{F}_t] - \mathbb{E}_Q[X_{t+1}\hat{Z} | \mathcal{F}_t], \\ \mathbb{E}_Q[2 - \hat{Z} | \mathcal{F}_t] &= 2 - \mathbb{E}_Q[\hat{Z} | \mathcal{F}_t]. \end{aligned}$$

Since $\hat{P}$ is a martingale measure, we know that: $\mathbb{E}_Q[X_{t+1} | \mathcal{F}_t] = X_t$, and we also know that: $\mathbb{E}_Q[X_{t+1}\hat{Z} | \mathcal{F}_t] = X_t\mathbb{E}_Q[\hat{Z} | \mathcal{F}_t]$. Now plugging these results into our original conditional expectation under $\check{P}$, we get:

$$\mathbb{E}_{\check{P}}[X_{t+1} | \mathcal{F}_t] = \frac{2X_t - X_t\mathbb{E}_Q[\hat{Z} | \mathcal{F}_t]}{2 - \mathbb{E}_Q[\hat{Z} | \mathcal{F}_t]} = X_t,$$

which means that $\check{P}$ satisfies the martingale property, and thus, $\check{P}$ is an equivalent martingale measure for the market $(\Omega, \mathcal{F}, P, \mathcal{F}, X)$.

□

SECTION A.3

| The Standard Model

Consider a frictionless financial market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ with two assets represented by $X = (S^{(0)}, S^{(1)})$. The process of the bank account is given as $S_t^{(0)} = S_0^{(0)} e^{rt} = S_0^{(0)} (1 + \tilde{r})^t$, $t = 0, 1, \dots, T$, with fixed interest rate $r \in \mathbb{R}$ and $\tilde{r} = e^r - 1$; and the risky asset's price follows the model:

$$S_t^{(1)} = S_0^{(1)} \exp(R_t) = S_0^{(1)} \prod_{j=1}^t (1 + \Delta \tilde{R}_j),$$

where $\tilde{R}_j = \sum_{n=1}^j (\exp(\Delta R_n) - 1)$, $R_0 = 0$, and the *daily logarithmic returns* $\Delta R_1, \Delta R_2, \dots$ of $(R_t)_{t \in T}$ (note that $\Delta R_t = \log \left(\frac{S_t^{(1)}}{S_{t-1}^{(1)}} \right)$ for $t = 1, \dots, T$) are assumed to be independent and identically distributed.

We call the market $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ the *standard model*, whenever the daily logarithmic returns ΔR_t , for $t = 1, \dots, T$ are normally distributed with mean μ and variance σ^2 . Note that according to *Proposition 2.15* in the lecture notes, the standard model cannot be complete as the normal distribution is a continuous distribution and hence, the underlying probability space cannot be decomposed into a finite number of atoms.

EXERCISE A.12 (European Call Option in the Standard Model). Let $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, X)$ be the *standard model* introduced above. Give a solution to the following problems.

- (i) Give an estimator for the parameters μ and σ^2 at time T given the historic data $S_0^{(1)}, S_1^{(1)}, \dots, S_{T-1}^{(1)}$.
- (ii) Consider a European call option with maturity T and strike K , i.e., a European contingent claim with pay-off at time T given by $H := (S_T^{(1)} - K)^+$.
 - (a) Construct a trading strategy φ such that $W_T(\varphi) \geq H$ $\mathbb{P}$ -almost surely.
 - (b) Let

$$\pi_L(H) := \sup \left\{ W_0 \in \mathbb{R} : \exists \text{ self-financing strategy } \varphi \text{ such that } W_0(\varphi) = W_0 \text{ and } W_T(\varphi) \geq H \right\}.$$

Show that

$$\pi_L(H) \geq \max \left(0, S_0^{(1)} - Ke^{-rT} \right) = \left(S_0^{(1)} - Ke^{-rT} \right)^+.$$

- (iii) Note that the two distributions $\mathcal{N}(\mu, \sigma^2)$ and $\mathcal{N}(r - \frac{1}{2}\tilde{\sigma}^2, \tilde{\sigma}^2)$ are equivalent to the Lebesgue measure and thus also equivalent to each other with density denoted by

$$f_{\tilde{\sigma}} := \frac{d\mathcal{N}(r - \frac{1}{2}\tilde{\sigma}^2, \tilde{\sigma}^2)}{d\mathcal{N}(\mu, \sigma^2)}.$$

Next, define a probability measure $\mathbb{Q}_{\tilde{\sigma}} \sim \mathbb{P}$ by

$$\frac{d\mathbb{Q}_{\tilde{\sigma}}}{d\mathbb{P}} := \prod_{n=1}^T f_{\tilde{\sigma}}(\Delta R_n).$$

Relative to the measure $\mathbb{Q}_{\tilde{\sigma}}$, the price process has the same structure as under $\mathbb{P}$, but with different parameters μ and σ^2 . Show that for all $\tilde{\sigma} > 0$ the following holds true:

- (a) $\mathbb{Q}_{\tilde{\sigma}}$ is a probability measure equivalent to $\mathbb{P}$ and the random variables $\Delta R_1, \dots, \Delta R_N$ are independent under $\mathbb{Q}_{\tilde{\sigma}}$ with law $\mathcal{N}\left(r - \frac{\tilde{\sigma}^2}{2}, \tilde{\sigma}^2\right)$;
- (b) $\mathbb{Q}_{\tilde{\sigma}}$ is an equivalent martingale measure.

Note that this then constructs a one-parametric family of equivalent martingale measures $\{\mathbb{Q}_{\tilde{\sigma}} : \tilde{\sigma} > 0\}$.

- (iv) Next, compute the option price that is obtained as an expectation under measure $\mathbb{Q}_{\tilde{\sigma}}$, i.e.,

$$\pi_{\mathbb{Q}_{\tilde{\sigma}}} := S_0^{(0)} \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}} \left(\frac{(S_T^{(1)} - K)^+}{S_T^{(0)}} \right) = \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}} \left((S_T^{(1)} e^{R_T - rT} - K e^{-rT})^+ \right).$$

- (v) The options price $\pi_{\mathbb{Q}_{\tilde{\sigma}}}$ depends via $\mathbb{Q}_{\tilde{\sigma}}$ on the parameter $\tilde{\sigma}$. Show that if $\tilde{\sigma} \rightarrow 0$ then $\pi_{\mathbb{Q}_{\tilde{\sigma}}} \rightarrow (S_0^{(1)} - K e^{-rT})^+$ and if $\sigma \rightarrow \infty$ then $\pi_{\mathbb{Q}_{\tilde{\sigma}}} \rightarrow S_0^{(1)}$. (Therefore, the price limits in the standard model coincide with the trivial ones. Put differently, absence of arbitrage does not provide much information on European call prices in the standard model.)

SOLUTION. (i) The daily logarithmic returns are defined as

$$\Delta R_t = \log \frac{S_t^{(1)}}{S_{t-1}^{(1)}}, \quad t = 1, \dots, T.$$

Given the historic data $S_0^{(1)}, S_1^{(1)}, \dots, S_{T-1}^{(1)}$, for each $t = 1, 2, \dots, T-1$, we calculate the daily logarithmic return $\Delta \hat{R}_t$ to get $\Delta \hat{R}_1, \Delta \hat{R}_2, \dots, \Delta \hat{R}_{T-1}$. Then the sample mean estimator $\hat{\mu}$ is calculated as

$$\hat{\mu} = \frac{1}{T-1} \sum_{t=1}^{T-1} \Delta \hat{R}_t.$$

Since ΔR_t are assumed to be *i.i.d.*, the sample mean estimator is the maximum likelihood estimator for μ . Subsequently, the sample variance estimator $\hat{\sigma}^2$ is calculated as

$$\hat{\sigma}^2 = \frac{1}{T-2} \sum_{t=1}^{T-1} (\Delta \hat{R}_t - \hat{\mu})^2,$$

where we adapted $n-1$ in the denominator to obtain an unbiased estimator.

- (ii) (a) We define the trading strategy $\varphi_t = (\varphi_t^{(0)}, \varphi_t^{(1)})$ as buying one unit of the risky asset at time 0 and holding it until time T , i.e., $\varphi_t^{(1)} = 1$. Since we are not investing in the bank account, $\varphi_t^{(0)} = 0$, and the wealth process is calculated as

$$W_t(\varphi) = \varphi_t^{(0)} S_t^{(0)} + \varphi_t^{(1)} S_t^{(1)} = S_t^{(1)},$$

where $S_t^{(1)} = S_0^{(1)} e^{R_t}$. The strategy is predictable because the holdings are constant. Then at time T , the wealth $W_T(\varphi) = S_T^{(1)}$. To verify that

$$W_T(\varphi) \geq H = (S_T^{(1)} - K)^+, \quad \mathbb{P}\text{-a.s.},$$

we consider two cases:

- $S_T^{(1)} \geq K$: then $H = S_T^{(1)} - K$, and $W_T(\varphi) - H = S_T^{(1)} - (S_T^{(1)} - K) = K \geq 0$;

- $S_T^{(1)} < K$: then $H = 0$, and $W_T(\varphi) - H = S_T^{(1)} \geq 0$.

Therefore, we have shown that $W_T(\varphi) \geq H$, $\mathbb{P}$ -a.s.

(b) The trading strategy defined in part (a) of (ii) is self-financing, since

$$\begin{aligned} W_{t-1}(\varphi) + \varphi_{t-1}^\top (X_t - X_{t-1}) &= S_{t-1}^{(1)} + \begin{pmatrix} \varphi_{t-1}^{(0)} \\ \varphi_{t-1}^{(1)} \end{pmatrix}^\top \begin{pmatrix} S_t^{(0)} - S_{t-1}^{(0)} \\ S_t^{(1)} - S_{t-1}^{(1)} \end{pmatrix} \\ &= S_{t-1}^{(1)} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}^\top \begin{pmatrix} S_t^{(0)} - S_{t-1}^{(0)} \\ S_t^{(1)} - S_{t-1}^{(1)} \end{pmatrix} \\ &= S_{t-1}^{(1)} + S_t^{(1)} - S_{t-1}^{(1)} \\ &= S_t^{(1)} = W_t(\varphi). \end{aligned}$$

Given that

$$\pi_L(H) := \sup \{W_0 \in \mathbb{R} : \exists \text{ self-financing strategy } \varphi \text{ s.t. } W_0(\varphi) = W_0 \text{ and } W_T(\varphi) \geq H\},$$

we consider two cases:

- $S_0^{(1)} \geq Ke^{-rT}$: $(S_0^{(1)} - Ke^{-rT})^+ = S_0^{(1)} - Ke^{-rT}$ and $\pi_L(H) \geq W_0(\varphi) = S_0^{(1)} \geq S_0^{(1)} - Ke^{-rT}$;
- $S_0^{(1)} < Ke^{-rT}$: $(S_0^{(1)} - Ke^{-rT})^+ = 0$ and $\pi_L(H) \geq W_0(\varphi) = S_0^{(1)} \geq 0$.

Therefore the trading strategy satisfies

$$\pi_L(H) \geq \max(0, S_0^{(1)} - Ke^{-rT}) = (S_0^{(1)} - Ke^{-rT})^+.$$

(iii) (a) The Radon–Nikodym derivative can be rewritten as

$$\begin{aligned} f_{\tilde{\sigma}}(x) &= \frac{d\mathcal{N}(r - \frac{1}{2}\tilde{\sigma}^2, \tilde{\sigma}^2)}{d\mathcal{N}(\mu, \sigma^2)} \\ &= \frac{\frac{1}{\sqrt{2\pi\tilde{\sigma}^2}} \exp\left(-\frac{(x - (r - \frac{1}{2}\tilde{\sigma}^2))^2}{2\tilde{\sigma}^2}\right)}{\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)} \\ &= \sqrt{\frac{\sigma^2}{\tilde{\sigma}^2}} \exp\left(-\frac{(x - (r - \frac{1}{2}\tilde{\sigma}^2))^2}{2\tilde{\sigma}^2} + \frac{(x - \mu)^2}{2\sigma^2}\right) > 0, \quad \forall x \in \mathbb{R}. \end{aligned}$$

Given that

$$\frac{d\mathbb{Q}_{\tilde{\sigma}}}{d\mathbb{P}} := \prod_{n=1}^T f_{\tilde{\sigma}}(\Delta R_n),$$

it is straightforward to recognise $\frac{d\mathbb{Q}_{\tilde{\sigma}}}{d\mathbb{P}} > 0$. Under $\mathbb{P}$, the joint probability density function of ΔR_t is

$$f_{\Delta R_t}^{\mathbb{P}}(x_t) = \prod_{n=1}^T f_{\mathcal{N}(\mu, \sigma^2)}(x_n), \quad t = 1, \dots, T.$$

Under $\mathbb{Q}_{\tilde{\sigma}}$, we calculate the joint probability density function as

$$\begin{aligned} f_{\Delta R_t}^{\mathbb{Q}_{\tilde{\sigma}}}(x_t) &= f_{\Delta R_t}^{\mathbb{P}}(x_t) \cdot \frac{d\mathbb{Q}_{\tilde{\sigma}}}{d\mathbb{P}}(x_t) \\ &= \prod_{n=1}^T f_{\mathcal{N}(\mu, \sigma^2)}(x_n) \cdot \prod_{n=1}^T f_{\tilde{\sigma}}(\Delta R_n) \\ &= \prod_{n=1}^T \left(f_{\mathcal{N}(\mu, \sigma^2)}(x_n) \cdot \frac{f_{\mathcal{N}(r - \frac{1}{2}\tilde{\sigma}^2, \tilde{\sigma}^2)}(x_n)}{f_{\mathcal{N}(\mu, \sigma^2)}(x_n)} \right) = \prod_{n=1}^T f_{\mathcal{N}(r - \frac{1}{2}\tilde{\sigma}^2, \tilde{\sigma}^2)}(x_n), \end{aligned}$$

where the joint density is the product of the marginal densities, and hence ΔR_n remain *i.i.d.* under $\mathbb{Q}_{\tilde{\sigma}}$. Then we calculate

$$\begin{aligned}
\mathbb{Q}(\Omega)_{\tilde{\sigma}} &= \mathbb{E}_{\mathbb{P}} \left[\frac{d\mathbb{Q}_{\tilde{\sigma}}}{d\mathbb{P}} \right] \\
&= \mathbb{E}_{\mathbb{P}} \left[\prod_{n=1}^T f_{\tilde{\sigma}}(\Delta R_n) \right] \\
&= \prod_{n=1}^T \mathbb{E}_{\mathbb{P}} [f_{\tilde{\sigma}}(\Delta R_n)] \quad (\Delta R_n \text{ i.i.d. under } \mathbb{Q}_{\tilde{\sigma}}) \\
&= \prod_{n=1}^T \int_{-\infty}^{\infty} \left(\sqrt{\frac{\sigma^2}{\tilde{\sigma}^2}} \exp \left(-\frac{(x - (r - \frac{1}{2}\tilde{\sigma}^2))^2}{2\tilde{\sigma}^2} + \frac{(x - \mu)^2}{2\sigma^2} \right) \right) \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right) \right) dx \\
&= \prod_{n=1}^T \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\tilde{\sigma}^2}} \exp \left(-\frac{(x - (r - \frac{1}{2}\tilde{\sigma}^2))^2}{2\tilde{\sigma}^2} \right) dx \\
&= \prod_{n=1}^T \int_{-\infty}^{\infty} f_{\mathcal{N}(r - \frac{1}{2}\tilde{\sigma}^2, \tilde{\sigma}^2)}(x_n) dx_n = 1.
\end{aligned}$$

In conclusion, we have shown that $\mathbb{Q}_{\tilde{\sigma}}$ is a probability measure and ΔR_n are *i.i.d.* under $\mathbb{Q}_{\tilde{\sigma}}$ with $\mathcal{N}(r - \frac{1}{2}\tilde{\sigma}^2, \tilde{\sigma}^2)$.

- (b) By Definition 1.15, we have to show that the discounted price process $\tilde{S}_t^{(1)}$ is a $\mathbb{Q}_{\tilde{\sigma}}$ -martingale with respect to the filtration $\mathbb{F}$. The discounted price of the risky asset is

$$\tilde{S}_t^{(1)} = \frac{S_t^{(1)}}{S_t^{(0)}} = \frac{S_0^{(1)} e^{R_t}}{S_0^{(0)} e^{rt}} = \tilde{S}_0^{(1)} \exp(R_t - rt) = \tilde{S}_0^{(1)} \exp \left(\sum_{n=1}^t \Delta R_n - rt \right).$$

Since $\tilde{S}_t^{(1)}$ depends on ΔR_n and $\mathcal{F}_t = \sigma(\Delta R_n)$, $\tilde{S}_t^{(1)}$ is adapted to the filtration $\mathbb{F}$. By part (a) of (iii), $\tilde{S}_t^{(1)}$ is an exponential of a normally distributed random variable under $\mathbb{Q}_{\tilde{\sigma}}$ with finite expectation, so $\tilde{S}_t^{(1)}$ is also integrable. We now verify the martingale property. From

$$\begin{aligned}
\tilde{S}_t^{(1)} &= \tilde{S}_0^{(1)} \exp(R_t - rt) \\
&= \tilde{S}_0^{(1)} \exp(\Delta R_t + R_{t-1} - r(t-1) - r) \\
&= \tilde{S}_0^{(1)} \exp(R_{t-1} - r(t-1)) \exp(\Delta R_t - r) = \tilde{S}_{t-1}^{(1)} \exp(\Delta R_t - r),
\end{aligned}$$

we calculate

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}} [\tilde{S}_t^{(1)} | \mathcal{F}_{t-1}] &= \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}} [\tilde{S}_{t-1}^{(1)} \exp(\Delta R_t - r) | \mathcal{F}_{t-1}] \\
&= \tilde{S}_{t-1}^{(1)} \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}} [\exp(\Delta R_t - r) | \mathcal{F}_{t-1}] \\
&= \tilde{S}_{t-1}^{(1)} \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}} [\exp(\Delta R_t - r)] \\
&= \tilde{S}_{t-1}^{(1)} \exp \left(\left(-\frac{1}{2}\tilde{\sigma}^2 \right) + \frac{1}{2}\tilde{\sigma}^2 \right) \\
&= \tilde{S}_{t-1}^{(1)} \cdot 1 = \tilde{S}_{t-1}^{(1)},
\end{aligned}$$

where we used the moment-generating function of $X := \Delta R_t - r \sim \mathcal{N}(-\frac{1}{2}\tilde{\sigma}^2, \tilde{\sigma}^2)$ to calculate the expectation. In conclusion, $\mathbb{Q}_{\tilde{\sigma}}$ is an equivalent martingale measure.

- (iv) We can rewrite the given formula as

$$\pi_{\mathbb{Q}_{\tilde{\sigma}}} = S_0^{(0)} \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}} \left[\frac{(S_T^{(1)} - K)^+}{S_T^{(0)}} \right] = S_0^{(0)} \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}} \left[\frac{(S_T^{(1)} - K)^+}{S_0^{(0)} e^{rT}} \right] = e^{-rT} \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}} [(S_T^{(1)} - K)^+].$$

From the previous parts, under $\mathbb{Q}_{\tilde{\sigma}}$, the daily logarithmic returns ΔR_n are *i.i.d.* with respect to $\mathcal{N}(r - \frac{1}{2}\tilde{\sigma}^2, \tilde{\sigma}^2)$. The cumulative return R_T is calculated as

$$R_T = \sum_{n=1}^T \Delta R_n \sim \mathcal{N}\left(T\left(r - \frac{1}{2}\tilde{\sigma}^2\right), T\tilde{\sigma}^2\right). \quad (R_0 = 0)$$

Since $S_T^{(1)}$ follows a log-normal distribution under $\mathbb{Q}_{\tilde{\sigma}}$, we use the *Black-Scholes formula* to calculate the price of the European call option. We can express the expected payoff as

$$\mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}}\left[\left(S_T^{(1)} - K\right)^+\right] = \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}}\left[S_T^{(1)} \mathbb{1}_{\{S_T^{(1)} > K\}}\right] - K \mathbb{Q}_{\tilde{\sigma}}\left(S_T^{(1)} > K\right),$$

where the first part is the expected value of $S_T^{(1)}$ when the option is exercised, and the second part is the probability that the option is exercised.

For the second part, notice that

$$S_T^{(1)} > K \iff e^{R_T} > \frac{K}{S_0^{(1)}} \iff R_T > \ln\left(\frac{K}{S_0^{(1)}}\right),$$

then

$$\mathbb{P} = \mathbb{Q}_{\tilde{\sigma}}\left(S_T^{(1)} > K\right) = \mathbb{Q}_{\tilde{\sigma}}\left(R_T > \ln\left(\frac{K}{S_0^{(1)}}\right)\right) = \mathbb{Q}_{\tilde{\sigma}}\left(Z > \frac{\ln\left(\frac{K}{S_0^{(1)}}\right) - T\left(r - \frac{1}{2}\tilde{\sigma}^2\right)}{\tilde{\sigma}\sqrt{T}}\right).$$

We define

$$d_2 = -\frac{\ln\left(\frac{K}{S_0^{(1)}}\right) - T\left(r - \frac{1}{2}\tilde{\sigma}^2\right)}{\tilde{\sigma}\sqrt{T}} = \frac{\ln\left(\frac{S_0^{(1)}}{K}\right) + T\left(r - \frac{1}{2}\tilde{\sigma}^2\right)}{\tilde{\sigma}\sqrt{T}},$$

then

$$\mathbb{P} = \mathbb{Q}_{\tilde{\sigma}}(Z > -d_2) = N(d_2),$$

where $N(\cdot)$ is the standard normal CDF.

For the first part, we calculate

$$\begin{aligned} E &= \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}}\left[S_T^{(1)} \mathbb{1}_{\{S_T^{(1)} > K\}}\right] \\ &= \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}}\left[S_0^{(1)} e^{R_T} \mathbb{1}_{\{S_T^{(1)} > K\}}\right] \\ &= S_0^{(1)} \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}}\left[e^{R_T} \mathbb{1}_{\left\{R_T > \ln\left(\frac{K}{S_0^{(1)}}\right)\right\}}\right] \\ &= S_0^{(1)} \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}}\left[e^{\mu_{R_T} + \sigma_{R_T} Z} \mathbb{1}_{\{Z > -d_2\}}\right] \quad (Z = \frac{R_T - \mu_{R_T}}{\sigma_{R_T}} \sim \mathcal{N}(0, 1)) \\ &= S_0^{(1)} e^{\mu_{R_T}} \mathbb{E}_{\mathbb{Q}_{\tilde{\sigma}}}\left[e^{\sigma_{R_T} Z} \mathbb{1}_{\{Z > -d_2\}}\right] \\ &= S_0^{(1)} e^{\mu_{R_T}} \int_{-d_2}^{\infty} e^{\sigma_{R_T} Z} \frac{1}{\sqrt{2\pi}} e^{-\frac{Z^2}{2}} dZ \\ &= S_0^{(1)} e^{\mu_{R_T}} \int_{-d_2}^{\infty} e^{\sigma_{R_T} Z - \frac{Z^2}{2}} \frac{1}{\sqrt{2\pi}} dZ \\ &= S_0^{(1)} e^{\mu_{R_T}} \frac{e^{\frac{\sigma_{R_T}^2}{2}}}{\sqrt{2\pi}} \int_{-d_2}^{\infty} e^{-\frac{(Z - \sigma_{R_T})^2}{2}} dZ \\ &= S_0^{(1)} e^{\mu_{R_T}} \frac{e^{\frac{\sigma_{R_T}^2}{2}}}{\sqrt{2\pi}} \int_{\xi}^{\infty} e^{-\frac{\xi^2}{2}} d\xi \quad (\xi = -d_2 - \sigma_{R_T}) \end{aligned}$$

$$= S_0^{(1)} e^{\mu_{R_T} + \frac{1}{2}\sigma_{R_T}^2} N(d_2 + \sigma_{R_T}) = S_0^{(1)} e^{\mu_{R_T} + \frac{1}{2}\sigma_{R_T}^2} N(d_1),$$

where

$$d_1 := d_2 + \sigma_{R_T} = \frac{\ln\left(\frac{S_0^{(1)}}{K}\right) + T(r + \frac{1}{2}\tilde{\sigma}^2)}{\tilde{\sigma}\sqrt{T}}.$$

Substituting back into E we get

$$E = S_0^{(1)} e^{T(r - \frac{1}{2}\tilde{\sigma}^2) + \frac{1}{2}T\tilde{\sigma}^2} N(d_1) = S_0^{(1)} e^{rT} N(d_1).$$

In conclusion, the option price is calculated by the *Black–Scholes formula*:

$$\pi_{Q_{\tilde{\sigma}}} = S_0^{(1)} N(d_1) - K e^{-rT} N(d_2),$$

where

$$d_1 = \frac{\ln\left(\frac{S_0^{(1)}}{K}\right) + T(r + \frac{1}{2}\tilde{\sigma}^2)}{\tilde{\sigma}\sqrt{T}}, \quad d_2 = d_1 - \tilde{\sigma}\sqrt{T}.$$

- (v) As $\tilde{\sigma} \rightarrow 0$, the variance of R_T is 0, and the mean of R_T becomes rT , i.e., $R_T \xrightarrow{\mathcal{D}} rT$. Then $S_T^{(1)} = S_0^{(1)} e^{R_T} \xrightarrow{\mathcal{D}} S_0^{(1)} e^{rT}$. Therefore the option price

$$\pi_{Q_{\tilde{\sigma}}} = e^{-rT} \mathbb{E}_{Q_{\tilde{\sigma}}} \left[\left(S_T^{(1)} - K \right)^+ \right] \xrightarrow{\mathcal{D}} e^{-rT} \left(S_0^{(1)} e^{rT} - K \right)^+ = \left(S_0^{(1)} - K e^{-rT} \right)^+.$$

If $\sigma \rightarrow \infty$, $\text{Var}(R_T) \rightarrow \infty$ and $\mu_{R_T} \rightarrow -\infty$. From (iv), the option price is

$$\pi_{Q_{\tilde{\sigma}}} = S_0^{(1)} N(d_1) - K e^{-rT} N(d_2),$$

where

$$d_1 = \frac{\ln\left(\frac{S_0^{(1)}}{K}\right) + T(r + \frac{1}{2}\tilde{\sigma}^2)}{\tilde{\sigma}\sqrt{T}} \approx \frac{\frac{1}{2}\tilde{\sigma}^2 T}{\tilde{\sigma}\sqrt{T}} = \frac{\tilde{\sigma}\sqrt{T}}{2} \rightarrow \infty,$$

$$d_2 = d_1 - \tilde{\sigma}\sqrt{T} \approx -\frac{\tilde{\sigma}\sqrt{T}}{2} \rightarrow -\infty.$$

Thus $N(d_1) \rightarrow 1$, $N(d_2) \rightarrow 0$, and $\pi_{Q_{\tilde{\sigma}}} \rightarrow S_0^{(1)}$.

In conclusion, we have proved that

$$\pi_{Q_{\tilde{\sigma}}} \rightarrow \left(S_0^{(1)} - K e^{-rT} \right)^+, \quad \tilde{\sigma} \rightarrow 0$$

and

$$\pi_{Q_{\tilde{\sigma}}} \rightarrow S_0^{(1)}, \quad \tilde{\sigma} \rightarrow 0.$$

□

SECTION A.4

| The Cox–Ross–Rubinstein Model

Let $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F}, (S^{(0)}, S^{(1)}))$ denote a canonical CRR model which is complete. Solve the following exercises.

EXERCISE A.13 (Forward-Start Options). Let $T_0 \in \{1, \dots, T-1\}$ and $K > 0$. The payoff of *forward starting call option* has the form

$$\left(\frac{S_T^{(1)}}{S_{T_0}^{(1)}} - K \right)^+.$$

Determine its arbitrage-free price and replicating strategy in the CRR model.

SOLUTION. In the CRR model, defined to ensure no-arbitrage in the model, the risk-neutral probability p is: $p = \frac{e^{r\Delta t} - d}{u - d}$, where u is the upward movement factor, and d is the downward movement factor, satisfying $u > e^{r\Delta t} > d$.

The payoff of the forward-start call option at the final time T is given by $\left(\frac{S(T)}{S(T_0)} - K \right)^+$. To find the arbitrage-free price of the forward-start call option at time T_0 , we need the discounted expected value of the payoff under the risk-neutral probability Q . The price at T_0 is:

$$\mathbb{E}^Q \left[e^{-r(T-T_0)} \left(\frac{S(T)}{S(T_0)} - K \right)^+ \mid \mathcal{F}_{T_0} \right].$$

Let j be the number of upward movements from T_0 to T . In the CRR model, we can express $S(T)$ in terms of $S(T_0)$ as:

$$S(T) = S(T_0) \cdot u^j \cdot d^{(T-T_0-j)}.$$

Therefore, the payoff becomes:

$$\left(u^j d^{(T-T_0-j)} - K \right)^+.$$

To obtain the arbitrage-free price, we calculate the expected value over all possible upward movements j :

$$\mathbb{E}^Q \left[\left(u^j d^{(T-T_0-j)} - K \right)^+ \right] = \sum_{j=0}^{T-T_0} \binom{T-T_0}{j} p^j (1-p)^{T-T_0-j} \cdot \left(u^j d^{(T-T_0-j)} - K \right)^+,$$

where $\binom{T-T_0}{j}$ represents the binomial coefficient, and $p^j (1-p)^{T-T_0-j}$ is the probability of having j upward movements and $T-T_0-j$ downward movements.

After calculating the expected payoff, we discount it to time T_0 to get the arbitrage-free price:

$$\text{Price at } T_0 = e^{-r(T-T_0)} \cdot \sum_{j=0}^{T-T_0} \binom{T-T_0}{j} p^j (1-p)^{T-T_0-j} \cdot \left(u^j d^{(T-T_0-j)} - K \right)^+.$$

To replicate this option in the CRR model, we use a dynamic hedging strategy by adjusting the portfolio at each time step between T_0 and T . Starting from T , we work backward through each node in the binomial tree to T_0 , determining the required holdings of the stock and risk-free bond at each step to replicate the option's payoff.

Next, we make portfolio adjustments, since the option only effectively starts at T_0 , our replication strategy is mainly adjusted from T_0 onwards to ensure that the portfolio value at T matches the option's payoff. This replication ensures that at expiration, the portfolio's value exactly matches the option's payoff.

□

EXERCISE A.14 (One-Period CRR). Assume that $T = 1$, i.e., a one-period CRR model. Suppose we want to determine the time-zero price of the derivative $H = S_1^{(1)}$, i.e., the derivative pays the stock price at time $T = 1$. What is its time-zero price W_0^H according to the risk-neutral pricing formula?

SOLUTION. In one-period CRR Model, S_0 be the current stock price (at time $t = 0$), S_1 be the stock price at time $T = 1$. In the one-period CRR model, the stock price S_1 can either move up to $S_1^u = u \cdot S_0$ or down to $S_1^d = d \cdot S_0$.

In the risk-neutral world, the probability of the stock price going up (denoted p) is chosen so that the expected value of S_1 in this risk-neutral measure matches the stock price compounded at the risk-free rate r . The risk-neutral probability p is given by:

$$p = \frac{e^{r \cdot T} - d}{u - d},$$

where $e^{r \cdot T}$ is the risk-free growth factor, d is the down-factor, and u is the up-factor.

The risk-neutral price of the derivative at time zero, W_H^0 is the discounted expected value of the payoff at time $T = 1$ under the risk-neutral measure. Thus,

$$W_H^0 = e^{-r} \mathbb{E}^Q[H].$$

Expanding this expectation, we get:

$$W_H^0 = e^{-r} (p \cdot S_1^u + (1 - p) \cdot S_1^d) = e^{-r} (p \cdot u \cdot S_0 + (1 - p) \cdot d \cdot S_0) = e^{-r} S_0 (p \cdot u + (1 - p) \cdot d).$$

Since $p = \frac{e^{r \cdot T} - d}{u - d}$, we can simplify $W_H^0 = S_0$.

□

EXERCISE A.15 (Asian Option). Consider the three-period CRR model in Figure A.1 below and take the interest rate $r = 0.25$. What is $D, U, \mathbb{Q}(R_t = U)$ in this case? For $n = 0, 1, 2, 3$ define

$$Y_n = \sum_{k=0}^n S_k^{(1)},$$

the sum of the stock prices between times zero and n . Consider an Asian call option, see Example 1.20, that expires at time three and has strike $K = 4$, i.e., whose payoff at time $T = 3$ is

$$H^{\text{asian}} = \left(\frac{1}{4} Y_3 - 4 \right)^+.$$

Let $W_n^{\text{asian}}(s, y)$ denote the price of this option at time n , if $S_n^{(1)} = s$ and $Y_n = y$. In particular, we have

$$W_3^{\text{asian}}(s, y) = \left(\frac{1}{4} y - 4 \right)^+.$$

- (i) Develop an algorithm for computing W_n^{asian} recursively. In particular, write a formula for W_n^{asian} in terms of W_{n+1}^{asian} .
- (ii) Apply the algorithm developed in (i) to compute $W_0^{\text{asian}}(4, 4)$, the price of the Asian option at time zero.

- (iii) Provide a formula for $\delta_n(s, y)$, the number of shares of stock that should be held by the replicating portfolio at time n if $S_n^{(1)} = s$ and $Y_n = y$.

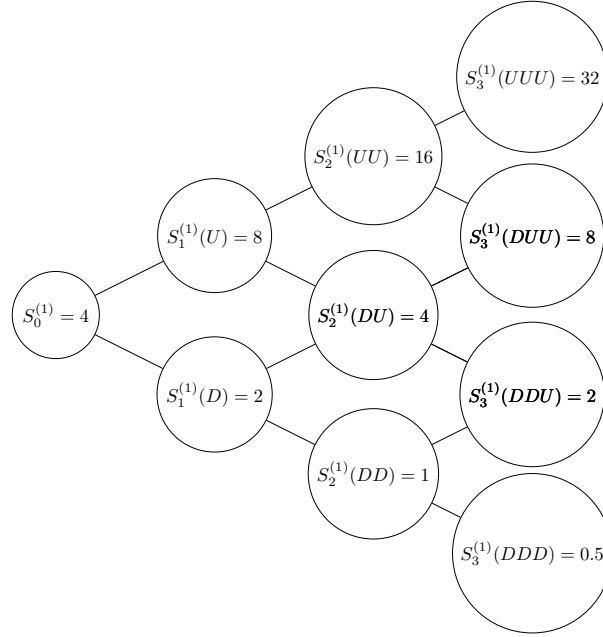


Figure A.1: Three-period binomial asset pricing model.

SOLUTION. (i) From the tree,

$$U = \frac{S_1^{(1)}(U)}{S_0} = \frac{8}{4} = 2, \quad D = \frac{S_1^{(1)}(D)}{S_0} = \frac{2}{4} = 0.5.$$

Hence, with $p := \mathbb{Q}(R_t = U)$,

$$p = \frac{e^r - D}{U - D} = \frac{1.284 - 0.5}{2 - 0.5} \approx 0.5227, \quad 1 - p \approx 0.4773.$$

The backward recursion is

$$W_n^{\text{asian}}(s, y) = e^{-r} \left[p W_{n+1}^{\text{asian}}(sU, y + sU) + (1 - p) W_{n+1}^{\text{asian}}(sD, y + sD) \right].$$

(ii) At maturity,

$$W_3^{\text{asian}}(s, y) = \left(\frac{y}{4} - 4 \right)^+.$$

The terminal values along the six distinct paths are

path	Y_3	$W_3^{\text{asian}}(S_3, Y_3)$
UUU	$4 + 8 + 16 + 32 = 60$	$(15 - 4)^+ = 11$
UUD	$4 + 8 + 16 + 8 = 36$	$(9 - 4)^+ = 5$
UDD	$4 + 8 + 4 + 2 = 18$	$(4.5 - 4)^+ = 0.5$
DUU	$4 + 2 + 4 + 8 = 18$	$(4.5 - 4)^+ = 0.5$
DDU	$4 + 2 + 1 + 2 = 9$	$(2.25 - 4)^+ = 0$
DDD	$4 + 2 + 1 + 0.5 = 7.5$	$(1.875 - 4)^+ = 0$

Applying the recursion gives

$$W_2^{\text{asian}}(16, 28) = 0.79(0.52 \cdot 11 + 0.48 \cdot 0.5) = 4.70,$$

$$W_2^{\text{asian}}(4, 14) = 0.79(0.52 \cdot 0.5 + 0.48 \cdot 0) = 0.20,$$

$$W_2^{\text{asian}}(1, 7) = 0.79(0.52 \cdot 0 + 0.48 \cdot 0) = 0,$$

$$W_1^{\text{asian}}(8, 12) = 0.79(0.52 \cdot 4.70 + 0.48 \cdot 0.20) = 1.97,$$

$$W_1^{\text{asian}}(2, 6) = 0.79(0.52 \cdot 0.20 + 0.48 \cdot 0) = 0.08,$$

$$W_0^{\text{asian}}(4, 4) = 0.79(0.52 \cdot 1.97 + 0.48 \cdot 0.08) = 0.83.$$

(iii) The number of shares in the replicating portfolio is

$$\delta_n(s, y) = \frac{W_{n+1}^{\text{asian}}(sU, y + sU) - W_{n+1}^{\text{asian}}(sD, y + sD)}{s(U - D)}.$$

□

SECTION A.5

| Variance-Optimal Hedging

EXERCISE A.16 (Variance-Optimal Hedge in an Affine GARCH Model). In this exercise we consider a univariate discrete-time stochastic volatility model of GARCH type given as follows: we model the discounted underlying asset price process $(\tilde{S}_t)_{t \in \mathbf{T}}$ as

$$\tilde{S}_t = \tilde{S}_{t-1} \exp \left(-\frac{1}{2} V_t + \sqrt{V_t} z_t^* \right), \quad (\text{A.1})$$

$$V_t = \omega + \beta V_{t-1} + \alpha \left(z_{t-1}^* - \gamma^* \sqrt{V_{t-1}} \right)^2, \quad (\text{A.2})$$

for some suitable parameters ω, α, β and γ such that $V_t \geq 0$ for all $t \in \mathbf{T}$ and where z_t^* is standard normal distributed. The process $(V_t)_{t \in \mathbf{T}}$ is called the *instantaneous variance process* of (the log price of) $\tilde{S}$. We also assume that the discounted asset price process $(\tilde{S}_t)_{t \in \mathbf{T}}$ is square-integrable with positive conditional variance process $(\sigma_t^2)_{t=1,2,\dots,T}$ and we denote by $\tilde{H}$ some discounted square-integrable contingent claim.

- (i) Argue why a variance-optimal strategy (W_0^*, ϕ^*) for $\tilde{H}$ exists and provide an expression of the strategy using [Theorem 2.6](#).
- (ii) Under the additional assumption that $H = f(\tilde{S}_T)$ for some function f , we have an integral representation for $f : \mathbb{C} \rightarrow \mathbb{C}$ of the form

$$f(x) = \int_{R-i\infty}^{R+i\infty} x^u l(u) du,$$

for some function l and $R \in \mathbb{R}$. For instance, the payoff of a European call option can be written as

$$f(x) = (x - K)^+ = \frac{1}{2\pi i} \int_{R-i\infty}^{R+i\infty} x^u \frac{K^{1-u}}{u(u-1)} du.$$

- (a) Assume that $\tilde{H}$ has an integral representation as above. Then show that the derivative prices $\tilde{W}_t^H$ for $t = 0, 1, \dots, T-1$ under some pricing measure $\mathbb{Q}$ and the variance-optimal hedge under the same measure can be expressed using such complex integrals as well.
- (b) Take as a fact that the model (A.1)–(A.2) is *affine*, which means that for any $t \leq T$ and $T \in \mathbf{T}$ the joint moment-generating function $g(t, T, u, v)$ of $(\tilde{S}_t, V_t)$ has the following exponential affine form:

$$g(t, T, u, v) = \mathbb{E}_{\mathbb{Q}} \left[\tilde{S}_T^u \exp(v V_{T+2}) \mid \mathcal{F}_t \right] = \tilde{S}_t^u \exp(A(t, T, u, v) + B(t, T, u, v) V_{t+1}),$$

for two deterministic functions A and B solving some associated difference equations. Use the representation in (a) and this fact to show that

$$\tilde{W}_t^H = \int_{R-i\infty}^{R+i\infty} g(t, T, u, 0) l(u) du,$$

and that the variance-optimal hedge is given by

$$\phi_{t+1}^* = \int_{R-i\infty}^{R+i\infty} \frac{\exp(A(t+1, T, u, 0)) g(t, t+1, u+1, B(t+1, T, u, 0)) - \tilde{S}_t g(t, T, u, 0)}{g(t, t+1, 2, 0) - \tilde{S}_t^2} l(u) du \mathbf{1}_{\{g(t, t+1, 2, 0) - \tilde{S}_t^2 > 0\}}.$$

SOLUTION. (i) Given that the discounted asset price process $(\tilde{S}_t)_{t \in T}$ is square-integrable, we notice that it is a martingale under the risk-neutral measure $\mathbb{Q}$:

$$\mathbb{E}_{\mathbb{Q}}[\tilde{S}_t | \mathcal{F}_{t-1}] = \tilde{S}_{t-1}.$$

By [Proposition 2.3](#), for all $t = 1, \dots, T$, the bounded mean-variance trade-off is

$$\mu_t^2 = \left(\mathbb{E}_{\mathbb{Q}}[\Delta \tilde{S}_t | \mathcal{F}_{t-1}] \right)^2 = 0 < K \sigma_t^2 = K \text{Var}_{\mathbb{Q}}(\Delta \tilde{S}_t | \mathcal{F}_{t-1}),$$

where $K > 0$ is a constant. Then by [Proposition 2.4](#), a variance-optimal strategy $\varphi^* = (W_0^*, \phi^*)$ exists. By [Theorem 2.6](#), the variance-optimal strategy is

$$\phi_t^* := \frac{\mathbb{E}_{\mathbb{Q}}[\Delta \hat{W}_t \Delta \tilde{S}_t | \mathcal{F}_{t-1}]}{\mathbb{E}_{\mathbb{Q}}[(\Delta \tilde{S}_t)^2 | \mathcal{F}_{t-1}]} \mathbb{1}_{\{\mathbb{E}_{\mathbb{P}}[(\Delta \tilde{S}_t)^2 | \mathcal{F}_{t-1}] > 0\}}, \quad t = 1, \dots, T,$$

where $\hat{W}_t = \mathbb{E}_{\mathbb{Q}}[\tilde{H} | \mathcal{F}_t]$ is a square-integrable contingent claim and

$$W_0^* := \hat{W}_0 = \mathbb{E}_{\mathbb{Q}}[\tilde{H}].$$

(ii) (a) Given that $H = f(\tilde{S}_T)$ with

$$f(x) = \int_{R-i\infty}^{R+i\infty} x^u l(u) du,$$

for some function l and $R \in \mathbb{R}$, we write the derivative prices $\tilde{W}_t^H$ and the martingale used in the hedging strategy $\hat{W}_t$ as

$$\begin{aligned} \tilde{W}_t^H &= \hat{W}_t = \mathbb{E}_{\mathbb{Q}}[\tilde{H} | \mathcal{F}_t] \\ &= \mathbb{E}_{\mathbb{Q}}[f(\tilde{S}_T) | \mathcal{F}_t] \\ &= \mathbb{E}_{\mathbb{Q}}\left[\int_{R-i\infty}^{R+i\infty} \tilde{S}_T^u l(u) du | \mathcal{F}_t\right] \\ &= \int_{R-i\infty}^{R+i\infty} \mathbb{E}_{\mathbb{Q}}[\tilde{S}_T^u l(u) | \mathcal{F}_t] du && \text{(Fubini's Theorem)} \\ &= \int_{R-i\infty}^{R+i\infty} \mathbb{E}_{\mathbb{Q}}[\tilde{S}_T^u | \mathcal{F}_t] l(u) du. \end{aligned}$$

And the variance-optimal strategy is

$$\phi_t^* := \frac{\mathbb{E}_{\mathbb{Q}}[(\hat{W}_t - \hat{W}_{t-1}) \Delta \tilde{S}_t | \mathcal{F}_{t-1}]}{\mathbb{E}_{\mathbb{Q}}[(\Delta \tilde{S}_t)^2 | \mathcal{F}_{t-1}]} \mathbb{1}_{\{\mathbb{E}_{\mathbb{P}}[(\Delta \tilde{S}_t)^2 | \mathcal{F}_{t-1}] > 0\}}, \quad t = 1, \dots, T,$$

which can be expressed using the integral representations. Therefore, both the derivative prices $\tilde{W}_t^H$ and the variance-optimal hedge ϕ_t^* under the risk-neutral measure $\mathbb{Q}$ can be expressed using complex integrals.

(b) When $v = 0$, the affine form simplifies to

$$g(t, T, u, 0) = \mathbb{E}_{\mathbb{Q}}[\tilde{S}_T^u | \mathcal{F}_t] = \tilde{S}_t^u \exp(A(t, T, u, 0) + B(t, T, u, 0)V_{t+1}).$$

We aim to compute the variance-optimal hedge:

$$\phi_{t+1}^* := \frac{\mathbb{E}_{\mathbb{Q}} \left[\left(\hat{W}_{t+1} - \hat{W}_t \right) \left(\tilde{S}_{t+1} - \tilde{S}_t \right) \mid \mathcal{F}_t \right]}{\mathbb{E}_{\mathbb{Q}} \left[\left(\tilde{S}_{t+1} - \tilde{S}_t \right)^2 \mid \mathcal{F}_t \right]} \mathbb{1}_{\left\{ \mathbb{E}_{\mathbb{P}} \left[\left(\tilde{S}_{t+1} - \tilde{S}_t \right)^2 \mid \mathcal{F}_t \right] > 0 \right\}}, \quad t = 1, \dots, T. \quad (\text{A.3})$$

The denominator is

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}} \left[\left(\tilde{S}_{t+1} - \tilde{S}_t \right)^2 \mid \mathcal{F}_{t-1} \right] &= \mathbb{E}_{\mathbb{Q}} \left[\tilde{S}_{t+1}^2 \mid \mathcal{F}_{t-1} \right] + \mathbb{E}_{\mathbb{Q}} \left[\tilde{S}_t^2 \mid \mathcal{F}_{t-1} \right] - 2\tilde{S}_t \mathbb{E}_{\mathbb{Q}} \left[\tilde{S}_{t+1} \mid \mathcal{F}_{t-1} \right] \\ &= \mathbb{E}_{\mathbb{Q}} \left[\tilde{S}_{t+1}^2 \mid \mathcal{F}_{t-1} \right] + \tilde{S}_t^2 - 2\tilde{S}_t^2 \\ &= g(t, t+1, 2, 0) - \tilde{S}_t^2. \end{aligned} \quad (\text{A.4})$$

And the numerator is

$$\begin{aligned} &\mathbb{E}_{\mathbb{Q}} \left[\left(\hat{W}_{t+1} - \hat{W}_t \right) \left(\tilde{S}_{t+1} - \tilde{S}_t \right) \mid \mathcal{F}_t \right] \\ &= \mathbb{E}_{\mathbb{Q}} \left[\hat{W}_{t+1} \tilde{S}_{t+1} - \hat{W}_{t+1} \tilde{S}_t - \hat{W}_t \tilde{S}_{t+1} + \hat{W}_t \tilde{S}_t \mid \mathcal{F}_t \right] \\ &= \mathbb{E}_{\mathbb{Q}} \left[\left(\hat{W}_{t+1} \tilde{S}_{t+1} - \hat{W}_t \tilde{S}_t \right) - \left(\hat{W}_{t+1} \tilde{S}_t - \hat{W}_t \tilde{S}_t \right) - \left(\hat{W}_t \tilde{S}_{t+1} - \hat{W}_t \tilde{S}_t \right) \mid \mathcal{F}_t \right] \\ &= \mathbb{E}_{\mathbb{Q}} \left[\hat{W}_{t+1} \tilde{S}_{t+1} - \hat{W}_t \tilde{S}_t \mid \mathcal{F}_t \right] - \hat{W}_t \left(\mathbb{E}_{\mathbb{Q}} \left[\tilde{S}_{t+1} \mid \mathcal{F}_t \right] - \tilde{S}_t \right) - \tilde{S}_t \left(\mathbb{E}_{\mathbb{Q}} \left[\hat{W}_{t+1} \mid \mathcal{F}_t \right] - \hat{W}_t \right) \\ &= \mathbb{E}_{\mathbb{Q}} \left[\hat{W}_{t+1} \tilde{S}_{t+1} \mid \mathcal{F}_t \right] - \hat{W}_t \tilde{S}_t. \end{aligned} \quad (\text{A.5})$$

Given that

$$\hat{W}_t = \tilde{W}_t^H = \int_{R-i\infty}^{R+i\infty} g(t, T, u, 0) l(u) du,$$

we have

$$\hat{W}_t \tilde{S}_t = \tilde{S}_t \int_{R-i\infty}^{R+i\infty} g(t, T, u, 0) l(u) du = \int_{R-i\infty}^{R+i\infty} \tilde{S}_t g(t, T, u, 0) l(u) du, \quad (\text{A.6})$$

and

$$\begin{aligned} &\mathbb{E}_{\mathbb{Q}} \left[\hat{W}_{t+1} \tilde{S}_{t+1} \mid \mathcal{F}_t \right] \\ &= \mathbb{E}_{\mathbb{Q}} \left[\tilde{S}_{t+1} \int_{R-i\infty}^{R+i\infty} g(t+1, T, u, 0) l(u) du \mid \mathcal{F}_t \right] \\ &= \mathbb{E}_{\mathbb{Q}} \left[\int_{R-i\infty}^{R+i\infty} \tilde{S}_{t+1} \tilde{S}_{t+1}^u \exp(A(t+1, T, u, 0) + B(t+1, T, u, 0) V_{t+2}) \mid \mathcal{F}_t \right] \\ &= \exp(A(t+1, T, u, 0)) \mathbb{E}_{\mathbb{Q}} \left[\int_{R-i\infty}^{R+i\infty} \tilde{S}_{t+1}^{u+1} \exp \left(\underbrace{B(t+1, T, u, 0) V_{t+2}}_{:=\mathcal{B}} \right) \mid \mathcal{F}_t \right] \\ &= \exp(A(t+1, T, u, 0)) \int_{R-i\infty}^{R+i\infty} \tilde{S}_t^{u+1} \exp(A(t, t+1, u+1, \mathcal{B}) + B(t, t+1, u+1, \mathcal{B}) V_{t+1}) \\ &= \exp(A(t+1, T, u, 0)) \int_{R-i\infty}^{R+i\infty} g(t, t+1, u+1, \mathcal{B}) \\ &= \int_{R-i\infty}^{R+i\infty} \exp(A(t+1, T, u, 0)) g(t, t+1, u+1, B(t+1, T, u, 0)). \end{aligned} \quad (\text{A.7})$$

Combining results (A.3)-(A.7), we obtain the expression of the variance-optimal hedge:

$$\phi_{t+1}^* = \int_{R-i\infty}^{R+i\infty} \frac{\exp(A(t+1, T, u, 0)) g(t, t+1, u+1, B(t+1, T, u, 0)) - \tilde{S}_t g(t, T, u, 0)}{g(t, t+1, 2, 0) - \tilde{S}_t^2} l(u) du \mathbb{1}_{\{g(t, t+1, 2, 0) - \tilde{S}_t^2 > 0\}}.$$

□

In the next exercise we construct an example of a financial market, where the bounded mean–variance trade-off condition is not satisfied and where the subspace $\mathcal{G}_T$ is indeed *not* closed.

EXERCISE A.17 (Counterexample for Closedness of the Space $\mathcal{G}_T$). Let $\Omega = [0, 1] \times \{-1, +1\}$ with its Borel σ -algebra $\mathcal{F}$. Outcomes are denoted by $\omega = (u, v)$ with $u \in [0, 1], v \in \{-1, +1\}$, and we define $U(\omega) = u$ the first and by $V(\omega) = v$ the second coordinate. Let $\mathcal{F}_0 = \mathcal{F}_1 = \sigma(U)$, $\mathcal{F}_2 = \mathcal{F}$ and let $\mathbb{P}$ be the measure on $(\Omega, \mathcal{F})$ such that U is distributed uniformly on $[0, 1]$ and the conditional distribution of V given U is $U^2\delta_{\{+1\}} + (1 - U^2)\delta_{\{-1\}}$. Let $X_0 = 0, \Delta X_1 = 1$ and

$$\Delta X_2 = V^+(1 + U) - 1 = V^+U - V^-,$$

so that

$$\Delta X_2(u, v) = u\delta_{\{v=+1\}} - \delta_{\{v=-1\}}.$$

Consider now the contingent claim $H = \frac{1}{U}V^+(1 + U)$.

(i) Show that $H \in L^2(\Omega, \mathcal{F}, \mathbb{P})$.

(ii) Let ϕ be a predictable process with terminal gain satisfying $G_2(\phi) = H$ $\mathbb{P}$ -almost surely. Show that

$$\frac{1}{U}V^+(1 + U) = H = \phi_1\Delta X_1 + \phi_2\Delta X_2 = \phi_1 + \phi_2(V^+(1 + U) - 1),$$

implies that $\phi_1 = \phi_2 = U^{-1}$ $\mathbb{P}$ -almost surely.

(iii) Show that ϕ is not in $\mathcal{S}^2$ and that therefore H is not in $\mathcal{G}_2$.

(iv) Next, set

$$\phi^n := \phi \mathbb{1}_{\{U \geq \frac{1}{n}\}} = U^{-1} \mathbb{1}_{\{U \geq \frac{1}{n}\}}, \quad (\text{A.8})$$

and show that $\phi^n \in \mathcal{S}^2$ for every $n \in \mathbb{N}$ and that

$$G_2(\phi^n) = U^{-1}V^+(1 + U) \mathbb{1}_{\{U \geq \frac{1}{n}\}} = H \mathbb{1}_{\{U \geq \frac{1}{n}\}}, \quad (\text{A.9})$$

converges to H in $L^2(\Omega, \mathcal{F}, \mathbb{P})$.

(v) Part i)-iv) shows that the space $\mathcal{G}_2$ is not closed in $L^2(\Omega, \mathcal{F}, \mathbb{P})$, so the variance optimization problem for H does not have a solution. To conclude this example, show that X as constructed above does not satisfy the bounded mean–variance trade-off condition.

SOLUTION. (i) Given that

$$H(\omega) = \frac{1}{U(\omega)}V^+(\omega)(1 + U(\omega)),$$

where $U(\omega) = u \in [0, 1]$ and $V(\omega) = v \in \{-1, +1\}$, we have $H(u, -1) = 0$ and $H(u, +1) = \frac{1+u}{u}$, then

$$\begin{aligned} \mathbb{E}[H^2] &= \mathbb{E}[H^2(u, V) \mid U = u] du \\ &= \mathbb{E}[H^2(u, 1) \cdot u^2 + H^2(u, -1) \cdot (1 - u^2)] du \\ &= \int_0^1 \left(\frac{1+u}{u} \right)^2 u^2 du \\ &= \int_0^1 (1+u)^2 du = \frac{7}{3} < \infty. \end{aligned}$$

Therefore we have shown that $H \in L^2(\Omega, \mathcal{F}, \mathbb{P})$.

(ii) Given that $\Delta X_1 = 1$ and $\Delta X_2 = V^+(1 + U) - 1$, we can write

$$H = \phi_1 \Delta X_1 + \phi_2 \Delta X_2 = \phi_1 + \phi_2(V^+(1 + U) - 1).$$

And we also have

$$H = \frac{1}{U} V^+(1 + U).$$

By solving

$$\begin{cases} \frac{1+U}{U} = \phi_1 + \phi_2 U, & V = 1 \implies V^+ = 1 \\ 0 = \phi_1 - \phi_2, & V = -1 \implies V^+ = 0 \end{cases}'$$

we obtain that $\phi_1 = \phi_2 = U^{-1}$, $\mathbb{P}$ -a.s.

(iii) By lecture notes, the subspace is defined by

$$\mathcal{G}_2 := \{G_2(\phi) : \phi \in \mathcal{S}^2\} \subset \mathcal{L}^2(\Omega, \mathcal{F}_T, \mathbb{P}).$$

From (ii), $\phi_1 = \phi_2 = U^{-1}$. By computing

$$\mathbb{E}[\phi^2] = \int_0^1 \left(\frac{1}{u}\right) du = \infty,$$

we claim that $\phi \notin \mathcal{S}^2$. Then $H \notin \mathcal{G}_2$ follows since $G_2(\phi) = H$ but $\phi \notin \mathcal{S}^2$.

(iv) Given that

$$\phi^n = U^{-1} \mathbb{1}_{\{U \geq \frac{1}{n}\}},$$

we compute

$$\mathbb{E}[(\phi^n)^2] = \int_0^1 \frac{1}{u^2} \mathbb{1}_{\{u \geq \frac{1}{n}\}} du = \int_{\frac{1}{n}}^1 \frac{1}{u^2} du = n - 1 < \infty,$$

then $\phi^n \in \mathcal{S}^2$ for every $n \in \mathbb{N}$.

For the second part, since

$$G_2(\phi^n) = H \mathbb{1}_{\{U \geq \frac{1}{n}\}},$$

we verify that

$$\begin{aligned} \mathbb{E}[(G_2(\phi^n) - H)^2] &= \mathbb{E}\left[\left(H \mathbb{1}_{\{U \geq \frac{1}{n}\}} - H\right)^2\right] \\ &= \mathbb{E}\left[\left(-H \mathbb{1}_{\{U \leq \frac{1}{n}\}}\right)^2\right] \\ &= \mathbb{E}\left[H^2 \mathbb{1}_{\{U \leq \frac{1}{n}\}}\right] \\ &= \int_0^{\frac{1}{n}} (1+u)^2 du \\ &\quad (\mathbb{E}[H^2 | U = u] = (1+u)^2 \text{ by part (i)}) \\ &= \frac{1}{n} + \frac{1}{n^2} + \frac{1}{3n^3} \rightarrow 0 \text{ as } n \rightarrow \infty. \end{aligned}$$

Therefore $G_2(\phi^n)$ converges to H in $\mathcal{L}^2(\Omega, \mathcal{F}, \mathbb{P})$.

(v) Given that $U = u$ and

$$\Delta X_2 = V^+(1 + U) - 1 = \begin{cases} u, & V = 1 \\ -1, & V = -1 \end{cases},$$

we compute

$$\mu_2 = \mathbb{E}[\Delta X_2 | U = u] = u \cdot u^2 + (-1)(1 - u^2) = u^3 + u^2 - 1,$$

and

$$\sigma_2^2 = \mathbb{E}[\Delta X_2^2 | U = u] - \mu_2^2 = u^2 \cdot u^2 + (1 - u^2) - \mu_2^2 = -u^6 - 2u^5 + 2u^3 + u^2.$$

Then

$$\frac{\mu_2^2}{\sigma_2^2} = \frac{u^6 + 2u^5 + u^4 - 2u^3 - 2u^2 + 1}{-u^6 - 2u^5 + 2u^3 + u^2} \rightarrow \infty$$

as $u \rightarrow 0$. Since for all $u \in [0, 1]$, $\mu_t^2 \leq K\sigma_t^2$ should be satisfied, and we have presented a contradiction. We conclude that X as constructed above does not satisfy the bounded mean-variance trade-off condition.

□

SECTION A.6

| The Semi-Static Variance-Optimal Hedging Problem

Consider the following extension of variance-optimal hedging, called *semi-static* variance-optimal hedging. The idea is, that in addition to the contingent claim H^0 which is to be hedged, we denote by $H = (H^1, \dots, H^n)^\top$ the vector of supplementary contingent claims, all assumed to be square-integrable random variables in $L^2(\Omega, \mathcal{F}_T, \mathbb{Q})$. Again, we associate to each H^i the martingale

$$H_t^i := \mathbb{E}[H^i | \mathcal{F}_t], \quad t = 0, 1, \dots, T, \quad i = 0, \dots, n. \quad (\text{A.10})$$

The static part of the strategy can be represented by an element v of $\mathbb{R}^n$, where v_i represents the quantity of claim H^i bought at time $t = 0$ and held until time $t = T$. The dynamic part ϑ of the strategy is again represented by an element of $\mathcal{S}^2$, the space of square-integrable predictable processes with respect to the price process S .

The variance-optimal semi-static hedge $(\vartheta, v) \in \mathcal{S}^2 \times \mathbb{R}^n$ and the optimal initial capital $c \in \mathbb{R}$ are the solution of the minimization problem

$$\epsilon^2 = \min_{(\vartheta, v) \in \mathcal{S}^2 \times \mathbb{R}^n, c \in \mathbb{R}} \mathbb{E} \left[\left(c - v^\top \mathbb{E}[H_T] + \sum_{t=1}^T \vartheta_t \Delta S_t - (H_T^0 - v^\top H_T) \right)^2 \right]. \quad (\text{A.11})$$

Note that $v^\top \mathbb{E}[H_T]$ is the cost of setting up the static part of the hedge, and its terminal value is $v^\top H_T$. The dynamic part is self-financing and results in the terminal value $\sum_{t=1}^T \vartheta_t \Delta S_t$. Adding the initial capital c and subtracting the target claim H_T^0 yields the above expression for the hedging problem.

To solve the variance-optimal semi-static hedging problem, we decompose it into an inner and an outer minimization problem and rewrite (A.11) as

$$\begin{cases} \epsilon^2(v) = \min_{\vartheta \in \mathcal{S}^2, c \in \mathbb{R}} \mathbb{E} \left[\left(c - v^\top \mathbb{E}[H_T] + \sum_{t=1}^T \vartheta_t \Delta S_t - (H_T^0 - v^\top H_T) \right)^2 \right], & \text{(inner problem)} \\ \epsilon^2 = \min_{v \in \mathbb{R}^n} \epsilon^2(v). & \text{(outer problem)} \end{cases} \quad (\text{A.12})$$

The inner problem is of the same form as the variance-optimal hedging problem without supplementary assets, while the outer problem turns out to be a finite-dimensional quadratic optimization problem. To formulate the solution, we write the Kunita–Watanabe decompositions of the claims $(H^0, \dots, H^n)$ with respect to S as

$$H_t^i = H_0^i + \sum_{s=1}^t \vartheta_s^i \Delta S_s + L_t^i, \quad t = 0, 1, \dots, T, \quad i = 0, \dots, n. \quad (\text{A.13})$$

As in the classical variance optimal hedging problem, we obtain the solution:

$$\vartheta_t^i = \frac{\mathbb{E}[\Delta H_t^i \Delta S_t | \mathcal{F}_{t-1}]}{\mathbb{E}[(\Delta S_t)^2 | \mathcal{F}_{t-1}]} \mathbb{1}_{\{\mathbb{E}[(\Delta S_t)^2 | \mathcal{F}_{t-1}] \neq 0\}}, \quad t = 1, \dots, T, \quad i = 0, \dots, n. \quad (\text{A.14})$$

We introduce the vector notation $\vartheta = (\vartheta^1, \dots, \vartheta^n)^\top$ for the strategies and $L = (L^1, \dots, L^n)^\top$ for the residuals in the Kunita–Watanabe decomposition.

EXERCISE A.18 (Semi-Static Variance-Optimal Hedging). Consider the variance-optimal semi-static hedging problem (A.11) and set

$$A := \text{Var} [L_T^0], \quad B := \text{Cov} [L_T, L_T^0], \quad C := \text{Cov} [L_T, L_T]. \quad (\text{A.15})$$

Assume that C is invertible. Show that the unique solution of the semi-static hedging problem is given by

$$c = \mathbb{E}[H_T^0], \quad v = C^{-1}B, \quad \vartheta_t^v = \vartheta_t^0 - v^\top \vartheta_t, \quad t = 1, \dots, T,$$

and that the minimal squared hedging error is given by

$$\epsilon^2 = A - B^\top C^{-1}B.$$

Moreover, show that the elements of A , B , and C can be expressed as

$$\mathbb{E}[L_T^i L_T^j] = \mathbb{E} \left[\sum_{t=1}^T \text{Var} (\Delta H_t^i | \mathcal{F}_{t-1}) - \sum_{t=1}^T \vartheta_t^i \vartheta_t^j \text{Var} (\Delta S_t | \mathcal{F}_{t-1}) \right], \quad i, j = 0, \dots, n.$$

SOLUTION. To minimise the objective function (A.11), we have to choose the initial capital c such that the expectation of the squared error is minimised. The squared error can be expressed as:

$$\begin{aligned} & \mathbb{E} \left[\left(c - v^\top \mathbb{E}[H_T] + \sum_{t=1}^T \vartheta_t \Delta S_t - (H_T^0 - v^\top H_T) \right)^2 \right] \\ &= \mathbb{E} \left[\left(c - \mathbb{E}[H_T^0] - H_T^0 + \mathbb{E}[H_T^0] - v^\top \mathbb{E}[H_T] + \sum_{t=1}^T \vartheta_t \Delta S_t + v^\top H_T \right)^2 \right] \\ &= \mathbb{E} \left[\left((c - \mathbb{E}[H_T^0]) - (H_T^0 - \mathbb{E}[H_T^0]) + v^\top (H_T - \mathbb{E}[H_T]) + \sum_{t=1}^T \vartheta_t \Delta S_t \right)^2 \right], \end{aligned}$$

where $c = \mathbb{E}[H_T^0]$ ensures the expected error term due to c is zero, which is standard in variance-optimal hedging.

For the optimal static hedging strategy v : the total hedging error at maturity T is:

$$\text{Error} = L_T^0 - v^\top L_T.$$

The variance of the total hedging error is:

$$\text{Var}(\text{Error}) = \text{Var}(L_T^0 - v^\top L_T) = \text{Var}(L_T^0) - 2v^\top \text{Cov}(L_T, L_T^0) + v^\top \text{Var}(L_T)v.$$

Substituting the definitions of A, B, C , we get:

$$\text{Var}(\text{Error}) = A - 2v^\top B + v^\top C v.$$

We need to minimize the function:

$$\begin{aligned} f(v) &= A - 2v^\top B + v^\top C v \\ \frac{\partial f}{\partial v} &= -2B + 2Cv = 0 \\ v &= C^{-1}B. \end{aligned}$$

For the hedging strategy: for each claim H_t^i the Kunita–Watanabe decomposition with respect to S gives:

$$H_t^i = H_0^i + \sum_{s=1}^t \vartheta_s^i \Delta S_s + L_t^i, \quad t = 0, \dots, T, \quad i = 0, \dots, n,$$

where ϑ_t^i is the optimal dynamic strategy for hedging H_i with respect to S , L_t^i is a residual term orthogonal to S .

For the primary claim H_0 , the decomposition is:

$$H_T^0 = H_0^0 + \sum_{t=1}^T \vartheta_t^0 \Delta S_t + L_T^0.$$

For each auxiliary claim H_i , we have:

$$H_T^i = H_0^i + \sum_{t=1}^T \vartheta_t^i \Delta S_t + L_T^i.$$

The hedging error at time T can be rewritten as:

$$\text{Error} = c - v^\top \mathbb{E}[H_T] + \sum_{t=1}^T \vartheta_t \Delta S_t - H_T^0 + v^\top H_T.$$

Substitute the Kunita–Watanabe decomposition for H_T^0 and H_T , the hedging error becomes:

$$\text{Error} = (c - v^\top \mathbb{E}[H_T]) + \sum_{t=1}^T (\vartheta_t - \vartheta_t^0 + v^\top \vartheta_t) \Delta S_t + (L_T^0 - v^\top L_T).$$

To minimize the hedging error variance, we focus on $\sum_{t=1}^T (\vartheta_t - \vartheta_t^0 + v^\top \vartheta_t) \Delta S_t$, $(\vartheta_t - \vartheta_t^0 + v^\top \vartheta_t) = 0$. Thus, the optimal dynamic hedging strategy becomes:

$$\vartheta_t^v = \vartheta_t^0 - v^\top \vartheta_t, \quad t = 1, \dots, T.$$

Substitute $v = C^{-1}B$ into $f(v)$:

$$\epsilon^2 = f(C^{-1}B) = A - 2(C^{-1}B)^\top B + (C^{-1}B)^\top C(C^{-1}B) = A - 2B^\top C^{-1}B + B^\top C^{-1}B = A - B^\top C^{-1}B.$$

Since $H_T^i = H_0^i + \sum_{t=1}^T \vartheta_t^i \Delta S_t + L_T^i$, we have:

$$\text{Var}(H_T^i | \mathcal{F}_{t-1}) = \text{Var}\left(\sum_{t=1}^T \vartheta_t^i \Delta S_t + L_T^i | \mathcal{F}_{t-1}\right),$$

Because L_T^i is orthogonal to S , the covariance of L_T^i and L_T^j is:

$$\begin{aligned} \mathbb{E}[L_T^i L_T^j] &= \text{Var}(H_T^i) - \text{Cov}\left(\sum_{t=1}^T \vartheta_t^i \Delta S_t, \sum_{t=1}^T \vartheta_t^j \Delta S_t\right) \\ &= \mathbb{E}[\text{Var}(H_T^i | \mathcal{F}_{t-1})] - \mathbb{E}\left[\text{Cov}\left(\sum_{t=1}^T \vartheta_t^i \Delta S_t, \sum_{t=1}^T \vartheta_t^j \Delta S_t | \mathcal{F}_{t-1}\right)\right]. \end{aligned}$$

The covariance of the hedging components:

$$\mathbb{E}\left[\text{Cov}\left(\sum_{t=1}^T \vartheta_t^i \Delta S_t, \sum_{t=1}^T \vartheta_t^j \Delta S_t | \mathcal{F}_{t-1}\right)\right] = \mathbb{E}\left[\sum_{t=1}^T \vartheta_t^i \vartheta_t^j \text{Var}(\Delta S_t | \mathcal{F}_{t-1})\right].$$

So we can get:

$$\mathbb{E}[L_T^i L_T^j] = \mathbb{E}\left[\sum_{t=1}^T \text{Var}(\Delta H_t^i | \mathcal{F}_{t-1}) - \sum_{t=1}^T \vartheta_t^i \vartheta_t^j \text{Var}(\Delta S_t | \mathcal{F}_{t-1})\right].$$

For $A = \text{Var}(L_T^0)$

$$A = \mathbb{E}\left[\sum_{t=1}^T \text{Var}(\Delta H_t^0 | \mathcal{F}_{t-1}) - \sum_{t=1}^T (\vartheta_t^0)^2 \text{Var}(\Delta S_t | \mathcal{F}_{t-1})\right].$$

For $B = \text{Cov}(L_T, L_T^0)$

$$B = \mathbb{E} \left[\sum_{t=1}^T \text{Cov}(\Delta H_t, \Delta H_t^0 \mid \mathcal{F}_{t-1}) - \sum_{t=1}^T \vartheta_t \vartheta_t^0 \text{Var}(\Delta S_t \mid \mathcal{F}_{t-1}) \right].$$

For $C = \text{Cov}(L_T, L_T)$

$$C = \mathbb{E} \left[\sum_{t=1}^T \text{Cov}(\Delta H_t, \Delta H_t \mid \mathcal{F}_{t-1}) - \sum_{t=1}^T \vartheta_t \vartheta_t^\top \text{Var}(\Delta S_t \mid \mathcal{F}_{t-1}) \right].$$

□

SECTION A.7

| Risk Measures

EXERCISE A.19 (Axiomatic Description of Risk). Let ρ be a normalised risk measure on $\mathcal{X}$. Show that any two of the following properties imply the remaining third.

- (i) *Convexity*, i.e., $\rho(\lambda X + (1 - \lambda)Y) \leq \lambda\rho(X) + (1 - \lambda)\rho(Y)$ for $\lambda \in [0, 1]$ and $X, Y \in \mathcal{X}$;
- (ii) *Positive Homogeneity*, i.e., $\rho(\lambda X) = \lambda\rho(X)$ for all $\lambda \geq 0$ and $X \in \mathcal{X}$;
- (iii) *Subadditivity*, i.e., $\rho(X + Y) \leq \rho(X) + \rho(Y)$ for $X, Y \in \mathcal{X}$.

SOLUTION. (i) & (ii) $\implies$ (iii): By convexity with $\lambda = \frac{1}{2}$ we have

$$\rho\left(\frac{1}{2}X + \frac{1}{2}Y\right) \leq \frac{1}{2}\rho(X) + \frac{1}{2}\rho(Y). \quad (\text{A.16})$$

Then

$$\rho(X + Y) = \rho\left(2\left(\frac{1}{2}X + \frac{1}{2}Y\right)\right) \stackrel{\text{(ii)}}{=} 2\rho\left(\frac{1}{2}X + \frac{1}{2}Y\right) \stackrel{(\text{A.16})}{\leq} 2\left(\frac{1}{2}\rho(X) + \frac{1}{2}\rho(Y)\right) = \rho(X) + \rho(Y).$$

(i) & (iii) $\implies$ (ii): By convexity with $Y = 0$ we have

$$\rho(\lambda X) \leq \lambda\rho(X) + (1 - \lambda)\rho(0) = \lambda\rho(X), \quad (\text{A.17})$$

where $\rho(0) = 0$ since ρ is a normalised risk measure on $\mathcal{X}$. Similarly we have

$$\rho((1 - \lambda)Y) \leq \lambda\rho(0) + (1 - \lambda)\rho(Y) = (1 - \lambda)\rho(Y). \quad (\text{A.18})$$

Then

$$\rho(X) = \rho(\lambda X + (1 - \lambda)X) \stackrel{\text{(iii)}}{\leq} \rho(\lambda X) + \rho((1 - \lambda)X) \stackrel{(\text{A.18})}{\leq} \rho(\lambda X) + (1 - \lambda)\rho(X),$$

which implies

$$\lambda\rho(X) \leq \rho(\lambda X) \stackrel{(\text{A.17})}{\leq} \lambda\rho(X).$$

Therefore, $\rho(\lambda X) = \lambda\rho(X)$.

(ii) & (iii) $\implies$ (i): Using subadditivity and positive homogeneity we have

$$\rho(\lambda X + (1 - \lambda)Y) \stackrel{\text{(iii)}}{\leq} \rho(\lambda X) + \rho((1 - \lambda)Y) \stackrel{\text{(ii)}}{=} \lambda\rho(X) + (1 - \lambda)\rho(Y).$$

□

EXERCISE A.20 (Value-at-Risk). Consider the Value-at-Risk risk measure VaR_λ from Example 4.9. This is not a convex risk measure, but it can be transformed to one as follows.

We define the *Average Value at Risk* at level $\lambda \in (0, 1]$ of a financial position $X \in \mathcal{X}$ as

$$\text{AVaR}_\lambda(X) := \frac{1}{\lambda} \int_0^\lambda \text{VaR}_\alpha(X) d\alpha.$$

The average value at risk $\text{AVaR}_\lambda(X)$ has the following representation, that you don't have to prove:

$$\text{AVaR}_\lambda(X) = \frac{1}{\lambda} \mathbb{E}_{\mathbb{P}}[(q - X)^+] - q = \frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\mathbb{E}_{\mathbb{P}}[(r - X)^+] - \lambda r),$$

for any λ -quantile q of X . Use this and Exercise 4.1 to show that the average value at risk is a coherent risk measure on $\mathcal{X}$.

SOLUTION.

LEMMA A.1.

For any real numbers a, b , and any $\alpha \in [0, 1]$, we have

$$(\alpha a + (1 - \alpha)b)^+ \leq (\alpha a)^+ + ((1 - \alpha)b)^+.$$

Proof. If $\alpha a + (1 - \alpha)b < 0$, then

$$(\alpha a + (1 - \alpha)b)^+ = 0 \leq (\alpha a)^+ + ((1 - \alpha)b)^+.$$

If $\alpha a + (1 - \alpha)b \geq 0$, then

$$(\alpha a + (1 - \alpha)b)^+ = \alpha a + (1 - \alpha)b \leq (\alpha a)^+ + ((1 - \alpha)b)^+,$$

where $a \leq a^+$ and $b \leq b^+$. □

- *Convexity:* For any $\alpha \in [0, 1]$, by the representation

$$\text{AVaR}_\lambda(X) = \frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\mathbb{E}_{\mathbb{P}}[(r - X)^+] - \lambda r),$$

we have

$$\begin{aligned} \text{AVaR}_\lambda(\alpha X + (1 - \alpha)Y) &= \frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\mathbb{E}_{\mathbb{P}}[(r - (\alpha X + (1 - \alpha)Y))^+] - \lambda r) \\ &= \frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\mathbb{E}_{\mathbb{P}}[(\alpha r + (1 - \alpha)r - \alpha X - (1 - \alpha)Y)^+] - \lambda r) \\ &\leq \frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\mathbb{E}_{\mathbb{P}}[\alpha(r - X)^+ + (1 - \alpha)(r - Y)^+] - \lambda r) \quad (\text{Lemma A.1}) \\ &= \frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\alpha \mathbb{E}_{\mathbb{P}}[(r - X)^+] + (1 - \alpha) \mathbb{E}_{\mathbb{P}}[(r - Y)^+] - \lambda r) \\ &\leq \alpha \frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\mathbb{E}_{\mathbb{P}}[(r - X)^+] - \lambda r) + (1 - \alpha) \frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\mathbb{E}_{\mathbb{P}}[(r - Y)^+] - \lambda r) \\ &\quad (\lambda r = \alpha \lambda r + (1 - \alpha) \lambda r) \\ &= \alpha \text{AVaR}_\lambda(X) + (1 - \alpha) \text{AVaR}_\lambda(Y). \end{aligned}$$

- *Positive homogeneity:* Let $r' = \gamma r$, then for all $\gamma \geq 0$ we have

$$\begin{aligned} \text{AVaR}_\lambda(\gamma X) &= \frac{1}{\lambda} \inf_{r' \in \mathbb{R}} (\mathbb{E}_{\mathbb{P}}[(r' - \gamma X)^+] - \lambda r') \\ &= \frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\mathbb{E}_{\mathbb{P}}[\gamma(r - X)^+] - \lambda \gamma r) \\ &= \frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\gamma \mathbb{E}_{\mathbb{P}}[(r - X)^+] - \gamma \lambda r) \\ &= \gamma \left(\frac{1}{\lambda} \inf_{r \in \mathbb{R}} (\mathbb{E}_{\mathbb{P}}[(r - X)^+] - \lambda r) \right) = \gamma \text{AVaR}_\lambda(X). \end{aligned}$$

- *Subadditivity:* From Exercise 4.1, we know that convexity and positive homogeneity imply subadditivity, it follows that

$$\text{AVaR}_\lambda(X + Y) \leq \text{AVaR}_\lambda(X) + \text{AVaR}_\lambda(Y).$$

In conclusion, we have shown that the average value at risk is a coherent risk measure on $\mathcal{X}$. □

EXERCISE A.21 (Gini's Mean Difference). For any $X \in L_1$ define $\Delta(X) := \mathbb{E}[|X - \tilde{X}|]$, where $\tilde{X}$ is an independent copy of X . The function $\Delta(\cdot)$ is often called *Gini's mean difference* and satisfies $\Delta(X) \in [0, 2\mathbb{E}[|X|]]$, with $\Delta(X) = 0$ if and only if X is $\mathbb{P}$ -a.s. constant. In analogy to the mean–standard deviation risk measure in *Example 4.13*, consider the functional on L_1 defined by

$$\rho_\lambda(X) := \mathbb{E}[-X] + \lambda\Delta(X), \quad \text{for some } \lambda \geq 0.$$

- (i) Show that $\rho_\lambda(X)$ is a coherent risk measure for any $\lambda \in [0, \frac{1}{2}]$. Note that a non-constant position X is acceptable if and only if $\mathbb{E}[X] > 0$ and the Gini coefficient of X , defined as

$$G(X) := \frac{\Delta(X)}{2\mathbb{E}[X]},$$

satisfies $G(X) \geq (2\lambda)^{-1}$.

- (ii) Show that for $\lambda > \frac{1}{2}$ there are nonnegative random variables X such that $\rho_\lambda(X) > 0$. In particular, ρ_λ cannot be monotone for $\lambda > \frac{1}{2}$.
 (iii) Show that

$$\Delta(X) = 2 \int_{-\infty}^{\infty} F(x)(1 - F(x))dx,$$

where F denotes the distribution function of X , and that

$$\Delta(X) = 4\text{cov}(X, F(X))$$

whenever X has a continuous distribution.

- (iv) Use the above equation to show that $\Delta(X) \leq \frac{2\sigma(X)}{\sqrt{3}}$ for all $X \in L_2$ and conclude that ρ_λ is dominated by the mean–standard deviation risk measure $\rho_{RC}(X) = -\mathbb{E}[X] + c\sigma(X)$ if $\lambda \leq \frac{c\sqrt{3}}{2}$. As in *Example 4.9*, we denote here by $\sigma(X)$ the square root of the variance of X .
 (v) Compute $\Delta(X)$ and $G(X)$ if X has a Pareto distribution with shape parameter $\alpha > 1$ and minimum 1, that is, $\log X$ is exponentially distributed with parameter α .
 (vi) Consider a log-normally distributed random variable $X = \exp(m + \sigma Z)$, where Z has a standard normal law $N(0, 1)$. Show that $G(X) = \text{erf}(\frac{\sigma}{2})$, where $\text{erf}(\cdot)$ is the Gaussian error function, that is,

$$\text{erf}(z) := \mathbb{P}[|Z| \leq z\sqrt{2}] = 2\Phi(z\sqrt{2}) - 1.$$

SOLUTION. (i) – *Monotonicity*: For all $X, Y \in \mathcal{X}$, if $X \leq Y$ a.s., then $\mathbb{E}[-X] \geq \mathbb{E}[-Y]$. $\Delta(X)$ depends on the distribution of X , $\Delta(X) \leq 2\mathbb{E}[|X|]$. Since $\mathbb{E}[|X|] \leq \mathbb{E}[|Y|]$, we have:

$$\Delta(X) \leq 2\mathbb{E}[|X|] \leq 2\mathbb{E}[|Y|], \quad \lambda\Delta(X) \leq 2\lambda\mathbb{E}[|X|] \leq 2\lambda\mathbb{E}[|Y|].$$

When $\lambda \in [0, \frac{1}{2}]$, the increase in $\Delta(X)$ is controlled,

$$\begin{aligned} \lambda\Delta(X) &\leq \mathbb{E}[|X|] \leq \mathbb{E}[|Y|] \\ \lambda\Delta(Y) &\leq \mathbb{E}[|Y|] \\ -\lambda\Delta(Y) &\geq -\mathbb{E}[|Y|] \\ \lambda\Delta(X) - \lambda\Delta(Y) &\leq \mathbb{E}[|X|] - \mathbb{E}[|Y|] \\ \rho_\lambda(X) &\geq \rho_\lambda(Y). \end{aligned}$$

– *Cash Invariance*: For all $m \in \mathbb{R}$, we have

$$\rho_\lambda(X + m) = \mathbb{E}[-(X + m)] + \lambda\Delta(X + m)$$

$$\begin{aligned}
&= \mathbb{E}[-X] + m + \lambda \mathbb{E}[|X + m - (\tilde{X} + m)|] \\
&= \mathbb{E}[-X] + m + \lambda \mathbb{E}[|X - \tilde{X}|] \\
&= \mathbb{E}[-X] + m + \lambda \Delta(X) = \rho_\lambda(X) + m.
\end{aligned}$$

- *Positive homogeneity*: For any $\mu \geq 0$, we have

$$\rho_\lambda(\mu X) = \mathbb{E}[-\mu X] + \lambda \Delta(\mu X) = \mu \mathbb{E}[-X] + \lambda \mathbb{E}[|\mu X - \mu \tilde{X}|] = \mu \mathbb{E}[-X] + \mu \lambda \Delta(X) = \mu \rho_\lambda(X).$$

- *Subadditivity*: For any $X, Y \in L_1$, we have

$$\begin{aligned}
\rho_\lambda(X + Y) &= \mathbb{E}[-(X + Y)] + \lambda \Delta(X + Y) \\
&= \mathbb{E}[-X] + \mathbb{E}[-Y] + \lambda \mathbb{E}[|X + Y - (\tilde{X} + \tilde{Y})|] \\
&\leq \mathbb{E}[-X] + \mathbb{E}[-Y] + \lambda (\mathbb{E}[|X - \tilde{X}|] + \mathbb{E}[|Y - \tilde{Y}|]) \\
&= \mathbb{E}[-X] + \mathbb{E}[-Y] + \lambda \Delta(X) + \lambda \Delta(Y) = \rho_\lambda(X) + \rho_\lambda(Y).
\end{aligned}$$

In conclusion, $\rho_\lambda(X)$ is a risk measure since it satisfies monotonicity and cash invariance. From *Exercise 4.1*, we know that positive homogeneity and subadditivity imply convexity, hence $\rho_\lambda(X)$ is also coherent.

(ii) Let $\Delta(X) = 2\mathbb{E}[X]$. Then

$$\rho_\lambda(X) = \mathbb{E}[-X] + \lambda \Delta(X) = (2\lambda - 1)\mathbb{E}[X].$$

If $\lambda > \frac{1}{2}$, then $2\lambda - 1 > 0$, so $\rho_\lambda(X) > 0$, and there are non-negative random variables X such that $\rho_\lambda(X) > 0$. Since

$$\rho_\lambda(0) = \mathbb{E}[-0] + \lambda \Delta(0) = 0,$$

we have $\rho_\lambda(X) > 0$ for some $X \geq 0$, which violates the monotonicity property. Therefore, ρ_λ cannot be monotone for $\lambda > \frac{1}{2}$.

(iii) We can write

$$\begin{aligned}
\Delta(X) &= \mathbb{E}[|X - \tilde{X}|] \\
&= \iint_{\mathbb{R}^2} |x - y| dF(x) dF(y) \\
&= 2 \iint_{x > y} (x - y) dF(x) dF(y) \quad (|x - y| \text{ is symmetric in } x \text{ and } y) \\
&= 2 \int_{-\infty}^{\infty} \int_{-\infty}^x (x - y) dF(x) dF(y) \\
&= 2 \int_{-\infty}^{\infty} \int_{-\infty}^x (x - y) dF(y) dF(x) \quad (\text{Fubini's theorem}) \\
&= 2 \int_{-\infty}^{\infty} \left([(x - y)F(y)]_{-\infty}^x + \int_{-\infty}^x F(y) dy \right) dF(x) \\
&= 2 \int_{-\infty}^{\infty} \int_{-\infty}^x F(y) dy dF(x) \\
&= 2 \int_{-\infty}^{\infty} F(y) \left(\int_y^{\infty} dF(x) \right) dy \\
&= 2 \int_{-\infty}^{\infty} F(y) (1 - F(y)) dy = 2 \int_{-\infty}^{\infty} F(x) (1 - F(x)) dx.
\end{aligned}$$

When X has a continuous distribution,

$$\mathbb{P}(F(X) \leq u) = \mathbb{P}(X \leq F^{-1}(u)) = F(F^{-1}(u)) = u, \quad u \in [0, 1].$$

Therefore, $F(X)$ is uniformly distributed on $[0, 1]$. Then, by definition,

$$\text{Cov}(X, F(X)) = \mathbb{E}[XF(X)] - \mathbb{E}[X]\mathbb{E}[F(X)] = \mathbb{E}[XF(X)] - \frac{1}{2}\mathbb{E}[X],$$

and

$$\begin{aligned}\mathbb{E}[X] &= \int_{-\infty}^{\infty} x dF(x) = [xF(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} F(x) dx = - \int_{-\infty}^{\infty} F(x) dx, \\ \mathbb{E}[XF(X)] &= \int_{-\infty}^{\infty} xF(x) dF(x) = \left[\frac{xF(x)^2}{2} \right]_{-\infty}^{\infty} - \frac{1}{2} \int_{-\infty}^{\infty} F(x)^2 dx = -\frac{1}{2} \int_{-\infty}^{\infty} F(x)^2 dx.\end{aligned}$$

We have

$$\begin{aligned}\Delta(X) &= 2 \int_{-\infty}^{\infty} F(x)(1 - F(x)) dx \\ &= 2 \left(\int_{-\infty}^{\infty} F(x) dx - \int_{-\infty}^{\infty} F(x)^2 dx \right) \\ &= 2(2\mathbb{E}[XF(X)] - \mathbb{E}[X]) \\ &= 4 \left(\mathbb{E}[XF(X)] - \frac{1}{2} \mathbb{E}[X] \right) = 4\text{Cov}(X, F(X)).\end{aligned}$$

(iv) From (iii), $\Delta(X) = 4\text{Cov}(X, F(X))$. Since $F(X)$ is uniformly distributed on $[0, 1]$,

$$\sigma_{F(X)}^2 = \frac{1}{12} \implies \sigma_{F(X)} = \frac{1}{\sqrt{12}}.$$

Using the *Cauchy–Schwarz inequality*:

$$|\text{Cov}(X, F(X))| \leq \sigma(X)\sigma_{F(X)} = \sigma(X) \cdot \frac{1}{\sqrt{12}}.$$

Therefore,

$$\Delta(X) = 4\text{Cov}(X, F(X)) \leq 4\sigma(X) \frac{1}{\sqrt{12}} = \frac{2\sigma(X)}{\sqrt{3}}.$$

If $\lambda \leq \frac{c\sqrt{2}}{3}$, then

$$\rho_{RC}(X) = -\mathbb{E}[X] + c\sigma(X) \geq -\mathbb{E}[X] + \frac{2\lambda\sigma(X)}{\sqrt{3}} \geq -\mathbb{E}[X] + \lambda\Delta(X) = \rho_{\lambda}(X).$$

hence ρ_{λ} is dominated by the mean–standard deviation risk measure $\rho_{RC}(X)$.

(v) We know the probability density function: $f_X(x) = \alpha x^{-\alpha-1}$, $x \geq 1$, CDF: $F_X(x) = 1 - x^{-\alpha}$, $x \geq 1$, the expectation:

$$\mathbb{E}[X] = \int_1^{\infty} x \cdot \alpha x^{-\alpha-1} dx = \frac{\alpha}{\alpha-1}$$

for $\alpha > 1$. Given that

$$\int_1^{\infty} x^{-\alpha} dx = \frac{1}{\alpha-1}, \quad \int_1^{\infty} x^{-2\alpha} dx = \frac{1}{2\alpha-1},$$

we have

$$\begin{aligned}\Delta(X) &= 2 \int_1^{\infty} F(x)(1 - F(x)) dx \\ &= 2 \int_1^{\infty} (1 - x^{-\alpha})x^{-\alpha} dx \\ &= 2 \int_1^{\infty} (x^{-\alpha} - x^{-2\alpha}) dx \\ &= 2 \left(\frac{1}{\alpha-1} - \frac{1}{2\alpha-1} \right) \\ &= 2 \left(\frac{(2\alpha-1) - (\alpha-1)}{(\alpha-1)(2\alpha-1)} \right) = \frac{2\alpha}{(\alpha-1)(2\alpha-1)}.\end{aligned}$$

Then

$$G(X) = \frac{\Delta(X)}{2\mathbb{E}[X]} = \frac{\frac{\alpha}{(\alpha-1)(2\alpha-1)}}{\frac{\alpha}{\alpha-1}} = \frac{1}{2\alpha-1}.$$

(vi) Since $\log X = m + \sigma Z$, $\log X$ is normally distributed with mean m and variance σ^2 . The CDF of X is:

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(\log X \leq \log x) = \Phi\left(\frac{\log x - m}{\sigma}\right).$$

$\Delta(X)$ calculated by:

$$\Delta(X) = E[|X - \tilde{X}|] = 2E[(X - \tilde{X})\mathbb{1}_{X > \tilde{X}}],$$

because in $\mathbb{E}[(X - \tilde{X})\mathbb{1}_{X > \tilde{X}}]$, X and $\tilde{X}$ are *i.i.d.* and the probability $\mathbb{P}(X > \tilde{X}) = \frac{1}{2}$, the expression for $\Delta(X)$ simplifies to an integral involving the density function of X :

$$\begin{aligned} \Delta(X) &= 4\text{cov}(X, F(X)) \\ &= 4(\mathbb{E}[XF(X)] - \mathbb{E}[X]\mathbb{E}[F(X)]) \\ &= 4(\mathbb{E}[\exp(m + \sigma Z)\Phi(Z)] - \mathbb{E}[X]\mathbb{E}[F(X)]) \\ &= 4(e^m \mathbb{E}[e^{\sigma Z}\Phi(Z)] - \mathbb{E}[X]\mathbb{E}[F(X)]) \\ &= 4\left(e^m \int_{-\infty}^{\infty} e^{\sigma z}\Phi(z)\phi(z)dz - \mathbb{E}[X]\mathbb{E}[F(X)]\right) \\ &= 4\left(e^{m+\frac{\sigma^2}{2}} \int_{-\infty}^{\infty} e^{\sigma z}\Phi(z)\phi(z-\sigma)dz - \mathbb{E}[X]\mathbb{E}[F(X)]\right) \\ &= 4\left(e^{m+\frac{\sigma^2}{2}}\Phi\left(\frac{\sigma}{\sqrt{2}}\right) - e^{m+\frac{\sigma^2}{2}}\mathbb{E}\Phi(Z)\right) = 4e^{m+\frac{\sigma^2}{2}}\left(\Phi\left(\frac{\sigma}{\sqrt{2}}\right) - \frac{1}{2}\right). \end{aligned}$$

Then

$$G(X) = \frac{\Delta(X)}{2\mathbb{E}[X]} = 2\left(\Phi\left(\frac{\sigma}{\sqrt{2}}\right) - \frac{1}{2}\right) = \text{erf}\left(\frac{\sigma}{2}\right),$$

where $\text{erf}(z) = 2\Phi(z\sqrt{2}) - 1$.

□

SECTION A.8

| Expected Utility

EXERCISE A.22 (An Utility Inequality). Show that for a utility function $u \in C^1(\mathbb{R})$ it holds that

$$m(\mu) > c(\mu) > \frac{\mathbb{E}[Xu'(X)]}{\mathbb{E}[u'(X)]},$$

where X has nondegenerate distribution μ and all expectations are assumed to be finite.

SOLUTION. (i) $m(\mu) > c(\mu)$: By contradiction, we suppose $m(\mu) \leq c(\mu)$. Since $\mathbb{P}(X \leq m(\mu)) \geq \frac{1}{2}$ and $m(\mu) \leq c(\mu)$, it follows that:

$$\mathbb{P}(X \leq c(\mu)) \geq \frac{1}{2}.$$

Define the function $\psi(x) = u(x) - u(c(\mu))$, when $x \leq c(\mu)$, since u is increasing, $\psi(x) \leq 0$. When $x \geq c(\mu)$, $\psi(x) \geq 0$.

We compute the expectation:

$$\mathbb{E}[\psi(X)] = \mathbb{E}[u(X)] - u(c(\mu)) = 0.$$

Separate the expectation over the regions:

$$\mathbb{E}[\psi(X)] = \int_{-\infty}^{c(\mu)} \psi(x) dF_X(x) + \int_{c(\mu)}^{\infty} \psi(x) dF_X(x) = 0.$$

The first integral is non-positive, since $\psi(x) \leq 0$ for $x \leq c(\mu)$. The second integral is non-negative, since $\psi(x) \geq 0$ for $x \geq c(\mu)$. Since $\mathbb{P}(X \leq c(\mu)) \geq \frac{1}{2}$, the negative contributions from $x \leq c(\mu)$ have at least half the total probability mass.

This would imply that the negative part outweighs the positive part unless the positive $\psi(x)$ values are significantly larger. However, due to the strict concavity of u , the increase in $u(x)$ for $x > c(\mu)$ is less than the decrease for $x < c(\mu)$. This means that the positive contributions cannot fully offset the negative contributions, leading to: $\mathbb{E}[\psi(X)] < 0$, which contradicts the fact that $\mathbb{E}[\psi(X)] = 0$. Therefore, our assumption must be false, and it must be that: $m(\mu) > c(\mu)$.

- (ii) $c(\mu) > \frac{\mathbb{E}[Xu'(X)]}{\mathbb{E}[u'(X)]}$: Since u is strictly increasing and strictly concave, its first derivative u' has the following properties: $u' > 0$ for all $x \in \mathbb{R}$, and because $u''(x) < 0$, u' is decreasing in x . This means that $u'(X)$ decreases as X increases, higher values of X are assigned smaller weights $u'(X)$, while lower values of X are assigned larger weights.

We consider the expression:

$$c(\mu) > \frac{\mathbb{E}[Xu'(X)]}{\mathbb{E}[u'(X)]} \implies c(\mu) \cdot \mathbb{E}[u'(X)] - \mathbb{E}[Xu'(X)] > 0.$$

In order to prove: $c(\mu) > \frac{\mathbb{E}[Xu'(X)]}{\mathbb{E}[u'(X)]}$, we can prove $c(\mu) \cdot \mathbb{E}[u'(X)] - \mathbb{E}[Xu'(X)] > 0$. Because $c(\mu)$ is a constant,

$$c(\mu)\mathbb{E}[u'(X)] - \mathbb{E}[Xu'(X)] = \mathbb{E}[u'(X)c(\mu)] - \mathbb{E}[Xu'(X)] = \mathbb{E}[u'(X)(c(\mu) - X)].$$

When $X < c(\mu)$, we have $c(\mu) - X > 0$. Since u' is decreasing, $u'(X) \geq u'(c(\mu))$, and hence $u'(X)(c(\mu) - X) > 0$.

- When $X = c(\mu)$, $c(\mu) - X = 0$ and $u'(X)(c(\mu) - X) = 0$.
- When $X > c(\mu)$, $c(\mu) - X < 0$ and $u'(X) \leq u'(c(\mu))$, because u' is decreasing, $u'(X)(c(\mu) - X)$ is negative but smaller in magnitude due to smaller $u'(X)$.

Overall, the positive contributions outweigh the negative ones, leading to:

$$\mathbb{E}[u'(X)(c(\mu) - X)] > 0.$$

Therefore:

$$c(\mu) > \frac{\mathbb{E}[Xu'(X)]}{\mathbb{E}[u'(X)]}.$$

□

EXERCISE A.23 (CARA Utility Function). Let $u(x) = 1 - \exp(-x)$, a CARA function. Consider an investor with utility function u who wants to invest an initial capital. There is one riskless asset, having value 1 and interest rate $r = 0$, and one risky asset with random pay-off S_1 having a normal $\mathcal{N}(m, \sigma^2)$ distribution with $\sigma^2 > 0$. Suppose she invests a fraction λ in the riskless asset and the remainder in the risky asset. The pay-off of this portfolio is thus $\lambda + (1 - \lambda)S_1$. The aim is to maximise her expected utility.

- (i) Show that $\mathbb{E}[\exp(uS_1)] = \exp\left(um + \frac{1}{2}u^2\sigma^2\right)$ ($u \in \mathbb{R}$).
- (ii) Compute for each λ the certainty equivalent of the portfolio.
- (iii) Let λ^* be the optimal value of λ . Give, by direct computations, conditions on the parameters such that each of the cases $\lambda^* = 0$, $\lambda^* = 1$ or $\lambda^* \in (0, 1)$ occurs.
- (iv) Compare the results of (iii) with the assertions of [Proposition 2.29](#).

SOLUTION. (i) For a normally distributed random variable $X \sim \mathcal{N}(\mu, \sigma^2)$, the moment-generating function (MGF) is given by

$$M_X(u) = \mathbb{E}[\exp(tX)] = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right), \quad t \in \mathbb{R}.$$

Apply the MGF to $S_1 \sim \mathcal{N}(m, \sigma^2)$,

$$\mathbb{E}[\exp(uS_1)] = \exp\left(mu + \frac{1}{2}u^2\sigma^2\right), \quad u \in \mathbb{R}.$$

- (ii) The certainty equivalent CE is the guaranteed amount of payoff that provides the same utility as the expected utility of the uncertain portfolio payoff $P = \lambda + (1 - \lambda)S_1$. Then

$$\begin{aligned} u(CE) &= \mathbb{E}[u(P)] \\ &= \mathbb{E}[1 - \exp(-P)] \\ &= 1 - \mathbb{E}[\exp(-P)] \\ &= 1 - \mathbb{E}[\exp(-\lambda) \cdot \exp(-(1 - \lambda)S_1)] \\ &= 1 - \exp(-\lambda) \cdot \mathbb{E}[\exp(-(1 - \lambda)S_1)] \\ &= 1 - \exp(-\lambda) \cdot \exp\left(-(1 - \lambda)m + \frac{1}{2}(1 - \lambda)^2\sigma^2\right) = 1 - \exp\left(-\lambda - (1 - \lambda)m + \frac{1}{2}(1 - \lambda)^2\sigma^2\right). \end{aligned}$$

Since $u(CE) = 1 - \exp(-CE)$, we have $\exp(-CE) = \mathbb{E}[\exp(-P)]$, so

$$\exp(-CE) = \exp\left(-\lambda - (1 - \lambda)m + \frac{1}{2}(1 - \lambda)^2\sigma^2\right).$$

Therefore, for each λ , the certainty equivalent of the portfolio is

$$CE = \lambda + (1 - \lambda)m - \frac{1}{2}(1 - \lambda)^2\sigma^2.$$

- (iii) We have

$$CE(\lambda) = \lambda + (1 - \lambda)m - \frac{1}{2}(1 - \lambda)^2\sigma^2 = m + \lambda(1 - m) - \frac{1}{2}(1 - \lambda)^2\sigma^2.$$

Differentiate $CE(\lambda)$ with respect to λ yields

$$CE'(\lambda) = \frac{d}{d\lambda} \left(m + \lambda(1-m) - \frac{1}{2}(1-\lambda)^2\sigma^2 \right) = 1 - m + (1-\lambda)\sigma^2.$$

Let $CE'(\lambda) = 0$, $\lambda^* = 1 - \frac{m-1}{\sigma^2}$. We consider the following cases:

- $\lambda^* = 0$ occurs when $\lambda^* \leq 0$:

$$\lambda^* \leq 0 \implies 1 - \frac{m-1}{\sigma^2} \leq 0 \implies \frac{m-1}{\sigma^2} \geq 1 \implies m-1 \geq \sigma^2,$$

when $m \geq 1 + \sigma^2$, the optimal $\lambda^* \leq 0$, but since $\lambda \in [0, 1]$, we set $\lambda^* = 0$.

- $\lambda^* = 1$ occurs when $\lambda^* \geq 1$:

$$\lambda^* \geq 1 \implies 1 - \frac{m-1}{\sigma^2} \geq 1 \implies \frac{m-1}{\sigma^2} \leq 0 \implies m-1 \leq 0,$$

when $m \leq 1$, the optimal $\lambda^* \geq 1$, but since $\lambda \in [0, 1]$, we set $\lambda^* = 1$.

- $\lambda^* \in (0, 1)$ occurs when $0 < \lambda^* < 1$:

$$0 < 1 - \frac{m-1}{\sigma^2} < 1 \implies 0 < \frac{m-1}{\sigma^2} < 1 \implies 1 < m < 1 + \sigma^2,$$

when $1 < m < 1 + \sigma^2$, the optimal λ^* lies strictly between 0 and 1.

(iv) Given that $\sigma^2 > 0$, by (iii),

$$CE''(\lambda) = \frac{d}{d\lambda} (1 - m + (1-\lambda)\sigma^2) = -\sigma^2 < 0,$$

which indicates $CE(\lambda)$ is strictly concave on $[0, 1]$, aligning with *Proposition 5.13 (i)*. Then we analyse the optimal investment strategies.

- When $m \leq 1$, by (iii), $\lambda^* = 1$, the investor should invest all in the risky asset. The expected portfolio payoff is

$$\mathbb{E}[P] = \lambda + (1-\lambda)m = 1 \leq CE(1) = 1,$$

which matches *Proposition 5.13 (iii)*.

- When $m \geq 1 + \sigma^2$, we have $\lambda^* = 0$, meaning the investor should invest all in the riskless asset. Now the portfolio payoff $P = S_1$, $c = m - \frac{1}{2}\sigma^2$, then

$$\frac{\mathbb{E}[u'(P)]}{\mathbb{E}[u(P)]} = \frac{\exp(-m + \frac{1}{2}\sigma^2)}{1 - \exp(-m + \frac{1}{2}\sigma^2)} \rightarrow 0,$$

as $m \rightarrow \infty$, so $c \geq \frac{\mathbb{E}[u'(P)]}{\mathbb{E}[u(P)]}$, which is different from *Proposition 5.13 (iii)*.

□

EXERCISE A.24 (Decreasing Risk Aversion). A utility function $u : \mathbb{R} \rightarrow \mathbb{R}$ is said to exhibit *decreasing risk aversion* if the function $x \mapsto \alpha(x)$ (the Arrow–Pratt coefficient) is decreasing. Show that this property is equivalent to saying that for every $x_1 < x_2$ there exists a concave function g such that $u(x_1 + z) = g(u(x_2 + z))$ for all z (for which the given expressions make sense).

SOLUTION. We need to prove the equivalence between two statements: the utility function u exhibits decreasing absolute risk aversion, i.e., $\alpha(x)$ is decreasing in x ; for every $x_1 < x_2$, there exists a concave function g such that $u(x_1 + z) = g(u(x_2 + z))$ for all z .

- (i) From exhibit decreasing risk aversion to existence of concave g . For fixed $x_1 < x_2$, we define g as:

$$g(y) = u(u^{-1}(y) + x_1 - x_2),$$

where u^{-1} denotes the inverse function of u .

$$g(u(x_2 + z)) = u(u^{-1}(u(x_2 + z)) + x_1 - x_2) = u((x_2 + z) + x_1 - x_2) = u(x_1 + z).$$

So the equation $g(u(x_2 + z)) = u(x_1 + z)$ holds for all z . And we need to prove g is a concave function, which means for all y , $g''(y) \leq 0$.

First, calculate the first order derivative of $g(y)$. We set $s = u^{-1}(y) + x_1 - x_2$,

$$g(y) = u(s),$$

$$g'(y) = u'(s) \frac{ds}{dy} = u'(s) \frac{d}{dy}(u^{-1}(y) + x_1 - x_2) = u'(s) \frac{1}{u'(u^{-1}(y))}.$$

Then calculate the second order derivative of $g(y)$,

$$A(y) = u'(s), \quad B(y) = u'(u^{-1}(y))$$

$$g'(y) = \frac{A(y)}{B(y)}$$

$$g''(y) = \frac{A'(y)B(y) - A(y)B'(y)}{[B(y)]^2}$$

$$A'(y) = u''(s) \cdot \frac{ds}{dy} = u''(s) \cdot \frac{1}{u'(u^{-1}(y))}$$

$$B'(y) = u''(u^{-1}(y)) \cdot \frac{du^{-1}(y)}{dy} = u''(u^{-1}(y)) \cdot \frac{1}{u'(u^{-1}(y))}$$

$$g''(y) = \frac{\left(\frac{u''(s)}{u'(u^{-1}(y))}\right)B(y) - A(y)\left(\frac{u''(u^{-1}(y))}{u'(u^{-1}(y))}\right)}{[B(y)]^2} = \frac{u''(s)u'(u^{-1}(y)) - u'(s)u''(u^{-1}(y))}{[u'(u^{-1}(y))]^2}.$$

Define the Arrow-Pratt absolute risk aversion factor: $\alpha(x) = -\frac{u''(x)}{u'(x)}$, so we can get:

$$u''(s) = -\alpha(s)u'(s), \quad u''(u^{-1}(y)) = -\alpha(u^{-1}(y))u'(u^{-1}(y)).$$

and we set $t = u^{-1}(y)$, so we can get: $u''(t) = -\alpha(t)u'(t)$, and put this back to $g''(y)$:

$$g''(y) = \frac{[-\alpha(s)u'(s)]u'(t) - u'(s)[- \alpha(t)u'(t)]}{[u'(t)]^2} = \frac{u'(s)u'(t)[\alpha(t) - \alpha(s)]}{[u'(t)]^2} = \left(\frac{u'(s)}{u'(t)}\right)[\alpha(t) - \alpha(s)].$$

Because u is an increasing and concave function, so $u' > 0$, because $x_1 < x_2$, so $x_1 - x_2 < 0$, so $s = t + x_1 - x_2 < t$. Under the assumption of decreasing absolute risk aversion (DARA), the $\alpha(x)$ decreases with x increases.

$$s < t \implies \alpha(s) > \alpha(t) \implies \alpha(t) - \alpha(s) < 0.$$

Because $u'(s) > 0$, $u'(t) > 0$, so $\frac{u'(s)}{u'(t)} > 0$,

$$g''(y) = \left(\frac{u'(s)}{u'(t)}\right)[\alpha(t) - \alpha(s)] < 0.$$

So g is a concave function, we can prove the result.

- (ii) From exhibit existence of concave to exhibit decreasing risk aversion. For $u(x_1 + z) = g(u(x_2 + z))$, for both sides of the equation about z find the first order derivative:

$$\begin{aligned} \frac{d}{dz}u(x_1 + z) &= \frac{d}{dz}g(u(x_2 + z)) = u'(x_1 + z) \\ \frac{d}{dz}g(u(x_2 + z)) &= g'(u(x_2 + z)) \cdot u'(x_2 + z). \end{aligned}$$

So we can get:

$$u'(x_1 + z) = g'(u(x_2 + z)) \cdot u'(x_2 + z).$$

The second order derivative relation is obtained by taking the derivative of Equation :

$$\begin{aligned} \frac{d}{dz} u'(x_1 + z) &= u''(x_1 + z) \\ \frac{d}{dz} [g'(u(x_2 + z)) \cdot u'(x_2 + z)] &= g''(u(x_2 + z)) \cdot [u'(x_2 + z)]^2 + g'(u(x_2 + z)) \cdot u''(x_2 + z) \\ u''(x_1 + z) &= g''(u(x_2 + z)) [u'(x_2 + z)]^2 + g'(u(x_2 + z)) u''(x_2 + z). \end{aligned}$$

Arrow–Pratt definition of Absolute Risk Aversion coefficients is $\alpha(x) = -\frac{u''(x)}{u'(x)}$ and we want to express the relation between $\alpha(x_1 + z)$ and $\alpha(x_2 + z)$.

From $u'(x_1 + z) = g'(u(x_2 + z)) \cdot u'(x_2 + z)$ we can get $g'(u(x_2 + z)) = \frac{u'(x_1 + z)}{u'(x_2 + z)}$ we use α to express:

$$\begin{aligned} u''(x_1 + z) &= -\alpha(x_1 + z) u'(x_1 + z) \\ u''(x_2 + z) &= -\alpha(x_2 + z) u'(x_2 + z), \end{aligned}$$

and we take the express into

$$\begin{aligned} u''(x_1 + z) &= g''(u(x_2 + z)) [u'(x_2 + z)]^2 + g'(u(x_2 + z)) u''(x_2 + z) \\ -\alpha(x_1 + z) u'(x_1 + z) &= g''(u(x_2 + z)) [u'(x_2 + z)]^2 + g'(u(x_2 + z)) (-\alpha(x_2 + z) u'(x_2 + z)) \\ &= g''(u(x_2 + z)) [u'(x_2 + z)]^2 - \left(\frac{u'(x_1 + z)}{u'(x_2 + z)} \right) \alpha(x_2 + z) u'(x_2 + z) \\ &= g''(u(x_2 + z)) [u'(x_2 + z)]^2 - u'(x_1 + z) \alpha(x_2 + z). \\ u'(x_1 + z) (\alpha(x_1 + z) - \alpha(x_2 + z)) &= g''(u(x_2 + z)) [u'(x_2 + z)]^2 \end{aligned}$$

We analyze the symbol of $g''(u(x_2 + z))$, because g is a concave function, so for all y , $g''(y) \leq 0$, so $g''(u(x_2 + z)) \leq 0$ and we use the result to prove $\alpha(x)$ is decrease. We have: $u'(x_1 + z) (\alpha(x_1 + z) - \alpha(x_2 + z)) = g''(u(x_2 + z)) [u'(x_2 + z)]^2$, and because $u' > 0$ and $[u'(x_2 + z)]^2 > 0$, so if: $\alpha(x_1 + z) - \alpha(x_2 + z) > 0$, then the right-hand side is non-positive, so we get:

$$\alpha(x_1 + z) - \alpha(x_2 + z) \leq 0 \implies \alpha(x_1 + z) \leq \alpha(x_2 + z).$$

Because $x_1 < x_2$, so $x_1 + z < x_2 + z$, this shows $\alpha(x)$ decreases as x increase.

□

EXERCISE A.25 (CARA Again). Consider the CARA utility function $u(x) = 1 - \exp(-\alpha x)$, $x \in \mathbb{R}$, with $\alpha > 0$, the constant Arrow–Pratt coefficient.

- (i) Show that the condition $\mathbb{E}[|u(\varphi \cdot Y)|] < \infty$ for all $\varphi \in \Xi$ of [Theorem 3.5](#) is equivalent to $\mathbb{E}[\exp(\varphi \cdot Y)] < \infty$, for all $\varphi \in \mathbb{R}^d$.
- (ii) Show that the risk-neutral measure $\mathbb{P}^*$ of [Proposition 3.6](#) is the same for all $\alpha > 0$.
- (iii) Suppose that Y has a d -dimensional multivariate normal distribution with mean vector m and invertible covariance matrix Σ . Compute the optimal $\varphi^* \in \mathbb{R}^d$.

SOLUTION. (i) Given that $u(x) = 1 - \exp(-\alpha x)$, we have $u(\varphi \cdot Y) = 1 - \exp(-\alpha \varphi \cdot Y)$, then

$$|u(\varphi \cdot Y)| = \begin{cases} 1 - \exp(-\alpha \varphi \cdot Y) \leq 1, & \varphi \cdot Y \geq 0 \\ \exp(-\alpha \varphi \cdot Y) - 1 \leq \exp(-\alpha \varphi \cdot Y), & \varphi \cdot Y < 0 \end{cases}.$$

Thus $|u(\varphi \cdot Y)| \leq \exp(-\alpha \varphi \cdot Y)$ since $\exp(-\alpha \varphi \cdot Y) \leq 1$ when $\varphi \cdot Y \geq 0$.

" $\Rightarrow$ ": since $|u(\varphi \cdot Y)| \geq \exp(-\alpha\varphi \cdot Y) - 1$, $\exp(-\alpha\varphi \cdot Y) \leq |u(\varphi \cdot Y)| + 1$, i.e.,

$$\mathbb{E}[\exp(-\alpha\varphi \cdot Y)] \leq \mathbb{E}[|u(\varphi \cdot Y)|] + 1 < \infty, \quad \forall \alpha > 0, \forall \varphi \in \Xi.$$

Then $\mathbb{E}[\exp(\varphi \cdot Y)] < \infty, \forall \varphi \in \mathbb{R}^d$.

" $\Leftarrow$ ": suppose $\mathbb{E}[\exp(\varphi \cdot Y)] < \infty, \forall \varphi \in \mathbb{R}^d$, then $\mathbb{E}[\exp((- \alpha\varphi) \cdot Y)] < \infty$. Therefore, $\mathbb{E}[|u(\varphi \cdot Y)|] \leq \mathbb{E}[\exp(-\alpha\varphi \cdot Y)] < \infty$.

In conclusion, we have shown that $\mathbb{E}[|u(\varphi \cdot Y)|] < \infty$ for all $\varphi \in \Xi$ is equivalent to $\mathbb{E}[\exp(\varphi \cdot Y)] < \infty$, for all $\varphi \in \mathbb{R}^d$.

- (ii) The investor aims to maximise the expected utility

$$\mathbb{E}_{\mathbb{P}}[u(\varphi \cdot Y)] = \mathbb{E}_{\mathbb{P}}[1 - \exp(-\alpha\varphi \cdot Y)] = 1 - \mathbb{E}_{\mathbb{P}}[\exp(-\alpha\varphi \cdot Y)],$$

which is equivalent to minimising $\mathbb{E}_{\mathbb{P}}[\exp(-\alpha\varphi \cdot Y)]$. By calculating the derivatives, the risk-neutral measure $\mathbb{P}^*$ is

$$\frac{d\mathbb{P}^*}{d\mathbb{P}} = \frac{u'(\varphi^* \cdot Y)}{\mathbb{E}_{\mathbb{P}}[u'(\varphi^* \cdot Y)]} = \frac{\exp(-\alpha\varphi^* \cdot Y)}{\mathbb{E}_{\mathbb{P}}[\exp(-\alpha\varphi^* \cdot Y)]}.$$

Since $\mathbb{P}^*$ is a martingale measure, $\mathbb{E}_{\mathbb{P}^*}[Y] = 0$, i.e.,

$$\mathbb{E}_{\mathbb{P}^*}[Y] = \int_{\Omega} Y d\mathbb{P}^* = \int_{\Omega} Y \frac{d\mathbb{P}^*}{d\mathbb{P}} d\mathbb{P} = \mathbb{E}_{\mathbb{P}} \left[\frac{Y \exp(-\alpha\varphi^* \cdot Y)}{\mathbb{E}_{\mathbb{P}}[\exp(-\alpha\varphi^* \cdot Y)]} \right] = 0,$$

which implies $\mathbb{E}_{\mathbb{P}}[Y \exp(-\alpha\varphi^* \cdot Y)] = 0$. Therefore, the risk-neutral measure $\mathbb{P}^*$ is the same for all $\alpha > 0$.

The form of the Radon–Nikodym derivative suggests that we can also verify it by assuming a distribution of $\varphi^* \cdot Y$. After calculating (iii), the risk-neutral measure $\mathbb{P}^*$ is

$$\begin{aligned} \frac{d\mathbb{P}^*}{d\mathbb{P}} &= \frac{u'(\varphi^* \cdot Y)}{\mathbb{E}_{\mathbb{P}}[u'(\varphi^* \cdot Y)]} \\ &= \frac{Y \exp(-\alpha\varphi^* \cdot Y)}{\mathbb{E}_{\mathbb{P}}[Y \exp(-\alpha\varphi^* \cdot Y)]} \quad (\varphi^* = \frac{m}{\alpha\Sigma} \text{ by (iii)}) \\ &= \frac{Y \exp(-\alpha \frac{m}{\alpha\Sigma} \cdot Y)}{\mathbb{E}_{\mathbb{P}}[Y \exp(-\alpha \frac{m}{\alpha\Sigma} \cdot Y)]} = \frac{Y \exp(-m\Sigma^{-1} \cdot Y)}{\mathbb{E}_{\mathbb{P}}[Y \exp(-m\Sigma^{-1} \cdot Y)]}, \end{aligned}$$

where α cancels out and the density depends only on $m\Sigma^{-1}$. Therefore, the risk-neutral measure $\mathbb{P}^*$ is the same for all $\alpha > 0$.

- (iii) The investor aims to minimise $\mathbb{E}_{\mathbb{P}}[\exp(-\alpha\varphi \cdot Y)]$. By the moment generating function of $Y \sim \mathcal{N}(m, \Sigma)$, we have

$$M_Y(\varphi) = \mathbb{E}_{\mathbb{P}}[\exp(-\alpha\varphi \cdot Y)] = \exp\left(-\alpha\varphi^\top m + \frac{1}{2}\alpha^2\varphi^\top \Sigma \varphi\right).$$

We minimise

$$f(\varphi) = -\alpha\varphi^\top m + \frac{1}{2}\alpha^2\varphi^\top \Sigma \varphi.$$

Let $f'(\varphi) = 0$, we have $-\alpha m + \alpha^2 \Sigma \varphi = 0$, then the optimal is $\varphi^* = \frac{m}{\alpha\Sigma}$.

□

EXERCISE A.26 (Normal and Inferior Goods). Given an arbitrage-free market and a utility function $\tilde{u}$ as at the beginning of this section, the transformed utility function u depends on the initial capital W_0 . In general, an optimal portfolio will also depend on W_0 . We study this for the case $d = 1$ and zero interest, i.e., $r = 0$. Assume that $\tilde{u}$ is a C^2 function and that everywhere below interchanging of expectation and differentiation is allowed. Put

$$f(w, \varphi) = \mathbb{E}[\tilde{u}'(W_0 + \varphi Y)Y].$$

- (i) Show that $\frac{\partial f}{\partial \varphi}(W_0, \varphi) < 0$.
- (ii) Conclude that locally for every $W_0 > 0$, there is a continuously differentiable function $x \mapsto \varphi^*(x)$ such that $f(W_0, \varphi^*(W_0)) = 0$.
- (iii) Show that

$$\frac{d\varphi^*(W_0)}{dW_0} = -\frac{\mathbb{E}[\tilde{u}''(W_0 + \varphi^*(W_0)Y)Y]}{\mathbb{E}[\tilde{u}''(\varphi^*(W_0)Y + W_0)Y^2]}.$$

- (iv) Assume that $\mathbb{E}[Y] > 0$ and that the Arrow–Pratt coefficient $\tilde{\alpha}(\cdot)$ of $\tilde{u}$ is a decreasing function. Show that $Y\tilde{\alpha}(W_0 + \varphi^*(W_0)Y) \leq Y\tilde{\alpha}(W_0)$.
- (v) Conclude, under the assumptions in (iv), that $\varphi^*(\cdot)$ is an increasing function of W_0 . (In Microeconomics, assets with the latter property are called *normal goods*. Assets with decreasing demand ξ^* are called *inferior goods*.)

SOLUTION. (i) Given that $f(W_0, \varphi) = \mathbb{E}[\tilde{u}'(W_0 + \varphi Y)Y]$, we compute

$$\frac{\partial f}{\partial \varphi}(W_0, \varphi) = \mathbb{E}\left[Y \frac{\partial}{\partial \varphi} \tilde{u}'(W_0 + \varphi Y)\right] = \mathbb{E}[\tilde{u}''(W_0 + \varphi Y)Y^2],$$

where $\tilde{u}''(x) < 0$ for all x since the utility function is strictly concave. The return of the risky asset Y should not be zero *a.s.*, otherwise, there is no randomness in the return, and the investment problem becomes trivial. Therefore, we conclude that $\frac{\partial f}{\partial \varphi}(W_0, \varphi) < 0$.

- (ii) Since $\tilde{u} \in C^2$, $f(w, \varphi)$ is a continuously differentiable function. By the definition of utility function, we have

$$f(w, \varphi) = \mathbb{E}[\tilde{u}'(W_0 + \varphi Y)Y] = \mathbb{E}[\alpha Y \exp(\alpha(W_0 + \varphi Y))].$$

Therefore we may have some $f(W_0, \varphi) = 0$ when $Y = 0$. From (i), $\frac{\partial f}{\partial \varphi}(W_0, \varphi) < 0$ for all φ , then $\frac{\partial f}{\partial \varphi}(W_0, \varphi^*(W_0)) \neq 0$. Then, by the *Implicit Function Theorem*, for every $W_0 > 0$, there exists an open interval containing W_0 and a unique continuously differentiable function $\varphi^*(\omega)$ defined on I such that $f(W_0, \varphi^*(W_0)) = 0$ for all $\omega \in I$.

- (iii) Differentiate both sides of $f(W_0, \varphi^*(W_0)) = 0$ with respect to W_0 , then

$$\frac{\partial f}{\partial W_0}(W_0, \varphi^*(W_0)) + \frac{\partial f}{\partial \varphi}(W_0, \varphi^*(W_0)) \cdot \frac{d\varphi^*(W_0)}{dW_0} = 0,$$

which implies

$$\frac{d\varphi^*(W_0)}{dW_0} = -\frac{\frac{\partial f}{\partial W_0}(W_0, \varphi^*(W_0))}{\frac{\partial f}{\partial \varphi}(W_0, \varphi^*(W_0))} = -\frac{\frac{\partial}{\partial W_0} \mathbb{E}[\tilde{u}'(W_0 + \varphi^*(W_0)Y)Y]}{\mathbb{E}[\tilde{u}''(W_0 + \varphi^*(W_0)Y)Y^2]} = -\frac{\mathbb{E}[\tilde{u}''(W_0 + \varphi^*(W_0)Y)Y]}{\mathbb{E}[\tilde{u}''(\varphi^*(W_0)Y + W_0)Y^2]}.$$

- (iv) The Arrow–Pratt coefficient $\tilde{\alpha}(x)$ is given as a decreasing function. Since the investor can invest a non-negative amount in the asset, $\varphi^*(W_0) \geq 0$. We consider the following two cases:

- $Y \geq 0$: then $W_0 + \varphi^*(W_0)Y \geq W_0$, and hence $Y\tilde{\alpha}(W_0 + \varphi^*(W_0)Y) \leq Y\tilde{\alpha}(W_0)$;
- $Y \leq 0$: now $\varphi^*(W_0)Y \leq 0$, so $W_0 + \varphi^*(W_0)Y \leq W_0$, then $Y\tilde{\alpha}(W_0 + \varphi^*(W_0)Y) \leq Y\tilde{\alpha}(W_0)$.

Therefore, in both cases, we have shown that $Y\tilde{\alpha}(W_0 + \varphi^*(W_0)Y) \leq Y\tilde{\alpha}(W_0)$.

(v) By (iii), we have

$$\frac{d\varphi^*(W_0)}{dW_0} = -\frac{\mathbb{E}[\tilde{u}''(W_0 + \varphi^*(W_0)Y)Y]}{\mathbb{E}[\tilde{u}''(\varphi^*(W_0)Y + W_0)Y^2]},$$

where the numerator is non-positive since $Y \geq 0$; and the denominator is negative since $\tilde{u}$ is concave, i.e., $\tilde{u}''(x) \leq 0$. Then $\frac{d\varphi^*(W_0)}{dW_0} > 0$ and hence $\varphi^*(\cdot)$ is an increasing function of W_0 . $\square$

EXERCISE A.27 (Relative Entropy and Poisson Distribution). Consider a market with one risky good, its value at $t = 1$ is S_1 and price S_0 (at $t = 0$). Assume that S_1 has under $\mathbb{P}$ a Poisson distribution with parameter $\alpha > 0$. Consider the family of probability measures $\mathbb{P}_\lambda$ on $(\Omega, \mathcal{F})$ with $\lambda \in \mathbb{R}^d$ given by

$$\frac{d\mathbb{P}_\lambda}{d\mathbb{P}} = \frac{e^{\lambda \cdot Y}}{\mathbb{E}[e^{\lambda \cdot Y}]},$$

where Y is the discounted net gains (see Section 6).

- (i) Show that $\mathbb{E}[e^{\lambda \cdot Y}] < \infty$ for all $\lambda \in \mathbb{R}$.
- (ii) Show that S_1 has a Poisson distribution with parameter αe^λ under $\mathbb{P}_\lambda$.
- (iii) Compute the minimizer of $\lambda \mapsto \mathbb{E}[e^{\lambda \cdot Y}]$ directly.
- (iv) Show that the minimizer λ^* satisfies $\mathbb{E}_{\mathbb{P}_{\lambda^*}}[Y] = 0$.

SOLUTION. (i) Since $Y = S_1$ and $S_1 \sim \text{Poisson}(\alpha)$, the probability mass function (PMF) of S_1 under $\mathbb{P}$ is

$$\mathbb{P}(S_1 = k) = \frac{e^{-\alpha} \alpha^k}{k!}, \quad k = 0, 1, 2, \dots$$

Then

$$\begin{aligned} \mathbb{E}_{\mathbb{P}}[e^{\lambda Y}] &= \sum_{k=0}^{\infty} e^{\lambda k} \mathbb{P}(S_1 = k) \\ &= \sum_{k=0}^{\infty} e^{\lambda k} \frac{e^{-\alpha} \alpha^k}{k!} \\ &= e^{-\alpha} \sum_{k=0}^{\infty} \frac{(e^\lambda \alpha)^k}{k!} \\ &= e^{-\alpha} e^{e^\lambda \alpha} \\ &= e^{\alpha(e^\lambda - 1)} < \infty, \end{aligned} \quad \left(\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x \right)$$

since $\alpha > 0$ and $e^\lambda > 0$ for all λ .

(ii) Under $\mathbb{P}_\lambda$, the probability of $S_1 = k$ is

$$\mathbb{P}_\lambda(S_1 = k) = \mathbb{E}_{\mathbb{P}} \left[\frac{d\mathbb{P}_\lambda}{d\mathbb{P}} \mathbf{1}_{\{S_1=k\}} \right] = \frac{\mathbb{E}_{\mathbb{P}}[e^{\lambda Y} \mathbf{1}_{\{S_1=k\}}]}{\mathbb{E}_{\mathbb{P}}[e^{\lambda Y}]} = \frac{e^{\lambda k}}{\mathbb{E}_{\mathbb{P}}[e^{\lambda Y}]} \mathbb{P}(S_1 = k).$$

Substitute $\mathbb{P}(S_1 = k)$ and $\mathbb{E}_{\mathbb{P}}[e^{\lambda Y}]$:

$$\mathbb{P}_\lambda(S_1 = k) = \frac{e^{\lambda k}}{e^{\alpha(e^\lambda - 1)}} \cdot \frac{e^{-\alpha} \alpha^k}{k!} = \frac{e^{-\alpha(e^\lambda - 1)} e^{\lambda k} e^{-\alpha} \alpha^k}{k!} = \frac{e^{-\alpha e^\lambda} \alpha^k e^{\lambda k}}{k!} = \frac{e^{-\alpha e^\lambda} (\alpha e^\lambda)^k}{k!},$$

which is exactly the PMF of a Poisson distribution with parameter αe^λ :

$$\mathbb{P}_\lambda(S_1 = k) = \frac{e^{-\alpha e^\lambda} (\alpha e^\lambda)^k}{k!}, \quad \text{for } k = 0, 1, 2, \dots$$

Therefore, S_1 has a Poisson distribution with parameter αe^λ under $\mathbb{P}_\lambda$.

(iii) By (i), we have $\mathbb{E}[e^{\lambda Y}] = e^{\alpha(e^\lambda - 1)}$. Let $f(\lambda) = e^{\alpha(e^\lambda - 1)}$, we calculate

$$f'(\lambda) = f(\lambda) \cdot \alpha e^\lambda > 0,$$

since $\alpha > 0$. Then $f(\lambda)$ is strictly increasing in λ . Therefore, the function $\mathbb{E}[e^{\lambda Y}]$ attains its minimum as $\lambda \rightarrow -\infty$, and the minimal value is $e^{-\alpha}$.

(iv) By (ii), under $\mathbb{P}_\lambda$, we have $\mathbb{E}_{\mathbb{P}_\lambda}[Y] = \mathbb{E}_{\mathbb{P}_\lambda}[S_1] = \alpha e^\lambda$. Then

$$\mathbb{E}_{\mathbb{P}_{\lambda^*}}[Y] = \alpha e^{\lambda^*} \rightarrow \alpha \cdot 0 = 0,$$

where $\lambda^* = \lambda \rightarrow -\infty$. Hence the minimiser λ^* satisfies $\mathbb{E}_{\mathbb{P}_{\lambda^*}}[Y] = 0$.

□

SECTION A.9

| Optimal Control

EXERCISE A.28 (Maximisation of Expected Utility from Terminal Wealth in the CRR Model). Consider a CRR model, in which the returns are *i.i.d.* with

$$\mathbb{P}(R_t = U) = p \quad (\text{where } p \text{ is not necessarily equal to the risk neutral value } p^*).$$

Consider the maximisation of the expected utility of terminal wealth $\mathbb{E}[u(W_T)]$, with $u(x) = \log x$. Compute the optimal trading strategy ϕ^* via dynamic programming.

Hint. Show that $\hat{v}_t(x) = \log x + k_t$ for some constants k_t and $\hat{v}_t$ as in equation (96) of the lecture notes.

SOLUTION. The CRR model specifies the evolution of wealth by

$$W_{t+1} = W_t + \phi_t(R_{t+1} - 1)W_t = W_t(1 + \phi_t(R_{t+1} - 1)),$$

where W_t is wealth at time t , ϕ_t is the fraction of wealth invested in the risky asset, and $R_{t+1} - 1$ is its net return. The utility function $u(x) = \log(x)$ gives a logarithmic-utility maximisation problem.

By dynamic programming, the value function satisfies

$$\hat{v}_t(x) = \max_{\phi_t} \mathbb{E}_t[\hat{v}_{t+1}(W_{t+1})].$$

Following the hint, suppose that

$$\hat{v}_t(x) = \log(x) + k_t,$$

where k_t is determined recursively. Substitution into the Bellman equation gives

$$\hat{v}_t(x) = \max_{\phi_t} \mathbb{E}_t[\log(W_{t+1}) + k_{t+1}] = \log(x) + \max_{\phi_t} \mathbb{E}_t[\log(1 + \phi_t(R_{t+1} - 1))] + k_{t+1}.$$

Since R_{t+1} takes the two values

$$R_{t+1} = \begin{cases} U, & \text{with probability } p, \\ D, & \text{with probability } 1 - p, \end{cases}$$

where $U > 1$ and $D < 1$, define

$$f(\phi_t) = p \log(1 + \phi_t(U - 1)) + (1 - p) \log(1 + \phi_t(D - 1)).$$

The first-order condition is

$$f'(\phi_t) = \frac{p(U - 1)}{1 + \phi_t(U - 1)} + \frac{(1 - p)(D - 1)}{1 + \phi_t(D - 1)} = 0.$$

Thus, ϕ_t^* is a solution of this equation, and the constants satisfy

$$k_t = k_{t+1} + \mathbb{E}_t[\log(1 + \phi_t^*(R_{t+1} - 1))], \quad k_T = 0.$$

Consequently, the optimal fraction is independent of current wealth, and the value function is $\hat{v}_t(x) = \log(x) + k_t$. $\square$

EXERCISE A.29 (Logarithmic Utility of Terminal Wealth). Assume we are in the same setting as in Section 3.3, with $d = 1$ and $u(x) = \log(x)$, but with $(R_t)_{t \in T}$ not necessarily *i.i.d.* Show using Proposition 3.22 that the optimal control $\alpha^*(t, \omega)$ in this situation is given as the maximiser of the mapping $\gamma \mapsto \mathbb{E}_{\mathbb{P}}[\log(1 + \gamma R_t) \mid \mathcal{F}_{t-1}](\omega)$.

SOLUTION. Let the utility function be

$$u(x) = \log(x),$$

and let T be the terminal time. Starting from initial wealth $W_0 = \omega$, we choose a predictable strategy (α_t) to maximise

$$\mathbb{E}_{\mathbb{P}}[u(W_T)] = \mathbb{E}_{\mathbb{P}}[\log(W_T)].$$

The dynamic-programming value functions $v_t(x)$, for $t = 0, 1, \dots, T$, satisfy

$$v_T(x) = u(x) = \log(x).$$

By Proposition 3.22, the Bellman equation for $t < T$ is

$$v_t(x) = \text{esssup}_{\alpha_t} \mathbb{E}_{\mathbb{P}}[v_{t+1}(G_t(x, \alpha_t, R_t)) \mid \mathcal{F}_{t-1}].$$

Here G_t describes the one-period wealth update. Since $d = 1$, there is one risky asset and one control variable α_t . The wealth dynamics are

$$W_t = W_{t-1}(1 + \alpha_t R_t).$$

Using the logarithmic ansatz in the Bellman equation gives

$$v_t(x) = \text{esssup}_{\alpha_t} \mathbb{E}_{\mathbb{P}}[\log(x(1 + \alpha_t R_t)) \mid \mathcal{F}_{t-1}] = \log(x) + \text{esssup}_{\alpha_t} \mathbb{E}_{\mathbb{P}}[\log(1 + \alpha_t R_t) \mid \mathcal{F}_{t-1}].$$

The term $\log(x)$ does not depend on α_t and therefore does not affect the maximiser. Hence,

$$\alpha_t^*(\omega) \in \arg \max_{\gamma} \mathbb{E}_{\mathbb{P}}[\log(1 + \gamma R_t) \mid \mathcal{F}_{t-1}](\omega).$$

This is precisely the claimed characterisation. No independence assumption on the returns is required. $\square$

EXERCISE A.30 (Optimal Control Variance-optimal Hedging). Let us consider the variance-optimal hedging problem, where $(S_t)_{t \in T}$ denotes a one-dimensional square-integrable asset price process and $H \in \mathcal{L}^2(\Omega, \mathcal{F}_T, \mathbb{P})$ a European contingent claim. Moreover, we assume that $(S_t)_{t \in T}$ is already a martingale under $\mathbb{P}$. We already know from Theorem 2.6 that the variance-optimal hedge (W_0^*, φ^*) is given by

$$\begin{cases} \varphi_{t+1}^* := \frac{\text{Cov}(\Delta \widehat{W}_{t+1}, \Delta X_{t+1} \mid \mathcal{F}_t)}{\sigma_{t+1}^2} \mathbb{1}_{\{\sigma_{t+1} \neq 0\}}, & t = 0, \dots, T-1, \\ W_0^* := \widehat{W}_0 = \mathbb{E}_{\mathbb{P}}[\widetilde{H}]. \end{cases}$$

Identify the variance-optimal hedging problem with a stochastic optimal control problem and show that the variance-optimal hedge (W_0^*, φ^*) solves the optimal control problem using Proposition 3.22.

SOLUTION. Let $(S_t)_{t \in T}$ be a one-dimensional square-integrable martingale price process under $\mathbb{P}$, and let $H \in \mathcal{L}^2(\Omega, \mathcal{F}_T, \mathbb{P})$ be a European contingent claim. The variance-optimal hedging problem is to find a self-financing strategy $\phi = (\phi_t)_{t=1}^T$ and initial wealth W_0 that minimise $\mathbb{E}[(H - W_T)^2]$. Since S is already discounted, a self-financing strategy satisfies

$$W_{t+1} = W_t + \phi_{t+1}(S_{t+1} - S_t).$$

The variance-optimal hedge (W_0^*, ϕ^*) given by [Theorem 2.6](#) is

$$\phi_{t+1}^* = \frac{\text{Cov}(\widehat{\Delta W_{t+1}}, \Delta X_{t+1} \mid \mathcal{F}_t)}{\sigma_{t+1}^2} \mathbb{1}_{\{\sigma_{t+1} \neq 0\}}, \quad W_0^* = \widehat{W}_0 = \mathbb{E}[\widetilde{H}],$$

where the quantities on the right-hand side come from the Galtchouk–Kunita–Watanabe decomposition. Define the state process by $V_t = W_t$ and the control by $\alpha_t = \phi_{t+1}^*$. The terminal cost is

$$g_T(V_T) = (H - V_T)^2.$$

There is no running cost. Hence, for $t < T$, the cost-to-go function is

$$v_t(x) := \text{essinf}_{\{\alpha_t, \dots, \alpha_{T-1}\}} \mathbb{E}[(H - V_T)^2 \mid V_t = x, \mathcal{F}_t].$$

The state dynamics are

$$V_{t+1} = G_t(V_t, \alpha_t, \varepsilon_t) = V_t + \alpha_t(S_{t+1} - S_t),$$

with $g_t \equiv 0$ for $t < T$. This matches the framework of [Proposition 3.22](#). The backward recursion is

$$v_T(x) = g_T(x) = (H - x)^2, \quad v_t(x) = \text{essinf}_{\alpha_t} \mathbb{E}[v_{t+1}(x + \alpha_t(S_{t+1} - S_t)) \mid \mathcal{F}_t], \quad t = 0, \dots, T-1.$$

The control $\alpha_t^* = \phi_{t+1}^*$ attains the conditional essential infimum at every step. Therefore, the strategy (W_0^*, ϕ^*) from [Theorem 2.6](#) solves this stochastic control problem. $\square$

EXERCISE A.31 (Maximising Expected Utility from Consumption). Consider the maximisation of $\mathbb{E}_{\mathbb{P}}[\sum_{t=0}^T \beta^t u(C_t)]$ in the setting of [Section 3.3](#), with the power utility function $u(x) = \frac{x^\gamma}{\gamma}$ for $\gamma < 1$ and $\gamma \neq 0$. Show that the optimal consumption is of the form

$$C_t^* = c \frac{\beta^{\frac{t}{1-\gamma}}}{N_t^{\frac{1}{1-\gamma}}},$$

determine c , and compute the maximal value of the objective function.

SOLUTION. The objective is

$$\max_{(C_t)_{t=0}^T} \mathbb{E}_{\mathbb{P}} \left[\sum_{t=0}^T \beta^t u(C_t) \right],$$

subject to $\sum_{t=0}^T \mathbb{E}_{\mathbb{P}}[N_t C_t] = W_0$, where N_t is the pricing process and $W_0 > 0$ is the initial wealth. The Lagrangian is

$$\mathcal{L} = \mathbb{E}_{\mathbb{P}} \left[\sum_{t=0}^T \beta^t \frac{C_t^\gamma}{\gamma} \right] - \lambda \left(\sum_{t=0}^T \mathbb{E}_{\mathbb{P}}[N_t C_t] - W_0 \right) = \sum_{t=0}^T \left(\beta^t \mathbb{E}_{\mathbb{P}} \left[\frac{C_t^\gamma}{\gamma} \right] - \lambda \mathbb{E}_{\mathbb{P}}[N_t C_t] \right) + \lambda W_0.$$

For each t and $\omega \in \Omega$, the first-order condition is

$$\frac{\partial \mathcal{L}}{\partial C_t(\omega)} = \beta^t C_t(\omega)^{\gamma-1} - \lambda N_t(\omega) = 0.$$

Therefore,

$$C_t(\omega) = \left(\frac{\beta^t}{\lambda N_t(\omega)} \right)^{\frac{1}{1-\gamma}} = \frac{1}{\lambda^{\frac{1}{1-\gamma}}} \frac{\beta^{\frac{t}{1-\gamma}}}{N_t(\omega)^{\frac{1}{1-\gamma}}} = c \frac{\beta^{\frac{t}{1-\gamma}}}{N_t(\omega)^{\frac{1}{1-\gamma}}},$$

where $c = \lambda^{-\frac{1}{1-\gamma}}$. Substitution into the budget constraint gives

$$W_0 = c \sum_{t=0}^T \beta^{\frac{t}{1-\gamma}} \mathbb{E}_{\mathbb{P}} \left[N_t^{-\frac{\gamma}{1-\gamma}} \right] \implies c = \frac{W_0}{\sum_{t=0}^T \beta^{\frac{t}{1-\gamma}} \mathbb{E}_{\mathbb{P}} \left[N_t^{-\frac{\gamma}{1-\gamma}} \right]}.$$

Set $M_t := \mathbb{E}_{\mathbb{P}} \left[N_t^{-\frac{\gamma}{1-\gamma}} \right]$. The maximal value is

$$\max_{(C_t)_{t=0}^T} \mathbb{E}_{\mathbb{P}} \left[\sum_{t=0}^T \beta^t u(C_t) \right] = \frac{c^\gamma}{\gamma} \sum_{t=0}^T \beta^{\frac{t}{1-\gamma}} M_t = \frac{W_0^\gamma}{\gamma \left(\sum_{t=0}^T \beta^{\frac{t}{1-\gamma}} M_t \right)^{\gamma-1}}.$$

□

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